Lecture notes. Lecture 11

Quantum information for newcomers
Quantum information theory and physical realisations of qubits

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These notes introduce the language of quantum information and quantum computing for bachelor students of physics, mathematics and computer science who want to learn quantum machine learning. They assume no course in quantum mechanics. We describe a register of N qubits by a vector of 2^N complex amplitudes and use gates, which are evolution operators, together with the Born rule to predict the bit strings recorded in different measurement bases. For the two-spin register we then study the joint correlations of the two spins and the state of either spin alone. Density matrices and the partial trace describe these reduced states, and they also describe how measurement back-action and noise change a state. Part I of the notes develops this language, which is the same for every physical system that carries qubits, and closes in Section 15 with what it allows and forbids, and Part II shows how atoms, superconducting circuits, spins and photons provide qubits and how interactions between them produce two-qubit gates. The standard textbook of the field is Ref. [1], and Ref. [20] is a free complement. The numerical values quoted below come from exact state vectors or from simulated measurement records with fixed random seeds, and no model is trained.

Prerequisites

Complex numbers and their modulus and phase. Vectors and matrices with complex entries, the matrix product, the conjugate transpose, eigenvalues and eigenvectors. Elementary probability, a probability distribution, the mean and the standard deviation. Some Python with NumPy for the numerical examples. Physical words such as spin, Hamiltonian, energy level or magnetic field are explained at their first appearance.

The same objects carry different names in physics, mathematics and computer science. Table 1 collects the names used in these notes.

physics	mathematics	computer science
qubit state $\|\psi\rangle$	unit vector in $\mathbb{C}^2$	two complex amplitudes
register of N qubits	unit vector in $\mathbb{C}^{2^N} = (\mathbb{C}^2)^{\otimes N}$	array of 2^N complex numbers
computational basis state $\|b_0 \dots b_{N-1}\rangle$	standard basis vector e_i	bit string, binary index i
ket $\|\psi\rangle$, bra $\langle\psi\|$	column vector, conjugate row vector	`psi`, `psi.conj()`
tensor product $\otimes$	Kronecker product	`np.kron`
observable	Hermitian matrix	quantity read out, with real values
Hamiltonian $\hat{\mathcal{H}}$	Hermitian generator of $e^{-it\hat{\mathcal{H}}}$	energy matrix of the device
gate, time evolution	unitary matrix	reversible instruction
circuit	product of unitary matrices	program
measurement, Born rule	projectors, $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$	sampling a bit string
expectation value $\langle\hat{O}\rangle$	$\mathrm{Tr}(\hat{O}\hat{\rho})$	mean over samples
density matrix $\hat{\rho}$	positive matrix of unit trace	generalised probability distribution
partial trace	linear map Tr_1 on matrices	marginalisation
noise channel	completely positive trace-preserving map	noisy instruction

Table 1: Physics, mathematics and computer-science names for the same objects.

Part I
Introduction to quantum information theory

The first part describes qubits by the rules of quantum mechanics alone. A qubit is a unit vector in a two-dimensional complex vector space, a gate is a unitary matrix generated by a Hamiltonian, and a measurement returns bit strings with the probabilities of the Born rule. None of these rules refers to the system that carries the qubit, so every statement of this part holds for an atom, a superconducting circuit, a spin or a photon alike. Quantum algorithms and quantum machine-learning methods are formulated in this platform-independent language.

1 Prepare, evolve, measure

A classical bit is a physical system with two distinguishable states, such as a voltage that is high or low or a magnetic domain that points up or down. We prepare it in one of its two states, transform it with logic gates and read it out. Quantum information keeps these three operations and replaces the two-state system by a quantum one, which carries one quantum bit, a *qubit*. Atoms, superconducting circuits, spins and photons provide such systems for different physical reasons, which Part II describes. A qubit can be prepared in a superposition of its two basis states. Its readout still returns one bit, 0 or 1, drawn at random with probabilities fixed by the state, and one run says little about the state. Quantum mechanics predicts probabilities, which we compare with records of many identically prepared runs.

A classical bit has the value 0 or 1. A *probabilistic bit*, a coin whose outcome we do not yet know, is described by the vector (p_0, p_1) of its two probabilities. The entries of this stochastic vector are non-negative and sum to one. Random operations act on it with stochastic matrices, whose columns are such vectors. A qubit is described by a vector of two *complex* numbers (a, b) with $|a|^2 + |b|^2 = 1$, a unit vector in $\mathbb{C}^2$, and unitary matrices change it. Reading it out samples a bit with the probabilities $|a|^2$ and $|b|^2$. Because a and b carry complex phases, two paths to the same outcome can cancel, which probabilities never do. Classical logic gates such as AND map two input bits to one output bit and cannot be undone. Quantum gates are unitary and therefore reversible, so a quantum computer runs reversible versions of the classical gates. A quantum program is a circuit, a sequence of gates, whose only output is a list of sampled bit strings.

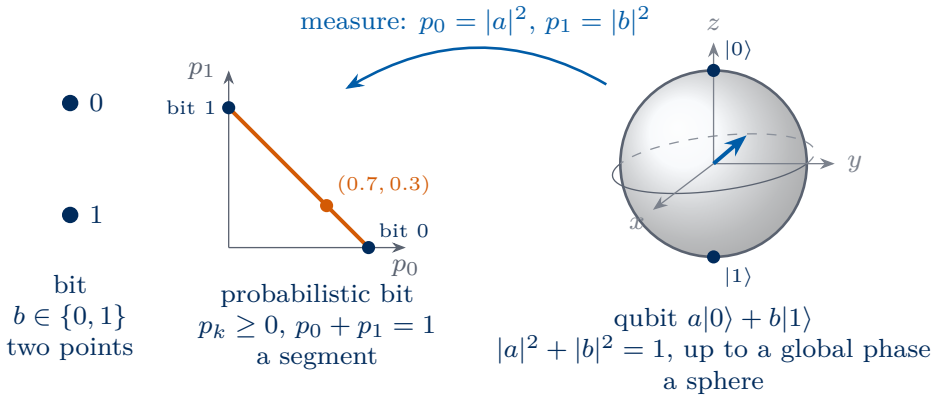


Figure 1: A bit takes one of two values, and a probabilistic bit is a point on the segment of probability vectors (p_0, p_1). A pure qubit state is a point on a sphere, the Bloch sphere.

Every experiment described here, and every quantum algorithm built on it, consists of three operations (Figure 2). We *prepare* a known initial state, usually all qubits in the state $|0\rangle$. We let the state *evolve* under a Hamiltonian, the matrix of energies of the system, switched on for a chosen time. Such a fixed evolution is a *gate*. We *measure* every qubit and record a bit string, with the

measurement basis chosen by the gates applied just before the readout. Our goal is to predict the measurement records of a register of two qubits, the *two-qubit register*, for any preparation, evolution and measurement basis. We also predict measurements performed on either qubit alone. Each concept appears first on one qubit. We then apply it to the register of two qubits, labelled qubit 0 and qubit 1, and finally state the rule for N qubits. Every quantum machine-learning method takes as input the objects defined along this route. They are Pauli strings and their expectation values estimated from bit strings, measurement settings, density matrices, Hamiltonians, time evolution and tensor contractions.

On paper every operation is linear algebra with vectors of length 2^N and matrices of size $2^N \times 2^N$ built with the Kronecker product. We write each operation first in this form. A computer evaluates the same index sums more efficiently as tensor contractions, written as `einsum` strings of the Python libraries NumPy and JAX and pictured as tensor-network diagrams. A short remark after each matrix form adds this view. The two forms agree number for number, and the contraction view leads to the tensor networks used to simulate many qubits.

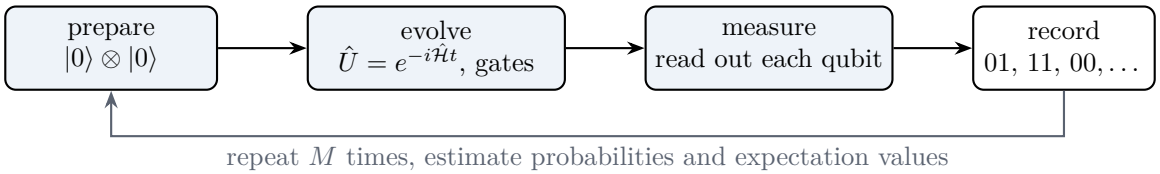


Figure 2: The three operations of every quantum experiment. A single run returns one bit string, so probabilities and expectation values are estimated from a record of M repeated runs, the shots.

The rules of quantum mechanics used here

Quantum mechanics enters through four rules, which we state as definitions and motivate physically where they are first used.

1. *State.* The state of N qubits is a unit vector $|\psi\rangle$ in $\mathbb{C}^{2^N}$, defined up to a global phase factor. When the preparation is uncertain, it is a density matrix.
2. *Evolution.* An isolated system evolves by a unitary matrix, $|\psi\rangle \to \hat{U}|\psi\rangle$, with $\hat{U} = e^{-it\hat{\mathcal{H}}}$ for a time-independent Hamiltonian $\hat{\mathcal{H}}$ switched on for a time t.
3. *Measurement.* Reading out the qubits returns the bit string b with the probability $|\langle b|\psi\rangle|^2$, the Born rule, and leaves the register in $|b\rangle$.
4. *Composition.* The state space of several qubits is the tensor product of their state spaces, computed with the Kronecker product.

2 One qubit

2.1 The qubit as a two-level system

A qubit is a quantum system whose states form a two-dimensional complex vector space spanned by two distinguishable basis states. Often these are two energy levels, the lower *ground state* $|g\rangle$ and the higher *excited state* $|e\rangle$, and we label them $|g\rangle = |0\rangle$ and $|e\rangle = |1\rangle$. Quantum mechanics allows superpositions of the two levels, although a measurement in the energy basis always finds the system in one of them. The energy difference $\Delta E = \hbar\omega_0$, with the reduced Planck constant $\hbar$, fixes the angular frequency ω_0 of the radiation that the qubit absorbs or emits. We use units with $\hbar = 1$, in which energies are measured as angular frequencies, so $\Delta E = \omega_0$. A register of N qubits is a set of N such systems that are prepared, transformed and measured individually and made to interact.

2.2 The state vector and Dirac notation

We choose the two levels as a basis. A spin-1/2 is the intrinsic angular momentum of an electron or a nucleus, whose two basis states have the spin pointing up or down along the z axis. Quantum information calls the basis states of any qubit $|0\rangle$ and $|1\rangle$. We label them $|0\rangle = |g\rangle = |\uparrow\rangle$ and $|1\rangle = |e\rangle = |\downarrow\rangle$ throughout, so the ground state and spin up are both $|0\rangle$, and represent them by the column vectors

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{2.1}$$

They form the *computational basis*. Dirac notation is shorthand for column and row vectors and the matrices built from them. The symbol $|\cdot\rangle$, a *ket*, denotes a column vector. Its conjugate transpose is a row vector, the *bra* $\langle\cdot| = (|\cdot\rangle)^\dagger$, where the dagger $\dagger$ means complex conjugation followed by transposition. A bra times a ket is the *inner product*

$$\langle\phi|\psi\rangle = \sum_k \phi_k^* \psi_k, \tag{2.2}$$

a complex number. Two states are orthogonal when their inner product vanishes, as $\langle 0|1\rangle = 0$, and a state is normalised when $\langle\psi|\psi\rangle = 1$. A general state of one qubit is a superposition

$$|\psi\rangle = a|0\rangle + b|1\rangle = \begin{pmatrix} a \\ b \end{pmatrix}, \qquad a, b \in \mathbb{C}, \qquad |a|^2 + |b|^2 = 1. \tag{2.3}$$

The complex numbers a and b are the *amplitudes*. On a computer the state of one qubit is an array of two complex numbers. The Born rule turns $|a|^2$ and $|b|^2$ into probabilities, and the normalisation condition takes its meaning from there. As a running example we take

$$|\psi_{\text{ex}}\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{e^{i\pi/4}}{2}|1\rangle, \tag{2.4}$$

whose normalisation is $3/4 + 1/4 = 1$.

A ket times a bra, in the opposite order, is the *outer product* $|u\rangle\langle v|$, a 2×2 matrix with the entries $u_j v_k^*$. Two of them recur constantly,

$$|0\rangle\langle 0| = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \qquad |1\rangle\langle 1| = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \tag{2.5}$$

Such matrices are *projectors*. Applying either one twice has the same effect as applying it once, and $|0\rangle\langle 0|\psi\rangle = a|0\rangle$ keeps only the $|0\rangle$ component of the state. Together the two projectors add up to the identity, $|0\rangle\langle 0| + |1\rangle\langle 1| = \hat{I}$. A matrix equal to its own conjugate transpose, $\hat{O}^\dagger = \hat{O}$, is *Hermitian* and has real eigenvalues. A *unitary* matrix, $\hat{U}^\dagger\hat{U} = \hat{I}$, preserves inner products and with them the normalisation.

2.3 Measurement in the computational basis and the Born rule

A measurement in the computational basis returns one bit, 0 or 1, and the *Born rule* gives the probabilities of the two outcomes,

$$p(0) = |\langle 0|\psi\rangle|^2 = |a|^2, \qquad p(1) = |\langle 1|\psi\rangle|^2 = |b|^2. \tag{2.6}$$

Normalisation is the statement $p(0) + p(1) = 1$. The measurement leaves the qubit in the basis state that was found, and a second measurement in the same basis repeats the result. We estimate the probabilities from many runs on identically prepared qubits, as in Figure 2. For the example state (2.4), $p(0) = |\sqrt{3}/2|^2 = 3/4$ and $p(1) = |e^{i\pi/4}/2|^2 = 1/4$. The phase factor $e^{i\pi/4}$ plays no role in this measurement.

2.4 Global and relative phase

Multiplying the whole state by a phase factor, $|\psi\rangle \to e^{i\alpha}|\psi\rangle$, changes no prediction of quantum mechanics. Every probability contains the amplitudes in pairs such as a^*a, b^*b or a^*b, where the factor $e^{-i\alpha}e^{i\alpha} = 1$ drops out. Such a *global phase* is unobservable, and we may always choose the amplitude a real and non-negative.

The phase between the two amplitudes, the *relative phase*, can be measured. The states

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \qquad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \tag{2.7}$$

differ only by the relative phase $e^{i\pi} = -1$ of the second amplitude. Their values of $|a|^2$ and $|b|^2$ coincide, and so do their outcome probabilities in the computational basis. Yet they are orthogonal, $\langle +|-\rangle = \frac{1}{2}(1-1) = 0$, and orthogonal states can be distinguished with certainty by a suitable measurement.

2.5 Observables and the Pauli matrices

A measurement that records $+1$ for the outcome 0 and -1 for the outcome 1 measures the spin along z in units of $\hbar/2$. The average of the recorded values over many runs is

$$\langle \hat{Z} \rangle = (+1)\, p(0) + (-1)\, p(1) = p(0) - p(1). \tag{2.8}$$

Quantum mechanics represents a measured quantity by a Hermitian matrix, an *observable*. Its eigenvalues are the values that can be recorded, and its eigenvectors, the *eigenstates*, are the states left behind by the measurement. By the spectral theorem, every Hermitian matrix has real eigenvalues and an orthonormal basis of eigenvectors. Recorded values are therefore real, and eigenstates of different outcomes are orthogonal. For the spin along z the matrix is $\hat{Z} = (+1)|0\rangle\langle 0| + (-1)|1\rangle\langle 1|$. The spin can equally be measured along x or along y. Such a measurement again records ± 1, and it leaves behind the equal superpositions $|\pm\rangle$ for the x axis and $|\pm i\rangle = (|0\rangle \pm i|1\rangle)/\sqrt{2}$ for the y axis. Building the matrices from these eigenstates as we built $\hat{Z}$, $\hat{X} = |+\rangle\langle +| - |-\rangle\langle -|$ and $\hat{Y} = |+i\rangle\langle +i| - |-i\rangle\langle -i|$, we obtain the three *Pauli matrices*

$$\hat{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \hat{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \hat{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.9}$$

Operators carry a hat throughout these notes. An array of numbers equated with an operator, as in equation (2.9), is the matrix of that operator in the computational basis and carries no hat. The Pauli matrices are also written $\hat{\sigma}_x \equiv \hat{X}$, $\hat{\sigma}_y \equiv \hat{Y}$, $\hat{\sigma}_z \equiv \hat{Z}$, with the vector $\hat{\boldsymbol{\sigma}} = (\hat{\sigma}_x, \hat{\sigma}_y, \hat{\sigma}_z)$. Each Pauli matrix is Hermitian and unitary, squares to the identity, $\hat{X}^2 = \hat{Y}^2 = \hat{Z}^2 = \hat{I}$, and has the eigenvalues ± 1. Two different Pauli matrices anticommute, for example $\hat{X}\hat{Z} = -\hat{Z}\hat{X}$. Their eigenvectors are

$$\hat{Z}:\ |0\rangle, |1\rangle, \qquad \hat{X}:\ |\pm\rangle = \frac{|0\rangle \pm |1\rangle}{\sqrt{2}}, \qquad \hat{Y}:\ |\pm i\rangle = \frac{|0\rangle \pm i|1\rangle}{\sqrt{2}}, \tag{2.10}$$

with the eigenvalue $+1$ for the first state of each pair. The average of an observable $\hat{O}$ with eigenvalues λ_k in the state $|\psi\rangle$ is the *expectation value*

$$\langle \hat{O} \rangle = \langle \psi | \hat{O} | \psi \rangle = \sum_k \lambda_k\, p(\lambda_k), \tag{2.11}$$

the eigenvalues weighted with their Born probabilities. Expanding $|\psi\rangle$ in the eigenvectors of $\hat{O}$ yields the second equality. For $\hat{Z}$ we recover $p(0) - p(1)$, and the example state has $\langle \hat{Z} \rangle = 3/4 - 1/4 = 1/2$.

2.6 Pauli algebra

Products of Pauli matrices appear when gates are combined, for example in the rotation formula of Section 3, and in the Pauli strings that describe measurements and Hamiltonians of many qubits. A product of two Pauli matrices is again a Pauli matrix up to a factor. With the indices $a, b, c \in \{x, y, z\}$, the Kronecker delta δ_{ab} and the Levi-Civita symbol ε_{abc}, which equals $+1$ when abc is a cyclic order of xyz (xyz, yzx, zxy), -1 for the other orders and 0 when an index repeats, the rule reads

$$\hat\sigma_a \hat\sigma_b = \delta_{ab}\,\hat I + i \sum_c \varepsilon_{abc}\,\hat\sigma_c, \tag{2.12}$$

as multiplying the 2×2 matrices confirms. For $a = b$ it yields $\hat\sigma_a^2 = \hat I$, and for different indices

$$\hat X\hat Y = i\hat Z, \qquad \hat Y\hat Z = i\hat X, \qquad \hat Z\hat X = i\hat Y, \qquad \hat Y\hat X = -i\hat Z, \quad \hat Z\hat Y = -i\hat X, \quad \hat X\hat Z = -i\hat Y. \tag{2.13}$$

We add and subtract (2.12) with a and b exchanged and find the anticommutators and commutators,

$$\{\hat\sigma_a, \hat\sigma_b\} = \hat\sigma_a\hat\sigma_b + \hat\sigma_b\hat\sigma_a = 2\delta_{ab}\,\hat I, \qquad [\hat\sigma_a, \hat\sigma_b] = \hat\sigma_a\hat\sigma_b - \hat\sigma_b\hat\sigma_a = 2i\sum_c \varepsilon_{abc}\,\hat\sigma_c. \tag{2.14}$$

Since the Pauli matrices are traceless, the trace of (2.12) is $\mathrm{Tr}(\hat\sigma_a\hat\sigma_b) = 2\delta_{ab}$. The four matrices $\hat I, \hat X, \hat Y, \hat Z$ are therefore orthogonal with respect to $\mathrm{Tr}(\hat A^\dagger \hat B)$ and form a basis of all 2×2 matrices,

$$\hat M = c_0\,\hat I + \mathbf{c}\cdot\hat{\boldsymbol\sigma}, \qquad c_0 = \tfrac12 \mathrm{Tr}\hat M, \qquad c_a = \tfrac12 \mathrm{Tr}(\hat\sigma_a \hat M). \tag{2.15}$$

The matrix $\hat M$ is Hermitian if and only if c_0 and $\mathbf{c}$ are real. For real coefficients the anticommutators yield $(\mathbf{c}\cdot\hat{\boldsymbol\sigma})^2 = |\mathbf{c}|^2 \hat I$. The eigenvalues of $\hat M$ are then $c_0 \pm |\mathbf{c}|$, and every unit-vector combination $\mathbf{n}\cdot\hat{\boldsymbol\sigma}$ has the eigenvalues ± 1.

The spin operators of a spin-1/2 are $\hat S_a = \hat\sigma_a/2$, the spin components in units of $\hbar$. Their combinations $\hat S^\pm = \hat S_x \pm i\hat S_y$, the *ladder operators*, are the transition operators between the two basis states,

$$\hat S^+ = |0\rangle\langle 1| = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \hat S^- = |1\rangle\langle 0| = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{2.16}$$

The operator $\hat S^+$ maps $|1\rangle$ to $|0\rangle$ and annihilates $|0\rangle$, so it raises the spin from down to up, and $\hat S^-$ lowers it. Many texts use the unnormalised combinations $\hat\sigma^\pm = \hat\sigma_x \pm i\hat\sigma_y = 2\hat S^\pm$.

2.7 The Bloch sphere

Up to the global phase, every normalised state (2.3) can be written with two angles,

$$|\psi\rangle = \cos\frac{\vartheta}{2}|0\rangle + e^{i\varphi}\sin\frac{\vartheta}{2}|1\rangle, \qquad 0 \le \vartheta \le \pi, \quad 0 \le \varphi < 2\pi. \tag{2.17}$$

The angles are the polar and azimuthal angles of a point on the unit sphere, the *Bloch sphere*. Its north pole is $|0\rangle$, its south pole $|1\rangle$, and its equator holds the equal superpositions $|\pm\rangle$ and $|\pm i\rangle$. The Cartesian coordinates of the point are the expectation values of the three Pauli matrices, obtained from (2.11) by matrix multiplication,

$$\mathbf{r} = (\langle\hat X\rangle, \langle\hat Y\rangle, \langle\hat Z\rangle) = (\sin\vartheta\cos\varphi,\ \sin\vartheta\sin\varphi,\ \cos\vartheta). \tag{2.18}$$

The vector $\mathbf{r}$ is the *Bloch vector*, and every pure state has $|\mathbf{r}| = 1$. The relative phase is the azimuth φ, so it drops out of $\langle\hat Z\rangle$ and appears in $\langle\hat X\rangle$ and $\langle\hat Y\rangle$. Orthogonal states sit at antipodal points, as $|+\rangle$ and $|-\rangle$ at $\pm\hat x$. In the example state, $\cos(\vartheta/2) = \sqrt3/2$ fixes $\vartheta = \pi/3$ and the phase fixes $\varphi = \pi/4$, so

$$\mathbf{r}_{\mathrm{ex}} = \left(\tfrac{\sqrt3}{2}\cos\tfrac{\pi}{4},\ \tfrac{\sqrt3}{2}\sin\tfrac{\pi}{4},\ \tfrac12\right) \approx (0.612,\ 0.612,\ 0.500). \tag{2.19}$$

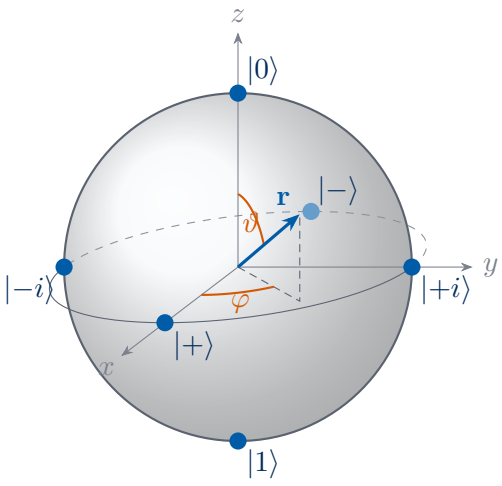


Figure 3: A pure state as a point on the Bloch sphere, shown for the example state with $\vartheta = \pi/3$ and $\varphi = \pi/4$. The polar angle ϑ sets the probabilities of the bits, and the azimuth φ is the relative phase. The six axis states $|0\rangle$, $|1\rangle$, $|\pm\rangle$ and $|\pm i\rangle$ are marked.

3 Evolution and gates as rotations

3.1 Evolution under a Hamiltonian

Gates move the state of a qubit on the Bloch sphere, and each gate is the evolution of the qubit under a Hamiltonian switched on for a fixed time. A closed quantum system evolves according to the Schrödinger equation, in units with $\hbar = 1$,

$$i\,\frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle, \tag{3.1}$$

where the *Hamiltonian* $\hat{\mathcal{H}}$ is a Hermitian matrix, 2×2 for one qubit. Its eigenvalues are the energies of the system, the *energy levels*, and its eigenvectors are the states of definite energy. Through (3.1) the same matrix generates the time evolution. A device is controlled by switching fields or couplings on and off, which changes $\hat{\mathcal{H}}$. For a Hamiltonian that does not depend on time the equation is a linear system of differential equations with constant coefficients, solved by the *matrix exponential*. For any square matrix $\hat{A}$ we define it by the power series

$$e^{\hat{A}} = \sum_{k\geq 0} \frac{\hat{A}^k}{k!} = \hat{I} + \hat{A} + \frac{\hat{A}^2}{2} + \frac{\hat{A}^3}{6} + \dots, \tag{3.2}$$

which converges for every matrix, as the series of the ordinary exponential does. For $\hat{A} = -it\hat{\mathcal{H}}$ we differentiate each term of the series. Every term contains only powers of the one matrix $\hat{\mathcal{H}}$, so the factors commute, and

$$\frac{d}{dt}e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 1} \frac{k\,(-i\hat{\mathcal{H}})^k\,t^{k-1}}{k!} = (-i\hat{\mathcal{H}}) \sum_{k\geq 1} \frac{(-it\hat{\mathcal{H}})^{k-1}}{(k-1)!} = -i\hat{\mathcal{H}}\,e^{-it\hat{\mathcal{H}}}. \tag{3.3}$$

The state

$$|\psi(t)\rangle = \hat{U}(t)|\psi(0)\rangle, \qquad \hat{U}(t) = e^{-it\hat{\mathcal{H}}}, \tag{3.4}$$

therefore satisfies $i\frac{d}{dt}|\psi(t)\rangle = i(-i\hat{\mathcal{H}})\hat{U}(t)|\psi(0)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle$, and at $t = 0$ it starts from the right initial state, since $\hat{U}(0) = \hat{I}$. The *evolution operator* $\hat{U}(t)$ is unitary. Its conjugate transpose is $\hat{U}^\dagger = e^{+it\hat{\mathcal{H}}^\dagger} = e^{it\hat{\mathcal{H}}}$ because $\hat{\mathcal{H}}$ is Hermitian, and $\hat{U}^\dagger\hat{U} = e^{it\hat{\mathcal{H}}}e^{-it\hat{\mathcal{H}}} = \hat{I}$, since the exponents commute. Evolution therefore preserves normalisation, and the Born probabilities sum to one at every time. Because the exponents commute, two evolutions in sequence also add their times, $\hat{U}(t_2)\hat{U}(t_1) = \hat{U}(t_1 + t_2)$.

The exponential is easiest to evaluate in the eigenbasis of the Hamiltonian. We diagonalise $\hat{\mathcal{H}} = \sum_k E_k |E_k\rangle\langle E_k|$, with the energies E_k and orthonormal eigenvectors $|E_k\rangle$. Every power is then $\hat{\mathcal{H}}^n = \sum_k E_k^n |E_k\rangle\langle E_k|$, and the series sums to the *spectral form*

$$\hat{U}(t) = \sum_k e^{-iE_k t} |E_k\rangle\langle E_k|. \tag{3.5}$$

An energy eigenstate $|E_k\rangle$ only acquires the phase $e^{-iE_k t}$. Its measurement predictions stay constant in time, and it is called a *stationary state*. A superposition of stationary states does change, since its components acquire different phases. On a computer $\hat{U}(t)$ is obtained either from this eigendecomposition or from a library routine for the matrix exponential, `scipy.linalg.expm` or `jax.scipy.linalg.expm`.

When the Hamiltonian changes in time the solution is no longer a single exponential. For a sequence of steps, each with a constant Hamiltonian $\hat{\mathcal{H}}_m$ for a duration t_m, the evolution is the product $\hat{U} = e^{-it_M \hat{\mathcal{H}}_M} \cdots e^{-it_2 \hat{\mathcal{H}}_2} e^{-it_1 \hat{\mathcal{H}}_1}$, with the latest step leftmost. The order matters whenever the Hamiltonians do not commute. A circuit is such a product of gates, and a smoothly varying $\hat{\mathcal{H}}(t)$ is the limit of many short steps, the time-ordered exponential.

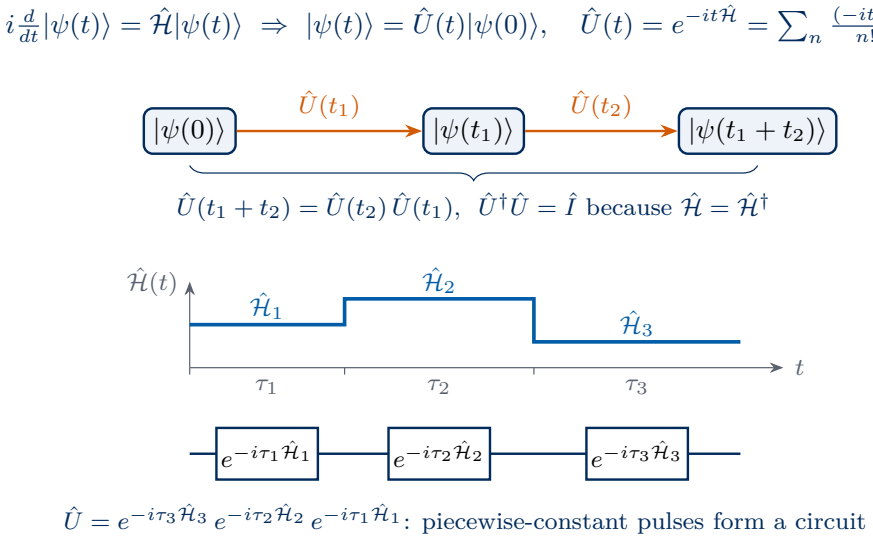


Figure 4: The evolution operator $\hat{U}(t) = e^{-it\hat{\mathcal{H}}}$ solves the Schrödinger equation for a constant Hamiltonian, and a sequence of steps with constant Hamiltonians gives a product of exponentials, a circuit, with the latest step leftmost.

3.2 The exponential of a Pauli matrix

For the Pauli matrices and their products the exponential needs no diagonalisation. The single-qubit gates and most two-qubit gates, which act on a pair of qubits (Section 7), are exponentials $e^{-ia\hat{P}}$ of a matrix $\hat{P}$ that squares to the identity. All even powers of such a matrix equal $\hat{I}$ and all odd powers equal $\hat{P}$. We split the exponential series into even and odd terms,

$$\begin{aligned} e^{-ia\hat{P}} &= \sum_{m\geq 0} \frac{(-ia)^{2m}}{(2m)!}\,\hat{I} + \sum_{m\geq 0} \frac{(-ia)^{2m+1}}{(2m+1)!}\,\hat{P} \\ &= \underbrace{\sum_m \frac{(-1)^m a^{2m}}{(2m)!}}_{\cos a}\,\hat{I} - i\underbrace{\sum_m \frac{(-1)^m a^{2m+1}}{(2m+1)!}}_{\sin a}\,\hat{P} = \cos a\,\hat{I} - i\sin a\,\hat{P}, \end{aligned} \tag{3.6}$$

the matrix version of Euler's formula $e^{-ia} = \cos a - i\sin a$. Every Pauli matrix squares to $\hat{I}$. So does $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} = n_x\hat{X} + n_y\hat{Y} + n_z\hat{Z}$ for a unit vector $\mathbf{n}$, because in its square the cross terms cancel in pairs by

anticommutation, $(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}})^2 = (n_x^2+n_y^2+n_z^2)\hat{I} = \hat{I}$. The rotation about the axis $\mathbf{n}$ by the angle θ is therefore

$$\hat{R}_{\mathbf{n}}(\theta) = e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2} = \cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}. \tag{3.7}$$

It turns the Bloch vector by the angle θ about $\mathbf{n}$, counterclockwise when $\mathbf{n}$ points towards the viewer. For example, the state $\hat{R}_x(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle - i\sin\frac{\theta}{2}|1\rangle$ has the polar angle $\vartheta = \theta$ and the azimuth $\varphi = -\pi/2$ in (2.17), so by (2.18) its Bloch vector is $(0, -\sin\theta, \cos\theta)$, the north pole turned by θ about the x axis. The rotations about the coordinate axes are $\hat{R}_x(\theta) = e^{-i\theta\hat{X}/2}$, $\hat{R}_y(\theta) = e^{-i\theta\hat{Y}/2}$ and

$$\hat{R}_z(\theta) = e^{-i\theta\hat{Z}/2} = \begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix} = e^{-i\theta/2}\begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}. \tag{3.8}$$

A Hamiltonian proportional to $\hat{Z}$, $\hat{\mathcal{H}} = \frac{\omega}{2}\hat{Z}$ with an angular frequency ω, generates $\hat{U}(t) = \hat{R}_z(\omega t)$. Up to a global phase this evolution multiplies the amplitude of $|1\rangle$ by $e^{i\omega t}$. In the form (2.17) the azimuth grows as $\varphi \to \varphi + \omega t$ while ϑ stays fixed, so the Bloch vector turns about the z axis with the angular frequency ω, counterclockwise seen from the north pole. This motion is called *precession*. In the same way $\hat{\mathcal{H}} = \frac{\Omega}{2}\hat{X}$ generates $\hat{R}_x(\Omega t)$. For any other axis we choose the coordinate axes of the Bloch sphere so that $\mathbf{n}$ becomes the new z axis. In these coordinates $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ plays the role of $\hat{Z}$, and the rotation is again a precession by θ about $\mathbf{n}$. Because of the factor $1/2$ in the exponent, the amplitudes acquire phases of only $\theta/2$ while the Bloch vector turns by θ. After a full turn, $\hat{R}_{\mathbf{n}}(2\pi) = -\hat{I}$ returns every Bloch vector to itself and multiplies the state by the unobservable global phase -1.

3.3 Single-qubit gates as rotations

Every single-qubit gate is, up to a global phase that no measurement can see, a rotation (3.7). A 2×2 unitary $\hat{U}$ has two orthonormal eigenvectors $|u_\pm\rangle$ with eigenvalues of modulus one, which we write as $e^{i\eta}e^{\mp i\theta/2}$. The difference of the two projectors, $|u_+\rangle\langle u_+| - |u_-\rangle\langle u_-|$, is Hermitian with the eigenvalues ± 1 and therefore equals $\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ for a unit vector $\mathbf{n}$ by (2.15), while their sum is $\hat{I}$. Hence $\hat{U} = e^{i\eta}\big(\cos\frac{\theta}{2}\,\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}\big) = e^{i\eta}\hat{R}_{\mathbf{n}}(\theta)$. The standard gates are named after their action. The gate $\hat{X}$ is the *bit flip*, $\hat{X}|0\rangle = |1\rangle$ and $\hat{X}|1\rangle = |0\rangle$. The gate $\hat{Z}$ is the *phase flip*, which leaves $|0\rangle$ alone and multiplies $|1\rangle$ by -1. The *phase gates* $\hat{S} = \mathrm{diag}(1, i)$ and $\hat{T} = \mathrm{diag}(1, e^{i\pi/4})$ multiply $|1\rangle$ by i and by $e^{i\pi/4}$. The *Hadamard gate* $\hat{H} = (\hat{X}+\hat{Z})/\sqrt{2}$ turns $|0\rangle$ into $|+\rangle$ and $|1\rangle$ into $|-\rangle$.

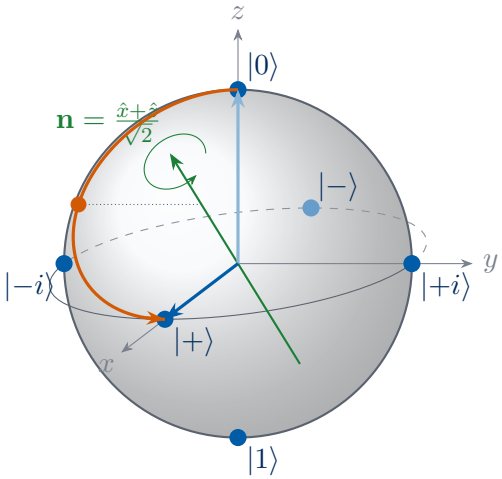


Figure 5: The Hadamard gate as a half turn about the axis $\mathbf{n} = (\hat{x}+\hat{z})/\sqrt{2}$ (green), which carries $|0\rangle$ at the north pole to $|+\rangle$ on the equator along the orange arc. Halfway, at the angle $\pi/2$, the arc passes through the Bloch vector $(\frac{1}{2}, -\frac{1}{\sqrt{2}}, \frac{1}{2})$ (dot).

Each gate is also the evolution under a Hamiltonian, its generator, switched on for the unit time. Evaluating (3.7) at the angle of each gate yields its rotation form. At $\theta = \pi$ the cosine vanishes and

the sine equals one, hence $\hat{R}_{\mathbf{n}}(\pi) = -i\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ and

$$\hat{X} = i\,\hat{R}_x(\pi), \qquad \hat{Z} = i\,\hat{R}_z(\pi), \qquad \hat{H} = \frac{\hat{X}+\hat{Z}}{\sqrt{2}} = i\,\hat{R}_{\mathbf{n}_H}(\pi) \quad \text{with} \quad \mathbf{n}_H = \frac{\hat{x}+\hat{z}}{\sqrt{2}}. \tag{3.9}$$

For the phase gates, $\hat{R}_z(\phi) = \mathrm{diag}(e^{-i\phi/2}, e^{i\phi/2}) = e^{-i\phi/2}\,\mathrm{diag}(1, e^{i\phi})$, which gives $\hat{S} = e^{i\pi/4}\hat{R}_z(\pi/2)$ and $\hat{T} = e^{i\pi/8}\hat{R}_z(\pi/4)$. The generator $\hat{\mathcal{H}}_G$ of a named gate $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$ is

$$\hat{\mathcal{H}}_G = \frac{\theta}{2}\,(\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \hat{I}), \tag{3.10}$$

where the term proportional to $\hat{I}$ produces the global phase $e^{i\theta/2}$, which equals $e^{i\eta}$ for the five named gates of Table 2. A general gate $\hat{G} = e^{i\eta}\hat{R}_{\mathbf{n}}(\theta)$ has the generator $\hat{\mathcal{H}}_G = \frac{\theta}{2}\mathbf{n}\cdot\hat{\boldsymbol{\sigma}} - \eta\hat{I}$, and the plain rotation in the last row of Table 2 has $\eta = 0$.

gate	matrix	rotation	axis $\mathbf{n}$	angle θ	generator $\hat{\mathcal{H}}_G$
$\hat{X}$	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	$i\,\hat{R}_x(\pi)$	$\hat{x}$	π	$\frac{\pi}{2}(\hat{X}-\hat{I})$
$\hat{Z}$	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	$i\,\hat{R}_z(\pi)$	$\hat{z}$	π	$\frac{\pi}{2}(\hat{Z}-\hat{I})$
$\hat{H}$	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	$i\,\hat{R}_{\mathbf{n}_H}(\pi)$	$(\hat{x}+\hat{z})/\sqrt{2}$	π	$\frac{\pi}{2}(\mathbf{n}_H\cdot\hat{\boldsymbol{\sigma}}-\hat{I})$
$\hat{S}$	$\begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}$	$e^{i\pi/4}\hat{R}_z(\pi/2)$	$\hat{z}$	$\pi/2$	$\frac{\pi}{4}(\hat{Z}-\hat{I})$
$\hat{T}$	$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$	$e^{i\pi/8}\hat{R}_z(\pi/4)$	$\hat{z}$	$\pi/4$	$\frac{\pi}{8}(\hat{Z}-\hat{I})$
$\hat{R}_{\mathbf{n}}(\theta)$	$\cos\frac{\theta}{2}\hat{I} - i\sin\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$	$\hat{R}_{\mathbf{n}}(\theta)$	$\mathbf{n}$	θ	$\frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$

Table 2: Single-qubit gates as rotations of the Bloch sphere and as evolution operators $\hat{G} = e^{-i\hat{\mathcal{H}}_G}$. For the five named gates the unobservable global phase equals $e^{i\theta/2}$.

We watch a gate act as a rotation by following the evolution that implements it. For $\hat{G} = e^{i\eta}\hat{R}_{\mathbf{n}}(\theta)$ we switch on $\hat{\mathcal{H}} = \frac{\theta}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ for the times $0 \le t \le 1$, and $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$ ends at the gate. Figure 6 shows the path of the Bloch vector of the state with $\vartheta = 2\pi/3$, $\varphi = \pi/3$, chosen away from every rotation axis.

Each path is an arc of a circle about the axis of the gate, spanning the angle of the gate. The component of the Bloch vector along the axis stays fixed. The gates $\hat{X}$, $\hat{Z}$ and $\hat{H}$ turn the state by half a circle, about x, about z and about the diagonal axis between x and z. The gates $\hat{S}$ and $\hat{T}$ turn it by a quarter and an eighth of a circle about z. Since the three phase gates share their axis, the path of $\hat{T}$ is the first half of the path of $\hat{S}$, which is the first half of the path of $\hat{Z}$.

3.4 A one-qubit circuit and interference

A *circuit* is a sequence of gates applied to a prepared state, followed by a measurement. It is drawn as a horizontal line for the qubit, with time running from left to right, gates as boxes and the measurement as a meter. Gates act on the state from the left, so the gate applied first stands rightmost in the matrix product. The circuit "first $\hat{H}$, then $\hat{S}$" is the matrix $\hat{S}\hat{H}$, and it differs from $\hat{H}\hat{S}$. Starting from $|0\rangle$, $\hat{S}\hat{H}|0\rangle = \hat{S}|+\rangle = |+i\rangle$ points along y, while $\hat{H}\hat{S}|0\rangle = \hat{H}|0\rangle = |+\rangle$ points along x, because $\hat{S}$ leaves $|0\rangle$ unchanged.

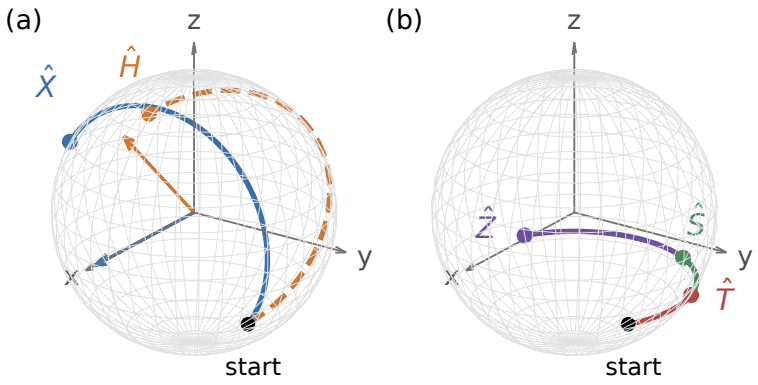


Figure 6: Bloch-vector paths during the evolutions $\hat{U}(t) = \hat{R}_{\mathbf{n}}(\theta t)$, $0 \le t \le 1$, that implement the gates $\hat{X}$, $\hat{Z}$, $\hat{H}$, $\hat{S}$ and $\hat{T}$, started from $\vartheta = 2\pi/3$, $\varphi = \pi/3$. (a) The half turns $\hat{X}$ and $\hat{H}$, with arrows marking their rotation axes. (b) The turns of $\hat{T}$, $\hat{S}$ and $\hat{Z}$ about the z axis, which share one arc, with the end point of each gate marked.

The circuit "first $\hat{H}$, then $\hat{Z}$, then $\hat{H}$" is the matrix $\hat{H}\hat{Z}\hat{H}$. The Hadamard gate exchanges the x and z axes of the Bloch sphere, so a half turn about z framed by two Hadamard gates is a half turn about x,

$$\hat{H}\hat{Z}\hat{H} = \hat{X}. \tag{3.11}$$

The circuit flips $|0\rangle$ to $|1\rangle$ with certainty. The first Hadamard gate prepares $|+\rangle$, the $\hat{Z}$ gate changes only the relative phase and gives $|-\rangle$, and the second Hadamard gate turns this phase difference into a different bit. With $\hat{R}_z(\phi)$ in place of $\hat{Z}$, the circuit becomes the *Ramsey interference* experiment of Figure 7. The amplitude of $|1\rangle$ after $\hat{H}\,\hat{R}_z(\phi)\,\hat{H}|0\rangle$ is

$$\langle 1|\hat{H}\,\hat{R}_z(\phi)\,\hat{H}|0\rangle = \tfrac{1}{2}(e^{-i\phi/2} - e^{i\phi/2}) = -i\sin\frac{\phi}{2}, \qquad \text{so} \qquad p(1) = \sin^2\frac{\phi}{2}. \tag{3.12}$$

Without the second Hadamard gate the state $\hat{R}_z(\phi)|+\rangle$ lies on the equator and $p(1) = 1/2$ for every ϕ. Interference converts a phase that is invisible in a $\hat{Z}$ measurement into a population.

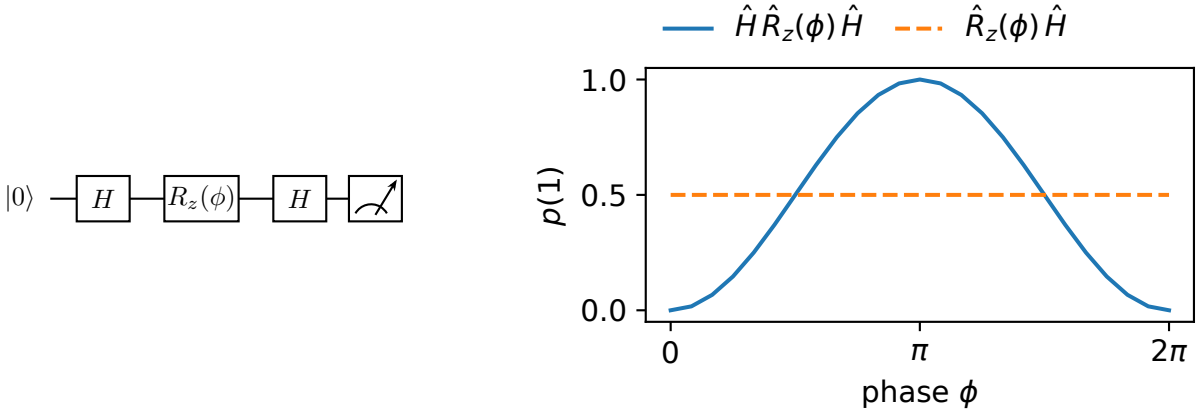


Figure 7: Ramsey interference. Left, the circuit. Right, the probability of the bit 1 against the phase ϕ, with the second Hadamard gate, $\sin^2(\phi/2)$, and without it, a constant $1/2$.

3.5 The Euler decomposition (optional)

A product of rotations is again a rotation, and every 2×2 unitary can be written as

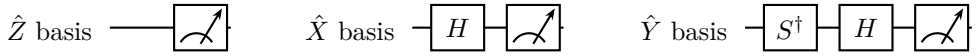
$$\hat{U} = e^{i\eta}\hat{R}_z(\theta_1)\,\hat{R}_y(\theta_2)\,\hat{R}_z(\theta_3), \tag{3.13}$$

the *Euler decomposition* with three angles $\theta_1, \theta_2, \theta_3$ and a global phase η. The set of single-qubit gates is therefore the set of rotations of the Bloch sphere, and two fixed rotation axes suffice to produce all of them. For the Hadamard gate, $\hat{H} = i\,\hat{R}_y(\pi/2)\,\hat{R}_z(\pi)$. For a diagonal gate such as $\hat{T}$ the angle θ_2 vanishes and only the sum $\theta_1 + \theta_3$ is determined, so the decomposition is not unique.

4 Other measurement bases and shots

4.1 Measurement along x and y

A laboratory reads a qubit in its computational basis. To measure the spin along x we first rotate the state so that the eigenvectors of $\hat{X}$ are carried to $|0\rangle$ and $|1\rangle$, and then measure $\hat{Z}$. The Hadamard gate, a half turn about the axis between x and z, performs this rotation, $\hat{H}|+\rangle = |0\rangle$ and $\hat{H}|-\rangle = |1\rangle$. For the y axis we first apply $\hat{S}^\dagger = \mathrm{diag}(1, -i)$, which turns $|\pm i\rangle$ into $|\pm\rangle$, and then $\hat{H}$. The three measurement circuits are



In every basis the outcome bit 0 stands for the eigenvalue $+1$ of the measured Pauli matrix and the bit 1 for -1. From $\langle\hat{\sigma}_\alpha\rangle = p_\alpha(0) - p_\alpha(1)$ and $p_\alpha(0) + p_\alpha(1) = 1$ the probabilities are

$$p_\alpha(0) = \frac{1 + \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad p_\alpha(1) = \frac{1 - \langle\hat{\sigma}_\alpha\rangle}{2}, \qquad \alpha = x, y, z. \tag{4.1}$$

For the example state (2.4) the Bloch vector gives $p_x(0) = 0.806$ and $p_y(0) = 0.806$, in addition to $p_z(0) = 0.750$.

The three measurements are incompatible. A state that gives a certain result in one basis gives random bits in the other two. The state $|0\rangle$ always returns 0 in the $\hat{Z}$ basis and random bits with probability one half each, a *fair coin*, in the $\hat{X}$ and $\hat{Y}$ bases, while $|+\rangle$ does the opposite (Figure 9). This pattern is called *complementarity*. Besides interference, it is the second way in which the relative phase shows up, since $|+\rangle$ and $|-\rangle$ have the same $\hat{Z}$ probabilities and opposite $\hat{X}$ outcomes. On the Bloch sphere a pure state has $|\mathbf{r}| = 1$, which allows at most one component to equal ± 1. A state with a certain outcome along one axis therefore lies on the equator of the other two.

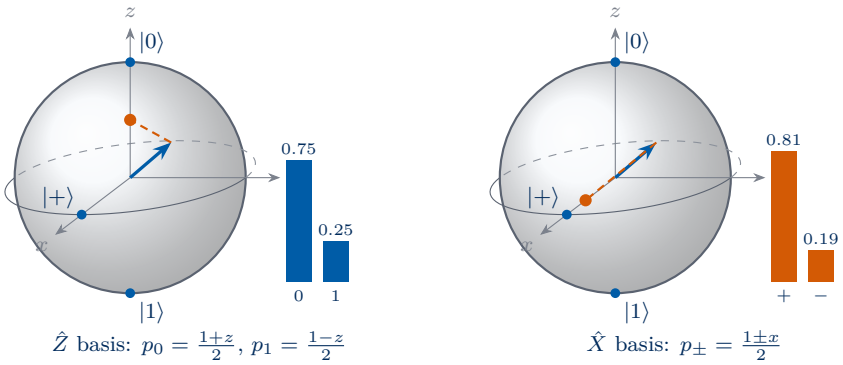


Figure 8: Measurement in the $\hat{Z}$ and $\hat{X}$ bases. The probability of the bit 0 is fixed by the projection of the Bloch vector on the measured axis, $p(0) = (1 + r_\alpha)/2$. After the measurement the state sits at the pole that was found.

4.2 Shots and the estimate of an expectation value

An experiment repeats preparation and measurement M times. Each repetition is a *shot* and returns a value $z_j = \pm 1$. We estimate $\langle\hat{Z}\rangle$ by the sample mean

$$\bar{z} = \frac{1}{M}\sum_{j=1}^{M} z_j. \tag{4.2}$$

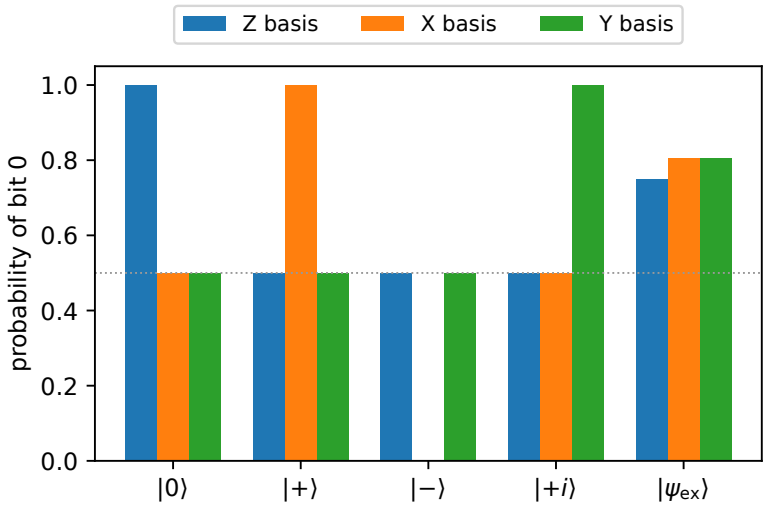


Figure 9: Probability of the bit 0 for five states measured in the $\hat{Z}$, $\hat{X}$ and $\hat{Y}$ bases. A bar at 1 or 0 is a certain outcome, a bar at 0.5 (dotted) a fair coin. No state has certain outcomes in two bases.

Its average over many records is $\langle \hat{Z} \rangle$, so the estimator is unbiased. Because $z_j^2 = 1$, one outcome has the variance $\langle \hat{Z}^2 \rangle - \langle \hat{Z} \rangle^2 = 1 - \langle \hat{Z} \rangle^2$. Since the shots are independent, the variance of the mean is this single-shot variance divided by M, and the standard error is

$$\mathrm{std}(\bar{z}) = \sqrt{\frac{1 - \langle \hat{Z} \rangle^2}{M}}. \tag{4.3}$$

With the error falling as $1/\sqrt{M}$, doubling the precision costs four times as many shots and one more decimal digit one hundred times as many. In practice $\langle \hat{Z} \rangle$ is unknown, and the standard error is estimated by the sample standard deviation of the record divided by $\sqrt{M}$.

A simulated record of 1000 shots of the example state, with the exact value $\langle \hat{Z} \rangle = 1/2$, gives the estimate 0.484 ± 0.028, in agreement with the standard error 0.027 predicted by (4.3). The spread of the estimates over 400 repeated records for each record length M follows the $1/\sqrt{M}$ law (Figure 10, left). The Ramsey experiment of Section 3.4, simulated with 200 shots for each phase, follows the curve $\sin^2(\phi/2)$ within about two error bars $\sqrt{p(1-p)/M}$ (Figure 10, right). Computed from the estimated p, these bars are largest near $p = 1/2$ and vanish where every shot gave the same bit. Every expectation value measured on a quantum device carries this statistical error, and variational algorithms have to budget it.

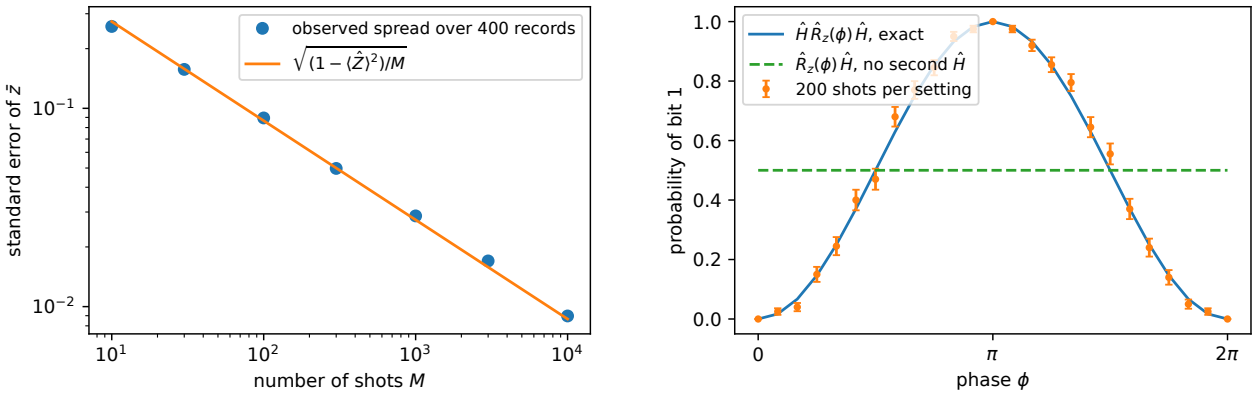


Figure 10: Left, the observed spread of the estimate of $\langle \hat{Z} \rangle$ over repeated records against the number of shots M, compared with formula (4.3). The slope $-1/2$ in the double-logarithmic plot is the $1/\sqrt{M}$ law. Right, the Ramsey fringe measured with 200 shots per phase, with the exact curve and the curve without the second Hadamard gate.

5 The two-qubit register in a computer

5.1 Two qubits and the tensor product

Two qubits, qubit 0 and qubit 1, form the smallest register in which the parts can be correlated. Suppose qubit 0 is prepared in $|\psi^{(0)}\rangle = a|0\rangle + b|1\rangle = (a, b)^T$ and, independently, qubit 1 in $|\psi^{(1)}\rangle = c|0\rangle + d|1\rangle = (c, d)^T$. The superscript names the qubit, and the first factor of a product always belongs to qubit 0. The joint state is the *tensor product*, computed by distributing the product over both sums,

$$|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle = (a|0\rangle + b|1\rangle) \otimes (c|0\rangle + d|1\rangle) = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}. \quad (5.1)$$

Every amplitude of the pair is a product of one amplitude of each qubit. Probabilities of bit strings factorise in the same way, for example $p(01) = |a|^2|d|^2$, as expected for independently prepared qubits. The symbol $\otimes$ is often omitted, and the shorthands $|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle = |\psi^{(0)}\rangle|\psi^{(1)}\rangle$ denote one vector, as in $|0\rangle \otimes |1\rangle = |0\rangle|1\rangle = |01\rangle$. We list the amplitudes of every register in the order $00, 01, 10, 11$ of the column vector in (5.1). In code the vector is `np.kron(psi0, psi1)`, and the unnormalised inputs $(a, b) = (1, 2)$ and $(c, d) = (3, 4)$ return $(3, 4, 6, 8)$.

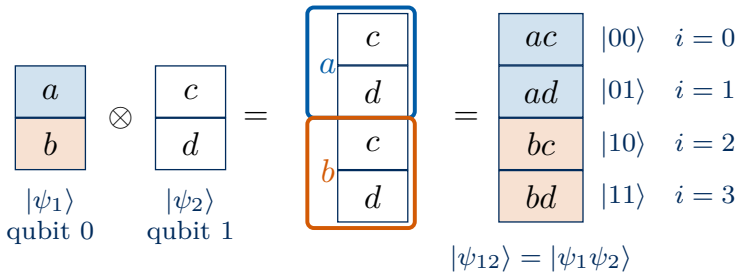


Figure 11: The tensor product $(a, b) \otimes (c, d) = (ac, ad, bc, bd)$. Each amplitude of the first qubit multiplies the whole vector of the second, which produces the order $00, 01, 10, 11$.

A state of the form $|a\rangle \otimes |b\rangle$ is a *product state*. The state space of the pair is the tensor product $\mathbb{C}^2 \otimes \mathbb{C}^2 = \mathbb{C}^4$, spanned by the four products of basis states

$$|00\rangle = |0\rangle \otimes |0\rangle, \quad |01\rangle = |0\rangle \otimes |1\rangle, \quad |10\rangle = |1\rangle \otimes |0\rangle, \quad |11\rangle = |1\rangle \otimes |1\rangle. \quad (5.2)$$

The label of a basis state is a *bit string*. Its first bit is the state of qubit 0 and its second bit the state of qubit 1. A general state of the register is any normalised superposition of the four basis states,

$$|\psi\rangle = \sum_{b_0, b_1 \in \{0,1\}} \psi_{b_0 b_1} |b_0 b_1\rangle, \qquad \sum_{b_0 b_1} |\psi_{b_0 b_1}|^2 = 1, \quad (5.3)$$

with four complex amplitudes. A state that cannot be written as a product of two single-qubit states is called *entangled*, and most states of the register are of this kind.

The spin components of the two qubits along z are $\hat{Z}_0$, the Pauli matrix $\hat{Z}$ acting on qubit 0, and $\hat{Z}_1$, acting on qubit 1. Every basis state is a simultaneous eigenstate of both, and the eigenvalue of $\hat{Z}_q$ on $|b_0 b_1\rangle$ is $(-1)^{b_q}$, equal to $+1$ for $b_q = 0$ and -1 for $b_q = 1$. A measurement of both spins along z reads out the bit string, and the computational basis takes its name from this readout. For N qubits the computational basis consists of the 2^N common eigenvectors of $\hat{Z}_0, \ldots, \hat{Z}_{N-1}$, labelled by the bit strings $b_0 b_1 \ldots b_{N-1}$.

5.2 The vector of 2^N amplitudes and its order

A computer stores the state of N qubits as a vector of 2^N complex numbers, the amplitudes $\psi_{b_0 \ldots b_{N-1}}$ of the computational basis states. To place them in one array we must agree on an order. We use the order produced by the *Kronecker product*, the matrix form of the tensor product computed in (5.1),

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a \begin{pmatrix} c \\ d \end{pmatrix} \\ b \begin{pmatrix} c \\ d \end{pmatrix} \end{pmatrix}. \tag{5.4}$$

Each entry of the left vector multiplies a copy of the right vector. Down the column the left factor, qubit 0, changes slowest and the right factor, qubit 1, fastest. The four entries are the amplitudes of 00, 01, 10 and 11, in binary counting order. The basis vector of a bit string is the Kronecker product $|b_0\rangle \otimes |b_1\rangle \otimes \cdots \otimes |b_{N-1}\rangle$ of single-qubit basis vectors, the unit vector with entry 1 at the index

$$i = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q}. \tag{5.5}$$

Qubit 0 is the most significant bit, and Table 3 lists the order for two and three qubits. For three qubits the bit string 101 sits at the index $1 \cdot 4 + 0 \cdot 2 + 1 \cdot 1 = 5$. Product states are Kronecker products of single-qubit states, and the running product state of the register is

$$|0\rangle \otimes |+\rangle = \tfrac{1}{\sqrt{2}}(1, 1, 0, 0)^T, \tag{5.6}$$

with amplitude $1/\sqrt{2}$ on 00 and on 01, the two bit strings in which qubit 0 is 0. A general state is any normalised vector of length 2^N.

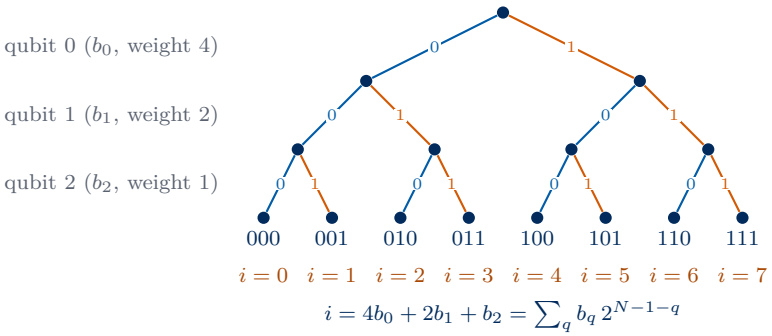


Figure 12: The storage order for $N = 3$. Reading the bits of qubits 0, 1 and 2 from the top of the tree to the bottom gives the binary index i, with qubit 0 as the most significant bit.

The memory grows exponentially with the number of qubits. With 16 bytes per complex number in double precision, the state of N qubits needs $16 \cdot 2^N$ bytes. Thirty qubits fill 16 GiB, the memory of a large workstation. Forty qubits need 16 TiB, and every further qubit doubles the requirement. Classical simulation of quantum systems is hard because of this cost, and tensor networks are one way around it.

Other software libraries may use the opposite convention. Qiskit, for example, treats qubit 0 as the least significant bit and prints bit strings with qubit 0 on the right. Before results from such a library are compared with ours, their bit strings must be reversed and their amplitudes reordered.

5.3 The state tensor

The flat index i hides the qubits, and a reshape of the vector of length 2^N into an array with N axes of length 2 recovers them,

$$\psi[b_0, b_1, \ldots, b_{N-1}] = \psi_i, \qquad i = \sum_q b_q 2^{N-1-q}. \tag{5.7}$$

$N = 2$			$N = 3$		
index i	bit string	Kronecker product	index i	bit string	Kronecker product
0	00	$\|0\rangle \otimes \|0\rangle$	0	000	$\|0\rangle \otimes \|0\rangle \otimes \|0\rangle$
1	01	$\|0\rangle \otimes \|1\rangle$	1	001	$\|0\rangle \otimes \|0\rangle \otimes \|1\rangle$
2	10	$\|1\rangle \otimes \|0\rangle$	2	010	$\|0\rangle \otimes \|1\rangle \otimes \|0\rangle$
3	11	$\|1\rangle \otimes \|1\rangle$	3	011	$\|0\rangle \otimes \|1\rangle \otimes \|1\rangle$
			4	100	$\|1\rangle \otimes \|0\rangle \otimes \|0\rangle$
			5	101	$\|1\rangle \otimes \|0\rangle \otimes \|1\rangle$
			6	110	$\|1\rangle \otimes \|1\rangle \otimes \|0\rangle$
			7	111	$\|1\rangle \otimes \|1\rangle \otimes \|1\rangle$

Table 3: Storage order of the amplitudes. The index of a bit string is its value as a binary number with qubit 0 as the most significant bit, equation (5.5).

The reshaped array is the *state tensor*, with one index per qubit. NumPy and JAX store arrays in row-major or C order, where the last axis changes fastest, as in equation (5.5). The call `psi.reshape((2,)*N)` produces the state tensor without moving any number in memory, and axis q of the result is qubit q. For two qubits the state tensor is a 2×2 matrix with qubit 0 labelling the rows,

$$\psi[b_0, b_1] = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}. \tag{5.8}$$

A reshape only regroups indices, whereas a transpose permutes the axes. Applied to the two-qubit state tensor, a transpose exchanges the roles of the qubits, and flattening the transposed array lists the amplitudes in the order $00, 10, 01, 11$.

5.4 An operator on one qubit of the register

Operators on the register are built with the Kronecker product, in the same way as states. For two 2×2 matrices the Kronecker product is a 4×4 block matrix. Every entry of the left factor multiplies a full copy of the right factor,

$$\hat{A} \otimes \hat{B} = \begin{pmatrix} A_{00}\hat{B} & A_{01}\hat{B} \\ A_{10}\hat{B} & A_{11}\hat{B} \end{pmatrix}, \qquad (\hat{A} \otimes \hat{B})_{(b_0 b_1),(b_0' b_1')} = A_{b_0 b_0'}\, B_{b_1 b_1'}, \tag{5.9}$$

where the row label $(b_0 b_1)$ is the bit string of the two qubits, at the index $2b_0 + b_1$ of Table 3, and likewise for the column label. By the element formula, a Kronecker product acts on product states factor by factor, $(\hat{A} \otimes \hat{B})(|u\rangle \otimes |v\rangle) = \hat{A}|u\rangle \otimes \hat{B}|v\rangle$, and two Kronecker products multiply in the same way, $(\hat{A} \otimes \hat{B})(\hat{C} \otimes \hat{D}) = \hat{A}\hat{C} \otimes \hat{B}\hat{D}$. An operator $\hat{U}$ on qubit 0 alone is $\hat{U} \otimes \hat{I}$, and on qubit 1 alone it is $\hat{I} \otimes \hat{U}$. Four examples written out are

$$\hat{Z} \otimes \hat{I} = \mathrm{diag}(1, 1, -1, -1), \qquad \hat{I} \otimes \hat{Z} = \mathrm{diag}(1, -1, 1, -1), \tag{5.10}$$

$$\hat{X} \otimes \hat{I} = \begin{pmatrix} 0 & \hat{I} \\ \hat{I} & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}, \qquad \hat{I} \otimes \hat{X} = \begin{pmatrix} \hat{X} & 0 \\ 0 & \hat{X} \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{5.11}$$

The matrix $\hat{X} \otimes \hat{I}$ flips qubit 0 by exchanging the amplitudes of $00 \leftrightarrow 10$ and $01 \leftrightarrow 11$, and $\hat{I} \otimes \hat{X}$ flips qubit 1 by exchanging $00 \leftrightarrow 01$ and $10 \leftrightarrow 11$. The diagonal of $\hat{Z}_0 = \hat{Z} \otimes \hat{I}$ holds $+1$ at the indices 0 and 1, the bit strings 00 and 01, and -1 at the indices 2 and 3. It reads bit 0 of the index. The alternating diagonal of $\hat{Z}_1 = \hat{I} \otimes \hat{Z}$ reads bit 1. Every operator on the register has its rows and columns labelled by bit strings in the order of Table 3.

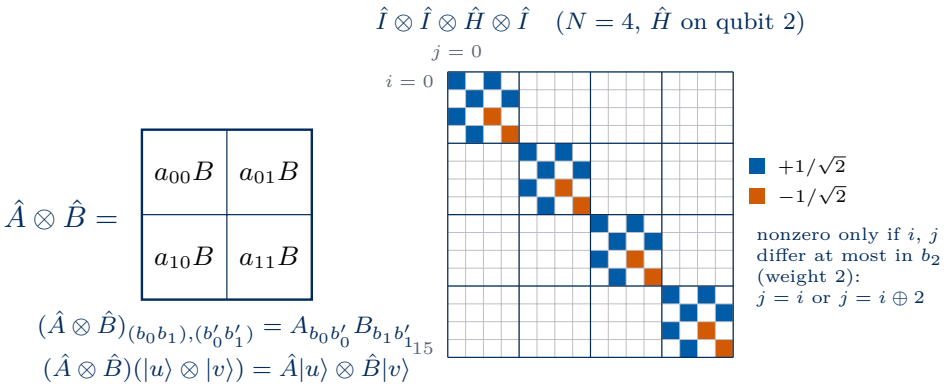


Figure 13: Left, the Kronecker product $\hat{A} \otimes \hat{B}$ as a block matrix in which every entry of $\hat{A}$ multiplies a copy of $\hat{B}$. Right, the pattern of nonzero entries of the 16×16 matrix $\hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, two in every row, a gate on one qubit of four.

In index notation, where a matrix–vector product reads $w_i = \sum_j A_{ij} v_j$, an operator on one qubit changes only the index of that qubit,

$$((\hat{U} \otimes \hat{I})\psi)[b_0', b_1] = \sum_{b_0} U_{b_0' b_0}\, \psi[b_0, b_1], \qquad ((\hat{I} \otimes \hat{U})\psi)[b_0, b_1'] = \sum_{b_1} U_{b_1' b_1}\, \psi[b_0, b_1]. \tag{5.12}$$

For three qubits the operator on qubit 1 changes only the middle index and equals $\hat{I} \otimes \hat{U} \otimes \hat{I}$. For N qubits the matrix $\hat{U}$ acts on index q of the state tensor alone.

Index sums on a computer

A computer evaluates the index sums of the Kronecker matrices without ever building a $2^N \times 2^N$ matrix. The function `einsum` of NumPy and JAX turns an index formula into code through a string of index letters. Letters that appear in two inputs and not in the output are summed over. Letters in the output are kept, in the order written after `->`. Table 4 lists the basic operations. The two sums of (5.12) are `"Aa,ab->Ab"` and `"Bb,ab->aB"`, where an upper-case letter labels the new index produced by the operator, the primed index of the formula. A *tensor-network diagram* pictures the same sum with a box for each tensor and a line, a *leg*, for each index, and joined legs are summed over (Figures 14 and 15). The contraction and the Kronecker matrix give the same vector. Its cost of about $2 \cdot 2^N$ multiplications per one-qubit gate, against 4^N for the full matrix, makes it the method used for many qubits.

The Kronecker matrix $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ of a one-qubit gate is mostly zeros. A computer stores it as a *sparse matrix*, a record of the nonzero entries and their positions. A common format is the compressed sparse row (CSR) format of `scipy.sparse`, and the function `scipy.sparse.kron` builds $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$ directly in this format. A one-qubit gate on N qubits has at most $2 \cdot 2^N$ nonzero entries, two in each row, out of 4^N. The cost of a sparse matrix–vector product is proportional to the number of nonzero entries.

operation	index notation	`einsum` string
matrix times vector	$w_i = \sum_j A_{ij} v_j$	`"ij,j->i"`
inner product $\langle a\|b\rangle$	$\sum_i a_i^* b_i$	`"i,i->"` on (a^*, b)
outer product $\|a\rangle\langle b\|$	$a_i b_j^*$	`"i,j->ij"` on (a, b^*)
trace	$\sum_i A_{ii}$	`"ii->"`
Kronecker product $a \otimes b$	$a_i b_j$, then reshape	`"i,j->ij"`, then `reshape(-1)`

Table 4: Index notation and `einsum` strings for basic operations.

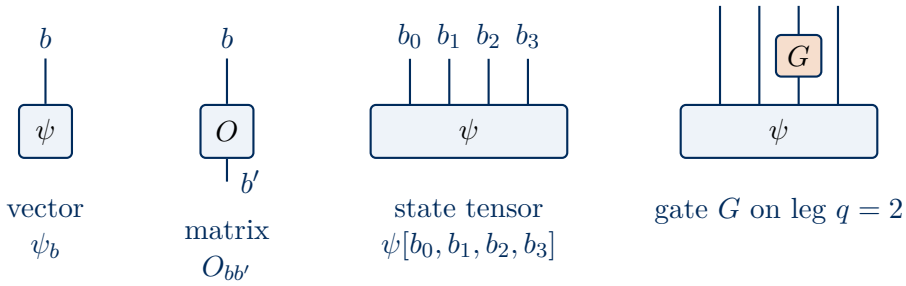


Figure 14: Tensor-network diagrams, the picture of index sums. A box is a tensor and a line, a leg, is one of its indices. A vector has one leg, a matrix two and the state tensor of N qubits N. A gate on qubit q is a matrix whose input leg is joined to leg q of the state tensor, and a joined leg is summed over.

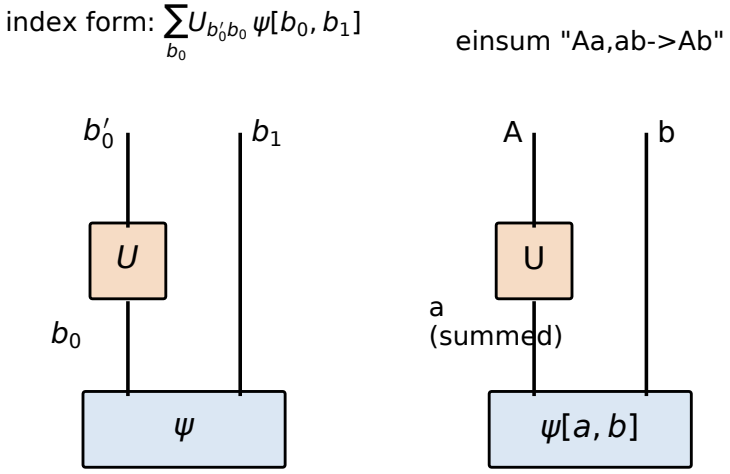


Figure 15: Tensor-network diagrams of a one-qubit operator $\hat{U}$ acting on qubit 0 (`"Aa,ab->Ab"`) and on qubit 1 (`"Bb,ab->aB"`) of the two-qubit state tensor ψ. The joined leg is summed, and the open legs are the indices of the new state.

6 Bit strings of two qubits

6.1 Probabilities, marginals and correlators

Measuring both qubits in the computational basis returns one of the four bit strings. By the Born rule the bit string $b_0 b_1$ appears with the probability

$$p(b_0 b_1) = |\psi[b_0, b_1]|^2, \tag{6.1}$$

the squared modulus of the corresponding entry of the state tensor. A record of M shots is a list of M bit strings drawn independently from these four probabilities, and from it we compute the following quantities.

- The *marginal* of qubit 0, $p_0(b_0) = \sum_{b_1} p(b_0 b_1)$, is what an observer who looks only at qubit 0 sees, and the marginal $p_1(b_1) = \sum_{b_0} p(b_0 b_1)$ of qubit 1 is defined in the same way.
- The single-qubit expectation values are $\langle \hat{Z}_q \rangle = p_q(0) - p_q(1)$.
- The *correlator*

$$\langle \hat{Z}_0 \hat{Z}_1 \rangle = \sum_{b_0 b_1} (-1)^{b_0 + b_1} \, p(b_0 b_1) \tag{6.2}$$

  is $+1$ when the two bits always agree, -1 when they always differ and 0 when equal and different bits are equally frequent. The factor $(-1)^{b_0+b_1}$ is the eigenvalue of $\hat{Z} \otimes \hat{Z}$ on $|b_0 b_1\rangle$, the *parity* of the bit string.

We estimate each quantity as the mean of the ± 1 values of the shots, the parities $(-1)^{b_0+b_1}$ in the case of the correlator. The standard error is the sample standard deviation divided by $\sqrt{M}$, as in Section 4.2.

The two bits are statistically independent when the joint probabilities factorise, $p(b_0 b_1) = p_0(b_0)\, p_1(b_1)$, and every correlator is then a product of single-qubit expectation values, $\langle \hat{Z}_0 \hat{Z}_1 \rangle = \langle \hat{Z}_0 \rangle \langle \hat{Z}_1 \rangle$. A vanishing correlator is weaker than independence. The distribution $p(00) = 1/2$, $p(01) = p(10) = 1/4$, $p(11) = 0$ has $\langle \hat{Z}_0 \hat{Z}_1 \rangle = 0$, while $p(00) = 1/2$ differs from $p_0(0)\, p_1(0) = 9/16$, so its bits are dependent. Product states always give independent bits. In the second example, the product state $\hat{R}_y(2\pi/3)|0\rangle \otimes |0\rangle$, qubit 0 is rotated by $2\pi/3$ about y and qubit 1 stays in $|0\rangle$. Qubit 1 always reads 0, so $\langle \hat{Z}_1 \rangle = 1$ and the correlator equals $\langle \hat{Z}_0 \rangle = \cos(2\pi/3) = -1/2$. A simulated record of 2000 shots reproduces this value within its standard error.

6.2 Bit strings in different bases

Each qubit of the register can be measured in its own basis by applying the rotation of Section 4.1 to that qubit before the $\hat{Z}$ readout. The rotation is $\hat{H}$ for the $\hat{X}$ basis and $\hat{H}\hat{S}^\dagger$ for the $\hat{Y}$ basis. A choice of basis for every qubit is a *measurement setting*, written as a string of letters such as $\hat{Z}\hat{Z}$, $\hat{Z}\hat{X}$ or $\hat{X}\hat{X}$, with the first letter for qubit 0. Outcome bit 0 means the eigenvalue $+1$ of the measured Pauli matrix.

The same state produces different records in different settings. Take $|0\rangle \otimes |+\rangle$, qubit 0 up along z and qubit 1 along $+x$. In the setting $\hat{Z}\hat{Z}$ qubit 0 always reads 0 and qubit 1 is a fair coin, so only 00 and 01 occur, with equal frequency. Every shot returns 00 in the setting $\hat{Z}\hat{X}$, where both qubits are measured along the axes of their own states. In the setting $\hat{X}\hat{X}$ qubit 0 is a fair coin and qubit 1 always reads 0, so 00 and 10 occur. Simulated records of 1000 shots in each setting show these patterns. A record in one setting reveals only part of the state, and quantum state tomography combines records from several settings to reconstruct it.

7 Two-qubit gates as rotations

7.1 Interactions switched on for a time

Single-qubit gates map product states to product states, and the bits stay uncorrelated. Correlated qubits need an interaction, a term of the Hamiltonian that acts on both qubits at once. A two-qubit gate is the evolution under such a coupling switched on for a time t. An Ising-type coupling is an energy proportional to the product of the same spin component of the two qubits. Its Hamiltonian $\hat{\mathcal{H}}_{\alpha\alpha} = \frac{J}{2}\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$, $\alpha \in \{x, y, z\}$, produces the evolution operator

$$\hat{R}_{\alpha\alpha}(\theta) = e^{-i\hat{\mathcal{H}}_{\alpha\alpha} t} = e^{-i\theta\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha / 2}, \qquad \theta = Jt. \tag{7.1}$$

The Kronecker product of two Pauli matrices squares to the identity, $(\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha)^2 = \hat{\sigma}_\alpha^2 \otimes \hat{\sigma}_\alpha^2 = \hat{I}$, and identity (3.6) applies unchanged,

$$\hat{R}_{\alpha\alpha}(\theta) = \cos\frac{\theta}{2}\, \hat{I} - i \sin\frac{\theta}{2}\, \hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha. \tag{7.2}$$

We also use three fixed two-qubit gates. With the projectors $|0\rangle\langle 0|$ and $|1\rangle\langle 1|$, the *controlled NOT* gate with control qubit 0 and target qubit 1 is

$$\widehat{\mathrm{CNOT}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{X} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \tag{7.3}$$

Qubit 1 is left alone when qubit 0 is 0 and flipped when qubit 0 is 1, which exchanges the amplitudes of 10 and 11. Circuit diagrams draw each qubit as a horizontal line, qubit 0 on top. CNOT is a

vertical line from a filled dot on the control qubit to the symbol $\oplus$ on the target qubit (Figure 17). The symbol $\oplus$ stands for addition of bits modulo two, the XOR operation, because CNOT maps $|b_0 b_1\rangle$ to $|b_0,\, b_1 \oplus b_0\rangle$. CNOT is a reversible XOR. It keeps the control bit and can therefore be undone, and applied twice it gives the identity. The *controlled* $\hat{Z}$ gate $\widehat{\mathrm{CZ}} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{Z} = \mathrm{diag}(1, 1, 1, -1)$ multiplies $|11\rangle$ by -1 and is symmetric in the two qubits. The $\widehat{\mathrm{SWAP}}$ gate exchanges the states of the qubits, $|01\rangle \leftrightarrow |10\rangle$,

$$\widehat{\mathrm{SWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{7.4}$$

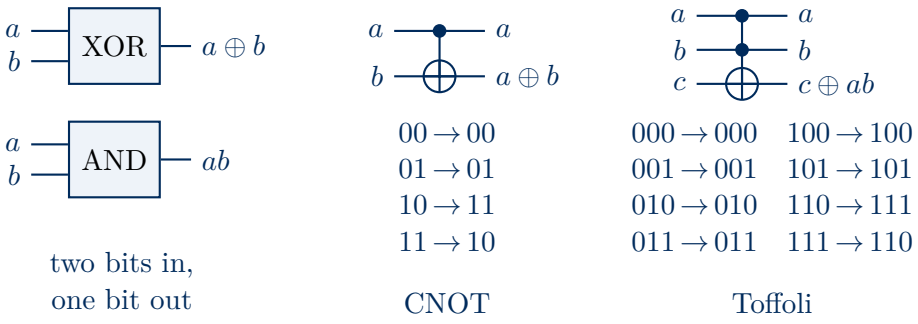


Figure 16: Classical XOR and AND lose information, since two input bits give one output bit and the inputs cannot be recovered. CNOT and Toffoli keep their inputs and write the XOR or the AND into the target bit. They permute the bit strings, as the truth tables show, so their matrices are permutation matrices and therefore unitary. With $c = 0$ the three-bit Toffoli gate, treated with many qubits in Section 14.3, computes the AND without losing the inputs.

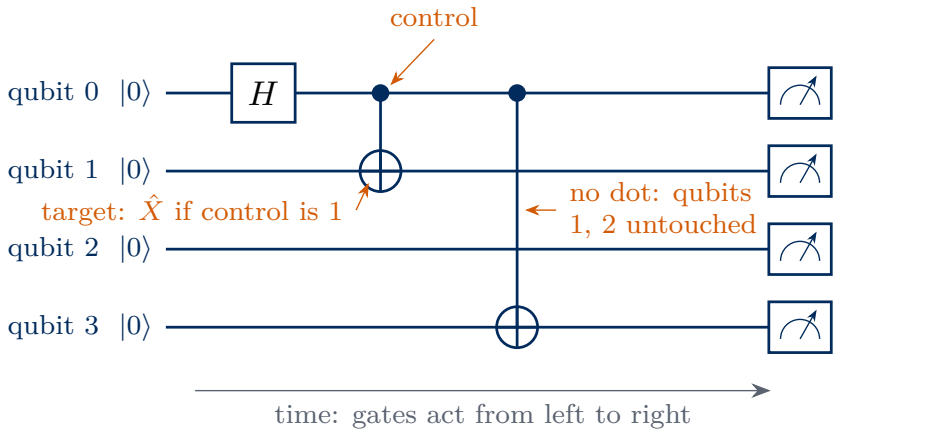


Figure 17: Circuit conventions. Each qubit is a wire, qubit 0 on top, and time runs from left to right. A box is a one-qubit gate, and CNOT joins a filled control dot to the target $\oplus$, also between qubits that are not neighbours on the diagram. A box with a meter needle at the end of a wire is a measurement of that qubit in the computational basis.

7.2 The rotation picture of each gate

Like the one-qubit gates, each two-qubit gate is a rotation or an evolution operator. This form identifies the coupling and the switching time that produce the gate, as worked out below for CZ and the Ising coupling.

$\hat{R}_{xx}$ as a rotation in a two-dimensional subspace

The operator $\hat{X} \otimes \hat{X}$ maps $|00\rangle \leftrightarrow |11\rangle$ and $|01\rangle \leftrightarrow |10\rangle$. By (7.2), in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\hat{R}_{xx}(\theta) = \begin{pmatrix} \cos\frac{\theta}{2} & 0 & 0 & -i\sin\frac{\theta}{2} \\ 0 & \cos\frac{\theta}{2} & -i\sin\frac{\theta}{2} & 0 \\ 0 & -i\sin\frac{\theta}{2} & \cos\frac{\theta}{2} & 0 \\ -i\sin\frac{\theta}{2} & 0 & 0 & \cos\frac{\theta}{2} \end{pmatrix}, \tag{7.5}$$

and $\hat{R}_{xx}(\theta)$ couples $|00\rangle$ only with $|11\rangle$. On the pair $\{|00\rangle, |11\rangle\}$ its matrix is that of an ordinary $\hat{R}_x(\theta)$, a rotation of a Bloch sphere with $|00\rangle$ at the north pole and $|11\rangle$ at the south pole. The pair $\{|01\rangle, |10\rangle\}$ rotates in the same way. At $\theta = \pi/2$ the state $|00\rangle$ reaches the equator of that sphere,

$$\hat{R}_{xx}(\pi/2)|00\rangle = \frac{|00\rangle - i|11\rangle}{\sqrt{2}}. \tag{7.6}$$

The equal superposition of $|00\rangle$ and $|11\rangle$ obtained at $\theta = \pi/2$ is a Bell state, studied in Section 8.

$\hat{R}_{zz}$ as a conditional rotation

The operator $\hat{Z} \otimes \hat{Z} = |0\rangle\langle 0| \otimes \hat{Z} - |1\rangle\langle 1| \otimes \hat{Z}$ is diagonal with the entries $(1, -1, -1, 1)$. The rotation $\hat{R}_{zz}(\theta) = \mathrm{diag}(e^{-i\theta/2}, e^{i\theta/2}, e^{i\theta/2}, e^{-i\theta/2})$ only attaches phases to the four basis states, and in block form

$$\hat{R}_{zz}(\theta) = |0\rangle\langle 0| \otimes \hat{R}_z(\theta) + |1\rangle\langle 1| \otimes \hat{R}_z(-\theta). \tag{7.7}$$

Qubit 1 rotates about z by $+\theta$ when qubit 0 is $|0\rangle$ and by $-\theta$ when qubit 0 is $|1\rangle$, a rotation of one qubit conditioned on the other. As an evolution operator, $\hat{R}_{zz}(\theta)$ is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ switched on for the time $t = \theta/J$.

CNOT as a conditional half turn

Since $\hat{X} = i\hat{R}_x(\pi)$, the conditional rotation $|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)$ equals CNOT up to the phase $-i$ in the branch where qubit 0 is 1. The phase gate $\hat{S}$ on qubit 0 supplies the missing factor i in that branch, so

$$\widehat{\mathrm{CNOT}} = (\hat{S} \otimes \hat{I})\,[|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_x(\pi)]. \tag{7.8}$$

CZ as a conditional half turn

In the same way CZ is the conditional half turn about z. It is also the evolution generated by the projector $\hat{P} = |11\rangle\langle 11| = \frac{\hat{I}-\hat{Z}}{2} \otimes \frac{\hat{I}-\hat{Z}}{2}$. Since $\hat{P}^k = \hat{P}$ for every $k \geq 1$, the exponential series sums to $e^{-ia\hat{P}} = \hat{I} + \sum_{k\geq 1} \frac{(-ia)^k}{k!}\hat{P} = \hat{I} + (e^{-ia} - 1)\hat{P}$, and at $a = \pi$ the operator $\hat{I} - 2\hat{P}$ flips the sign of $|11\rangle$ alone,

$$\widehat{\mathrm{CZ}} = (\hat{S} \otimes \hat{I})\,[|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_z(\pi)] = \exp\Big[-\frac{i\pi}{4}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{Z})\Big]. \tag{7.9}$$

The projector form shows how the Ising coupling makes CZ. Expanding $(\hat{I} - \hat{Z}_0)(\hat{I} - \hat{Z}_1) = \hat{I} - \hat{Z}_0 - \hat{Z}_1 + \hat{Z}_0\hat{Z}_1$, which is four times the projector $|11\rangle\langle 11|$, turns the exponential into $\hat{R}_{zz}(\pi/2)$ times one z rotation on each qubit and a global phase,

$$\widehat{\mathrm{CZ}} = e^{-i\pi/4}\,[\hat{R}_z(-\tfrac{\pi}{2}) \otimes \hat{R}_z(-\tfrac{\pi}{2})]\,\hat{R}_{zz}(\tfrac{\pi}{2}), \qquad \widehat{\mathrm{CNOT}} = (\hat{I} \otimes \hat{H})\,\widehat{\mathrm{CZ}}\,(\hat{I} \otimes \hat{H}). \tag{7.10}$$

CZ is therefore the Ising coupling switched on for the time $t = \pi/(2J)$, followed by one z rotation on each qubit. The second relation follows from $\hat{H}\hat{Z}\hat{H} = \hat{X}$, which turns the conditional sign flip into a conditional bit flip, so CNOT is CZ between two Hadamard gates on the target.

CNOT and SWAP as evolution operators (optional)

CNOT is also the evolution operator of one Hamiltonian. The matrix $\hat{P} = \frac{\hat{I}-\hat{Z}}{2} \otimes \frac{\hat{I}-\hat{X}}{2}$, a product of the projector on $|1\rangle$ of qubit 0 and the projector on $|-\rangle$ of qubit 1, is itself a projector, $\hat{P}^2 = \hat{P}$. Its exponential is $e^{-ia\hat{P}} = \hat{I} + (e^{-ia} - 1)\hat{P}$ as for CZ, and for $a = \pm\pi$

$$e^{\mp i\pi\hat{P}} = \hat{I} - 2\hat{P} = \exp\Big[\mp \frac{i\pi}{4}(\hat{I} - \hat{Z}) \otimes (\hat{I} - \hat{X})\Big] = \widehat{\mathrm{CNOT}}. \tag{7.11}$$

The operator $\hat{I} - 2\hat{P}$ flips the sign of $|1\rangle|-\rangle$ and leaves the three orthogonal states $|0\rangle|\pm\rangle$ and $|1\rangle|+\rangle$ unchanged. Since $\hat{X}|\pm\rangle = \pm|\pm\rangle$, this sign flip is the action of CNOT written in the $\hat{X}$ basis of the target.

SWAP is generated by the Heisenberg exchange coupling $\hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z} = 4\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$, proportional to the scalar product of the two spin vectors with the spin operators $\hat{S}_a = \hat{\sigma}_a/2$. The three symmetric states $|00\rangle$, $|11\rangle$ and $(|01\rangle + |10\rangle)/\sqrt{2}$ do not change when the qubits are exchanged and are eigenstates of this coupling with the eigenvalue $+1$. The antisymmetric *singlet* $(|01\rangle - |10\rangle)/\sqrt{2}$ changes sign under the exchange and has the eigenvalue -3, since $\hat{X} \otimes \hat{X}$, $\hat{Y} \otimes \hat{Y}$ and $\hat{Z} \otimes \hat{Z}$ each multiply it by -1. SWAP has the eigenvalue $+1$ on the symmetric states and -1 on the singlet. On these eigenvectors the operator $\frac{1}{2}(\hat{I} + \hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z})$ has the same eigenvalues as SWAP, $\frac{1}{2}(1 + 1) = 1$ and $\frac{1}{2}(1 - 3) = -1$, and the two operators are therefore equal,

$$\widehat{\mathrm{SWAP}} = \tfrac{1}{2}(\hat{I} + \hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z}) = e^{i\pi/4} \exp\Big[-\frac{i\pi}{4}(\hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z})\Big]. \tag{7.12}$$

In the exponential form the symmetric states acquire the phase $e^{i\pi/4}e^{-i\pi/4} = 1$ and the singlet the phase $e^{i\pi/4}e^{3i\pi/4} = -1$. Since $\hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z} = 4\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$, the Heisenberg coupling $J_{\mathrm{ex}}\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1 = \frac{J_{\mathrm{ex}}}{4}(\hat{X} \otimes \hat{X} + \hat{Y} \otimes \hat{Y} + \hat{Z} \otimes \hat{Z})$ applies SWAP at $J_{\mathrm{ex}}t = \pi$ and a square root of SWAP at $J_{\mathrm{ex}}t = \pi/2$. We write its strength as J_{ex} to keep it apart from the coefficient J of $\hat{X}\otimes\hat{X}+\hat{Y}\otimes\hat{Y}+\hat{Z}\otimes\hat{Z}$ used for the register in Section 9, which equals $J_{\mathrm{ex}}/4$. A related coupling moves one excitation between the qubits. Its operator $|01\rangle\langle 10| + |10\rangle\langle 01| = \frac{1}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$ follows from the Pauli algebra, and the coupling $\frac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$ swaps $|01\rangle$ and $|10\rangle$ with the phase $-i$ at $Jt = \pi/2$. The result is the inverse of the $\widehat{\mathrm{iSWAP}}$ gate, and the opposite sign of J produces $\widehat{\mathrm{iSWAP}}$ itself. In the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$ the two gates are

$$\widehat{\mathrm{iSWAP}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \qquad \sqrt{\widehat{\mathrm{SWAP}}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1+i}{2} & \frac{1-i}{2} & 0 \\ 0 & \frac{1-i}{2} & \frac{1+i}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{7.13}$$

both act only in the pair $\{|01\rangle, |10\rangle\}$, and $\sqrt{\widehat{\mathrm{SWAP}}}$ applied twice gives SWAP.

7.3 Parametrised gates and universality

A *parametrised gate* is a family of unitary operators $\hat{U}(\theta)$ labelled by a continuous angle θ, such as $\hat{R}_x(\theta)$, $\hat{R}_y(\theta)$, $\hat{R}_z(\theta)$, $\hat{R}_{xx}(\theta)$ and $\hat{R}_{zz}(\theta)$. A *fixed gate*, such as $\hat{H}$, $\hat{S}$, $\hat{T}$, CNOT or CZ, is one member of such a family at a fixed value of its angle, together with fixed one-qubit phases. The angle of a two-qubit rotation is set by the coupling and the time for which it acts. A coupling $\frac{J(t)}{2}\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$ with a time-dependent strength commutes with itself at all times, and switching it on for the duration T produces

$$\hat{U} = \hat{R}_{\alpha\alpha}(\theta), \qquad \theta = \int_0^T J(t')\,dt', \tag{7.14}$$

and $\theta = JT$ for a constant coupling. Any angle is reached by choosing the duration. Any entangling two-qubit gate of this section, together with arbitrary single-qubit rotations, is *universal*, meaning that every unitary operation on the register decomposes into it and single-qubit gates [1].

7.4 Gates and expectation values on a computer

On paper a two-qubit gate is the 4×4 matrix $\hat{G}$ multiplying the vector of amplitudes. An expectation value is the product $\langle\psi|\hat{O}|\psi\rangle$ of a row vector, a matrix and a column vector. A computer evaluates the same sums as contractions. For this the 4×4 gate is reshaped into a tensor $G[b_0', b_1', b_0, b_1]$ with two output and two input legs, which acts on the register as

$$(\hat{G}\psi)[b_0', b_1'] = \sum_{b_0, b_1} G[b_0', b_1', b_0, b_1]\, \psi[b_0, b_1], \qquad \texttt{"ABab,ab->AB"}, \tag{7.15}$$

drawn as a box joined to both legs of the state (Figure 18, left). The expectation value of an observable $\hat{O}$ is a closed network,

$$\langle\psi|\hat{O}|\psi\rangle = \sum_{a,b,a',b'} \psi^*[a', b']\, O[a', b', a, b]\, \psi[a, b], \qquad \texttt{"AB,ABab,ab->"}, \tag{7.16}$$

with the conjugate state ψ^* on top, the operator in the middle and ψ below (Figure 18, right). With every leg joined, the network evaluates to one number. For an operator on one qubit the box on the other qubit is a plain line, the identity.

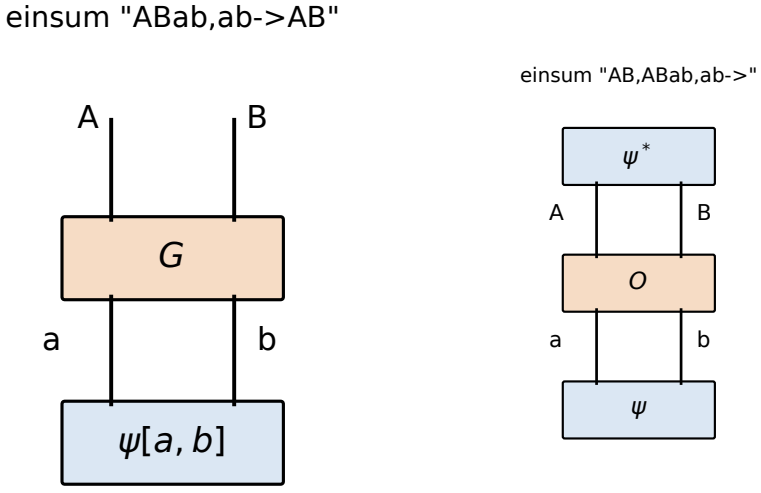


Figure 18: Left, a two-qubit gate as a box with two input and two output legs contracted with both legs of the state, `"ABab,ab->AB"`. Right, the expectation value $\langle\psi|\hat{O}|\psi\rangle$ as a closed network, `"AB,ABab,ab->"`.

7.5 Pauli strings and gates on N qubits

The Kronecker product of Section 5.4 extends to a register of N qubits with one factor per qubit. Factor q acts on qubit q, qubit 0 being the leftmost, and an operator that leaves a qubit alone carries the identity in that position.

Pauli strings

A *Pauli string* is a Kronecker product of one Pauli matrix or identity for each qubit,

$$\hat{P} = \hat{P}_0 \otimes \hat{P}_1 \otimes \cdots \otimes \hat{P}_{N-1}, \qquad \hat{P}_q \in \{\hat{I}, \hat{X}, \hat{Y}, \hat{Z}\}. \tag{7.17}$$

We write only the factors that differ from the identity and label them by their qubit. For three qubits $\hat{X}_0\hat{Z}_2 = \hat{X} \otimes \hat{I} \otimes \hat{Z}$, and $\hat{Z}_0\hat{Z}_1$ of Section 6 is $\hat{Z} \otimes \hat{Z}$. There are 4^N Pauli strings, and their properties follow from the rule $(\hat{A} \otimes \hat{B})(\hat{C} \otimes \hat{D}) = \hat{A}\hat{C} \otimes \hat{B}\hat{D}$ applied factor by factor.

- Every Pauli string is Hermitian and squares to the identity, $\hat{P}^2 = \hat{P}_0^2 \otimes \cdots \otimes \hat{P}_{N-1}^2 = \hat{I}$, and its eigenvalues are therefore ± 1. One shot returns one of these two values.
- Two Pauli strings either commute or anticommute. Every qubit on which they carry two different non-identity Pauli matrices contributes a sign -1, because the two factors anticommute. An even number of such qubits makes the strings commute, and an odd number makes them anticommute. For example $\hat{X}_0\hat{X}_1$ and $\hat{Z}_0\hat{Z}_1$ commute, while $\hat{X}_0$ and $\hat{Z}_0\hat{Z}_1$ anticommute.
- Different strings are orthogonal, $\mathrm{Tr}(\hat{P}\hat{Q}) = 2^N \delta_{PQ}$, because the trace of a Kronecker product is the product of the traces and every Pauli matrix is traceless. The 4^N strings therefore form a basis of all $2^N \times 2^N$ matrices, and every Hermitian matrix is a real combination

$$\hat{\mathcal{H}} = \sum_{\hat{P}} c_P \, \hat{P}, \qquad c_P = \frac{\mathrm{Tr}(\hat{P}\hat{\mathcal{H}})}{2^N}. \tag{7.18}$$

  The Hamiltonian $J(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h(\hat{Z}_0 + \hat{Z}_1)$ of two coupled spins in a field has $c_{XX} = c_{YY} = c_{ZZ} = J$, $c_{ZI} = c_{IZ} = h$ and all eleven other coefficients zero.
- A string made of $\hat{Z}$ factors and identities is diagonal in the computational basis. Its entry on $|b_0 b_1 \ldots b_{N-1}\rangle$ is the parity $(-1)^{b_{q_1}+\cdots+b_{q_k}}$ of the bits of the qubits $q_1, \ldots, q_k$ that carry a $\hat{Z}$. For three qubits $\hat{Z}_0\hat{Z}_2$ has the entry $(-1)^{b_0+b_2}$, a generalisation of the parity of equation (6.2).

A one-qubit gate on qubit q

A gate $\hat{G}$ on qubit q of an N-qubit register is the Kronecker product with $\hat{G}$ in position q and identities elsewhere, $\hat{I} \otimes \cdots \otimes \hat{G} \otimes \cdots \otimes \hat{I}$. For $N = 4$ the Hadamard gate on qubit 2 is the 16×16 matrix $\hat{I} \otimes \hat{I} \otimes \hat{H} \otimes \hat{I}$, whose pattern of nonzero entries is drawn in Figure 13. By the element formula (5.9) it changes only the bit b_2 and mixes each amplitude with the one whose bit string differs in bit 2 alone. The two amplitudes sit at the indices i and $i \oplus 2^{N-1-q}$, where $\oplus$ flips the bit of value $2^{N-1-q} = 2$ in the binary index. The pair 0000 and 0010, at the indices 0 and 2, becomes

$$\psi'_{0000} = \frac{\psi_{0000} + \psi_{0010}}{\sqrt{2}}, \qquad \psi'_{0010} = \frac{\psi_{0000} - \psi_{0010}}{\sqrt{2}}, \tag{7.19}$$

and the same holds for the seven other pairs, for example 1101 and 1111 at the indices 13 and 15. In index notation the update reads $\psi'[b_0, b_1, b'_2, b_3] = \sum_{b_2} H_{b'_2 b_2} \psi[b_0, b_1, b_2, b_3]$, a box on the third leg of the state tensor. The gate touches each of the 2^N amplitudes once and costs $2 \cdot 2^N$ multiplications, against 4^N for the full matrix, as in Section 5.4.

Every gate on a chain of N qubits is the exponential of its generator, a Pauli string padded with identities on the untouched qubits. The rotation $\hat{R}_j^\alpha(\theta) = e^{-i\theta\hat{\sigma}_j^\alpha/2}$, the two-qubit rotation $e^{-i\theta\hat{X}_i\hat{X}_j/2}$ and the controlled gates of Section 7.2 are built this way for any qubits i and j of the chain.

A two-qubit gate on a distant pair

CNOT with control qubit i and target qubit j is a sum of two Kronecker products of single-qubit operators, with identities on all other qubits,

$$\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i + |1\rangle\langle 1|_i \, \hat{X}_j, \tag{7.20}$$

where a subscript names the qubit of each factor. For $N = 4$,

$$\widehat{\mathrm{CNOT}}_{0\to 3} = |0\rangle\langle 0| \otimes \hat{I} \otimes \hat{I} \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{I} \otimes \hat{I} \otimes \hat{X}. \tag{7.21}$$

The gate permutes the basis states by flipping b_3 whenever $b_0 = 1$. The four exchanged pairs are $1000 \leftrightarrow 1001$, $1010 \leftrightarrow 1011$, $1100 \leftrightarrow 1101$ and $1110 \leftrightarrow 1111$, at the indices $8 \leftrightarrow 9$, $10 \leftrightarrow 11$, $12 \leftrightarrow 13$ and $14 \leftrightarrow 15$, and the eight bit strings with $b_0 = 0$ stay in place. With the control below the target,

$\widehat{\mathrm{CNOT}}_{3\to0} = \hat I\otimes\hat I\otimes\hat I\otimes|0\rangle\langle0| + \hat X\otimes\hat I\otimes\hat I\otimes|1\rangle\langle1|$ flips b_0 whenever $b_3 = 1$ and exchanges $0001 \leftrightarrow 1001$, $0011 \leftrightarrow 1011$, $0101 \leftrightarrow 1101$ and $0111 \leftrightarrow 1111$.

A general 4×4 gate on the qubits i and j acts in the same way, through its matrix elements on the bits b_i, b_j, with the other bits as spectators. On a computer the gate tensor $G[b_0', b_1', b_0, b_1]$ of Section 7.4 is contracted with the legs i and j of the state tensor while the other legs pass through (Figure 19), and no 16×16 matrix is built.

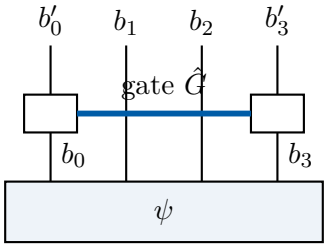


Figure 19: A two-qubit gate on the distant qubits 0 and 3 of a four-qubit state tensor. The gate box joins the legs b_0 and b_3, which are summed and replaced by the output legs b_0' and b_3'. The legs b_1 and b_2 pass the gate unchanged. The two boxes joined by a bar are one tensor with four legs.

When only neighbouring qubits can be coupled directly, a distant gate is compiled with SWAP gates that move the qubits next to each other and back, for example $\widehat{\mathrm{CNOT}}_{0\to3} = \widehat{\mathrm{SWAP}}_{23}\widehat{\mathrm{SWAP}}_{12}\,\widehat{\mathrm{CNOT}}_{0\to1}\,\widehat{\mathrm{SWAP}}_{12}\widehat{\mathrm{SWAP}}_{23}$ (optional remark).

A chain of CNOT gates prepares the *Greenberger–Horne–Zeilinger* (GHZ) state, the equal superposition of all qubits in 0 and all qubits in 1,

$$|\mathrm{GHZ}_N\rangle = \frac{|0\ldots0\rangle + |1\ldots1\rangle}{\sqrt2} = \widehat{\mathrm{CNOT}}_{N-2\to N-1}\cdots\widehat{\mathrm{CNOT}}_{1\to2}\,\widehat{\mathrm{CNOT}}_{0\to1}\,\hat H_0|0\ldots0\rangle. \tag{7.22}$$

The Hadamard gate creates $(|0\rangle+|1\rangle)|0\ldots0\rangle/\sqrt2$, and each CNOT copies the bit of the previous qubit into the next one in both branches. The fan-out $\widehat{\mathrm{CNOT}}_{0\to N-1}\cdots\widehat{\mathrm{CNOT}}_{0\to2}\,\widehat{\mathrm{CNOT}}_{0\to1}\,\hat H_0$, with qubit 0 as the control of every gate, prepares the same state.

8 Entanglement

8.1 The Bell state and its predictions

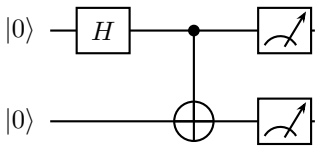
The CNOT and $\hat R_{xx}$ gates of Section 7 correlate the two qubits, and the simplest state they produce from $|00\rangle$ is the *Bell state*

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt2}. \tag{8.1}$$

Its predictions split into local ones, which concern one qubit, and joint ones, which concern both. Measured alone in any basis, each qubit gives a fair coin, since its Bloch vector, computed with $\hat X\otimes\hat I$, $\hat Y\otimes\hat I$ and $\hat Z\otimes\hat I$, vanishes. The parity of each joint measurement is certain. In the settings $\hat Z\hat Z$ and $\hat X\hat X$ the two bits agree in every shot, and in the setting $\hat Y\hat Y$ they disagree in every shot. The Bell state is an eigenstate of $\hat Z\otimes\hat Z$ and $\hat X\otimes\hat X$ with the eigenvalue $+1$ and of $\hat Y\otimes\hat Y$ with the eigenvalue -1. The last sign comes from $\hat Y|0\rangle = i|1\rangle$ and $\hat Y|1\rangle = -i|0\rangle$, whence $\hat Y\otimes\hat Y|00\rangle = -|11\rangle$ and $\hat Y\otimes\hat Y|11\rangle = -|00\rangle$.

No product state has these predictions, because for a product state the bits are independent in every setting and $\langle\hat Z_0\hat Z_1\rangle = \langle\hat Z_0\rangle\langle\hat Z_1\rangle$. The value $\langle\hat Z_0\hat Z_1\rangle = 1$ would require $|\langle\hat Z_0\rangle| = |\langle\hat Z_1\rangle| = 1$, with both qubits at a pole of the Bloch sphere. Their $\hat X$ outcomes would then be independent fair coins with $\langle\hat X_0\hat X_1\rangle = \langle\hat X_0\rangle\langle\hat X_1\rangle = 0$, against the value one of the Bell state. The Bell state is the simplest example of an entangled state.

We prepare it from $|00\rangle$ by a Hadamard gate on qubit 0 followed by CNOT with qubit 0 as control,



$|\Phi^+\rangle$ measured in a setting

The Hadamard gate gives $(|00\rangle + |10\rangle)/\sqrt{2}$, and CNOT flips qubit 1 in the second branch,

$$\widehat{\mathrm{CNOT}}\,(\hat{H} \otimes \hat{I})|00\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} \equiv |\Phi^+\rangle. \tag{8.2}$$

The interaction gate of Section 7.2 offers a second route. Acting on $|00\rangle$, $\hat{R}_{xx}(\pi/2)$ produces $(|00\rangle - i|11\rangle)/\sqrt{2}$, and the phase gate $\hat{S}$ on qubit 0 multiplies the $|11\rangle$ term by i and turns the state into $|\Phi^+\rangle$.

8.2 The product test and the Schmidt decomposition

We decide whether a state of the register is a product state $|a\rangle \otimes |b\rangle$ or entangled, the two cases defined in Section 5, by reading the state tensor as the 2×2 matrix

$$\Psi = \begin{pmatrix} \psi_{00} & \psi_{01} \\ \psi_{10} & \psi_{11} \end{pmatrix}, \tag{8.3}$$

with qubit 0 labelling the rows. A product state has the amplitudes $\psi[b_0, b_1] = a_{b_0} b_{b_1}$, so $\Psi = ab^T$ is an outer product and has rank one. For the running product state $|0\rangle \otimes |+\rangle$,

$$\Psi = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} = 1 \cdot \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix}, \tag{8.4}$$

whose second row vanishes. The Bell state has $\Psi = \hat{I}/\sqrt{2}$ of rank two, which no outer product reproduces, and is therefore entangled. For two qubits the test reduces to $\det \Psi = \psi_{00}\psi_{11} - \psi_{01}\psi_{10} = 0$ for a product state.

The *singular value decomposition* (SVD), which generalises diagonalisation to any matrix, makes the test quantitative. Every complex matrix can be written as

$$\Psi = U \Sigma V^\dagger = \sum_{k=1}^{2} s_k \, u_k v_k^\dagger, \qquad \Sigma = \mathrm{diag}(s_1, s_2), \qquad s_1 \geq s_2 \geq 0, \tag{8.5}$$

with two unitary matrices U and V, whose columns u_1, u_2 and v_1, v_2 form two orthonormal sets of vectors, and the non-negative *singular values* s_k. Geometrically the matrix first changes the basis unitarily and stretches the two new basis directions by the factors s_1 and s_2. A second unitary change of basis follows. The squares s_k^2 are the eigenvalues of the Hermitian matrix $\Psi\Psi^\dagger$, so the singular values follow from an ordinary eigenvalue problem. In components, $\psi[b_0, b_1] = \sum_k U_{b_0 k} \, s_k \, V^*_{b_1 k}$. Read back as a state, the decomposition is the *Schmidt decomposition*

$$|\psi\rangle = \sum_k s_k \, |u_k\rangle \otimes |v_k^*\rangle, \qquad s_1^2 + s_2^2 = 1, \tag{8.6}$$

where the Schmidt vectors $|u_k\rangle$ of qubit 0 are the columns of U and the Schmidt vectors $|v_k^*\rangle$ of qubit 1 are the rows of $V^\dagger$, the complex conjugates of the columns of V. a sum of at most two product terms, normalised because $\sum_k s_k^2 = \sum_{ab} |\psi_{ab}|^2 = 1$. The *Schmidt coefficients* s_k quantify entanglement. The vectors $|u_k\rangle$ are orthonormal, and so are the $|v_k^*\rangle$. A measurement of qubit 0 in the basis $\{|u_k\rangle\}$ therefore returns the outcome k with the probability s_k^2 and leaves qubit 1 in $|v_k^*\rangle$. The number of nonzero Schmidt coefficients, the *Schmidt rank*, equals the rank of Ψ, one for a product state and two for an entangled state. Among the entangled states of two qubits the equal spread $s_1 = s_2 = 1/\sqrt{2}$ carries the largest possible entanglement, since the local outcome is then a fair coin.

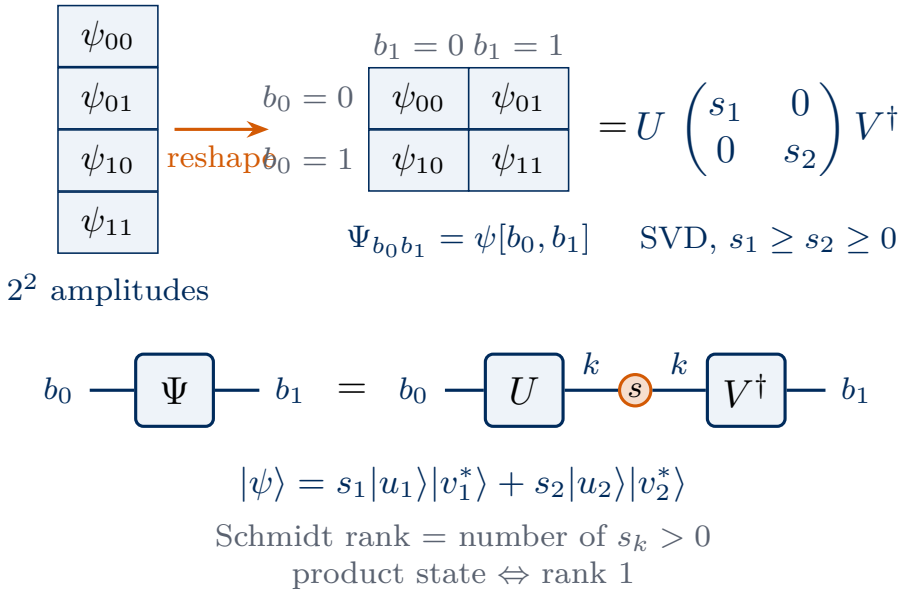


Figure 20: The product test and the Schmidt decomposition. The four amplitudes are reshaped into the 2×2 matrix Ψ, and its SVD gives the Schmidt coefficients $s_1 \geq s_2$. The number of nonzero coefficients, the Schmidt rank, tells a product state from an entangled one.

For $|0\rangle \otimes |+\rangle$ the decomposition has $s_1 = 1$, $s_2 = 0$, $|u_1\rangle = |0\rangle$ and $|v_1^*\rangle = |+\rangle$. For the Bell state $\Psi = \hat{I}/\sqrt{2}$ is already diagonal with orthonormal rows and columns, so $s_1 = s_2 = 1/\sqrt{2} = 0.7071$ with $|u_k\rangle = |v_k^*\rangle$ the two basis vectors.

The function `np.linalg.svd` returns U, the singular values and $V^\dagger$, so the Schmidt vectors of qubit 1 are the rows of its third output.

A gate $\hat{A}$ on qubit 0 and a gate $\hat{B}$ on qubit 1 change the matrix to $\hat{A}\Psi\hat{B}^T$, and multiplication by unitary matrices leaves the singular values unchanged. Local gates can change individual correlators while the Schmidt coefficients stay fixed. Entanglement is therefore unchanged by local gates, and only an interaction between the qubits creates it. The two preparations of the Bell state, by CNOT and by $\hat{R}_{xx}(\pi/2)$ followed by $\hat{S}$, accordingly produce equally entangled states.

The family

$$|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle \tag{8.7}$$

interpolates between the product state $|00\rangle$ at $\theta = 0$ and the Bell state at $\theta = \pi/4$. Its matrix is $\mathrm{diag}(\cos\theta, \sin\theta)$, and the Schmidt coefficients are $|\cos\theta|$ and $|\sin\theta|$ (Figure 21).

Concurrence (optional)

For two qubits the Schmidt coefficients follow from the determinant without an SVD. With the amplitudes $\psi = (\alpha, \beta, \gamma, \delta)^T$ the matrix is $\Psi = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}$. The determinant of a product is the product of the determinants, $\det\Psi = \det U \, s_1 s_2 \, \det V^\dagger$ by (8.5), and a unitary matrix has a determinant of modulus one. The product of the singular values is therefore the modulus of the determinant, and their squares sum to one,

$$s_1 s_2 = |\alpha\delta - \beta\gamma|, \qquad s_1^2 + s_2^2 = 1 \qquad \Longrightarrow \qquad s_{1,2}^2 = \frac{1 \pm \sqrt{1 - C^2}}{2}, \qquad C = 2|\alpha\delta - \beta\gamma|. \tag{8.8}$$

The number C, between 0 and 1, is the *concurrence* of the pure state. It vanishes for a product state, where $\alpha\delta = \beta\gamma$ as for the amplitudes ac, ad, bc, bd of (5.1), and equals one for the Bell state, where $\alpha = \delta = 1/\sqrt{2}$ and $\beta = \gamma = 0$.

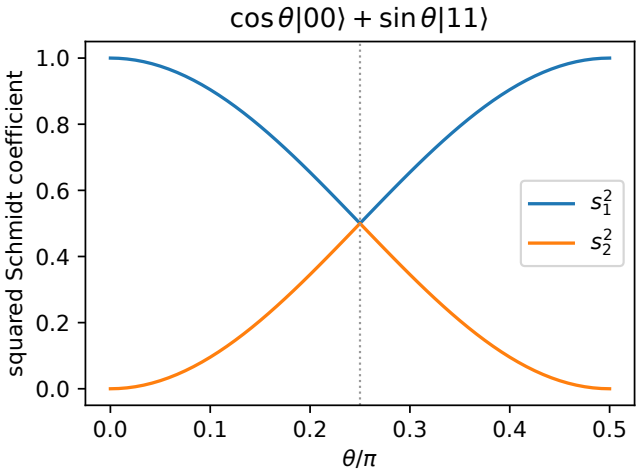


Figure 21: Squared Schmidt coefficients of $\cos\theta|00\rangle + \sin\theta|11\rangle$. They start at 1 and 0 for the product state and meet at 1/2 at $\theta = \pi/4$ (dotted), the Bell state. Beyond it the two terms exchange their roles, and the state returns to the product $|11\rangle$ at $\theta = \pi/2$.

8.3 Bell correlations against a classical coin

A *correlated coin*, a source that prepares $|00\rangle$ or $|11\rangle$ with probability one half each, imitates the Bell state classically and reproduces its $\hat{Z}\hat{Z}$ record. In each run, however, the coin holds a definite product state along z. Its bits in the $\hat{X}$ and $\hat{Y}$ bases are then independent and random, and its correlators $\langle\hat{X}\hat{X}\rangle$ and $\langle\hat{Y}\hat{Y}\rangle$ vanish. Simulated records of 4000 shots of both preparations in each setting confirm this (Table 5 and Figure 22). The coin matches the Bell state in $\hat{Z}\hat{Z}$ and fails in $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$, and no mixture of $|00\rangle$ and $|11\rangle$ reproduces all three records. To treat the coin and the Bell state as different objects with the same $\hat{Z}\hat{Z}$ statistics we need a description of incompletely known preparations, the density matrix.

preparation	$\langle\hat{Z}\hat{Z}\rangle$	$\langle\hat{X}\hat{X}\rangle$	$\langle\hat{Y}\hat{Y}\rangle$
Bell state $\|\Phi^+\rangle$	$+1.000 \pm 0.000$	$+1.000 \pm 0.000$	-1.000 ± 0.000
correlated coin	$+1.000 \pm 0.000$	-0.006 ± 0.016	-0.002 ± 0.016

Table 5: Correlators estimated from 4000 simulated shots per setting, with standard errors. The Bell state has definite parities in all three settings, so its standard errors vanish.

8.4 Single-qubit Bloch vectors during an entangling rotation

The Bloch vector of qubit 0, computed with $\hat{X}\otimes\hat{I}$, $\hat{Y}\otimes\hat{I}$ and $\hat{Z}\otimes\hat{I}$ as for the Bell state, shows how entanglement builds up during an entangling rotation. We write the rotation angle of $\hat{R}_{xx}$ as θ_{xx}, to keep it apart from the angle θ of the family (8.7). During the rotation

$$\hat{R}_{xx}(\theta_{xx})|00\rangle = \cos\tfrac{\theta_{xx}}{2}|00\rangle - i\sin\tfrac{\theta_{xx}}{2}|11\rangle \tag{8.9}$$

the Bloch vector of qubit 0 is $(0, 0, \cos\theta_{xx})$. Its length shrinks from one to 0.000 at the Bell point $\theta_{xx} = \pi/2$ and grows again, pointing down, towards $|11\rangle$ at $\theta_{xx} = \pi$ (Figure 23). A vector inside the sphere belongs to no pure state of one qubit. While the qubits are entangled, qubit 0 alone is therefore in no pure state, and its description needs a density matrix.

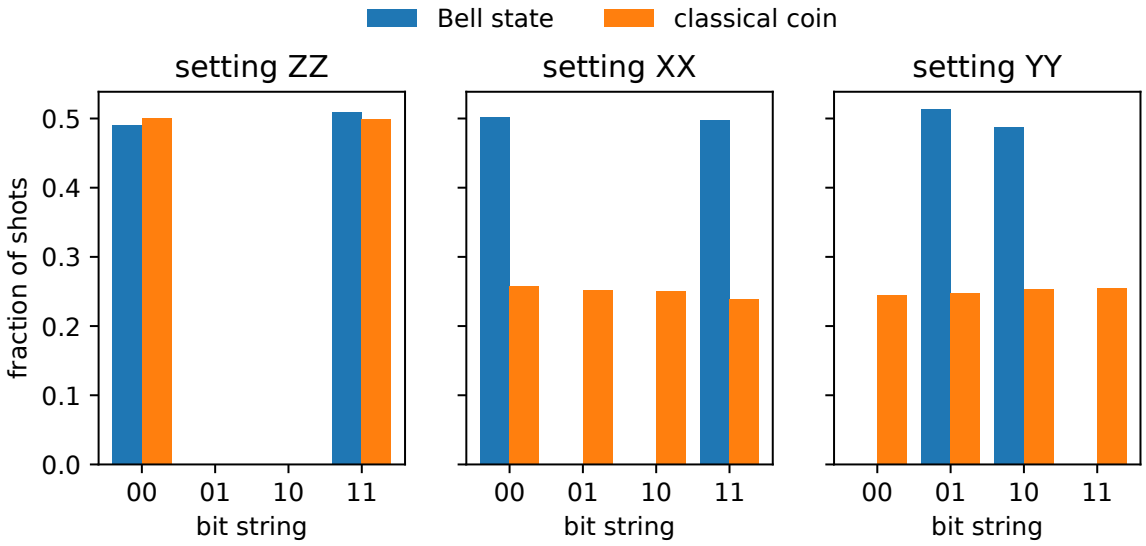


Figure 22: Frequencies of the four bit strings for the Bell state and the correlated coin in the settings $\hat{Z}\hat{Z}$, $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$. The Bell state produces only 00 and 11 in $\hat{Z}\hat{Z}$ and $\hat{X}\hat{X}$, and only 01 and 10 in $\hat{Y}\hat{Y}$. The coin produces all four bit strings in $\hat{X}\hat{X}$ and $\hat{Y}\hat{Y}$.

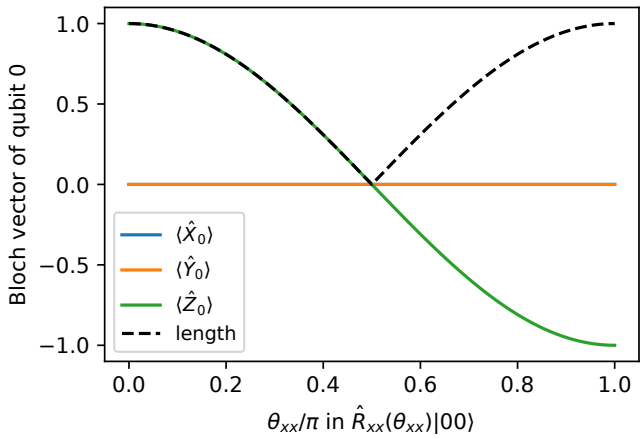


Figure 23: Components and length (dashed) of the Bloch vector of qubit 0 during $\hat{R}_{xx}(\theta_{xx})|00\rangle$, against the rotation angle θ_{xx}. The length $|\cos\theta_{xx}|$ equals one only for the product states $|00\rangle$ and $|11\rangle$ and vanishes at the Bell point.

9 The register's Hamiltonian

9.1 Pauli strings and exact diagonalisation

Physical qubits interact through a Hamiltonian, and quantum machine-learning methods prepare its ground state, learn it from data and map its phases. For two spins a standard model Hamiltonian couples them by the Heisenberg exchange of Section 7.2 and adds a field h along z that acts on each spin,

$$\hat{\mathcal{H}} = J\,(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1) + h\,(\hat{Z}_0 + \hat{Z}_1), \qquad \hat{X}_0\hat{X}_1 \equiv \hat{X}\otimes\hat{X}, \quad \hat{Z}_0 \equiv \hat{Z}\otimes\hat{I}, \quad \hat{Z}_1 \equiv \hat{I}\otimes\hat{Z}. \tag{9.1}$$

Each term is a Pauli string of Section 7.5 with a real coefficient. By (7.18) every Hamiltonian of two qubits is a real combination of the sixteen Pauli strings, and (9.1) uses five of them. The Hamiltonian is a 4×4 Hermitian matrix built from Kronecker products. Its eigenvalues are the allowed energies and its eigenvectors the stationary states, and *exact diagonalisation* computes all of them numerically.

For longer spin chains the Hamiltonian is stored as a sparse matrix (Section 5.4). Each Pauli string has one nonzero entry in each row, so L terms give at most $L\cdot 2^N$ nonzero entries, and sparse

routines find the ground state and apply $e^{-it\hat{\mathcal{H}}}$ to a vector from matrix–vector products alone.

The rotation picture of Section 7.2 predicts the spectrum. The exchange term has the eigenvalue $+1$ on the three symmetric states and -3 on the singlet, so it contributes J and $-3J$. The field term counts the spins along z, since $\hat{Z}_0 + \hat{Z}_1$ equals $+2$ on $|00\rangle$, -2 on $|11\rangle$ and 0 on the two states with a single 1. The symmetric and singlet states are eigenvectors of both terms, and the four energies are

$$E_{|00\rangle} = J + 2h, \qquad E_{|11\rangle} = J - 2h, \qquad E_{(|01\rangle+|10\rangle)/\sqrt{2}} = J, \qquad E_{\text{singlet}} = -3J. \tag{9.2}$$

For $J > 0$ the coupling is *antiferromagnetic*, with the exchange energy lowest when the two spins point in opposite directions, whereas a coupling with $J < 0$ favours parallel spins and is ferromagnetic. At $J > 0$ and zero field the ground state is the singlet, an entangled state with $s_1 = s_2 = 1/\sqrt{2}$, at the energy $-3J$ found by exact diagonalisation. The singlet level is flat (Figure 24), because each of its two components has one spin up and one spin down, so $\hat{Z}_0 + \hat{Z}_1$ vanishes on it.

The level of $|11\rangle$ falls with h and crosses the singlet where $J - 2h = -3J$, at $h = 2J$. A strong positive field lowers the energy of spins in $|1\rangle$ and makes the product state $|11\rangle$ the ground state for $h > 2J$, while for $h < -2J$ the ground state is $|00\rangle$. Between the two crossings the ground state is entangled, and outside them it is a product state. One parameter of the Hamiltonian thus switches the ground state between an entangled and a product state, the smallest example of the ground-state problems addressed by variational quantum algorithms.

A single qubit with the ground state $|0\rangle$ and the gap ω_0 of Section 2.1 has the Hamiltonian $-\frac{\omega_0}{2}\hat{Z}$, which corresponds to a negative field $h = -\omega_0/2$ in (9.1).

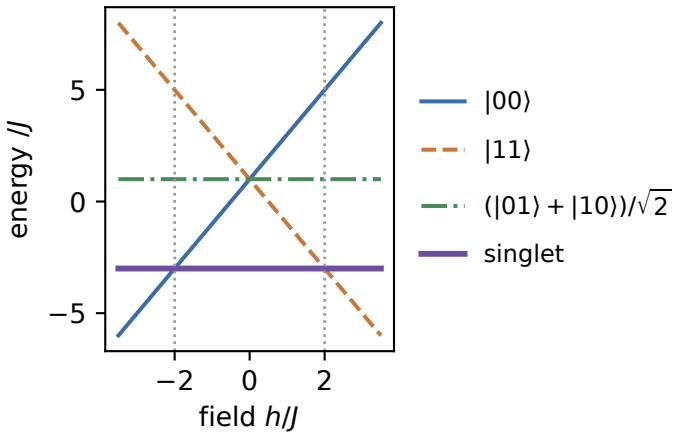


Figure 24: Energy levels of the two-spin Hamiltonian (9.1) against the field h for $J = 1$. The flat singlet level is the ground state for $|h| < 2J$, and the product states $|11\rangle$ and $|00\rangle$ take over beyond the crossings at $h = \pm 2J$ (dotted).

9.2 An excitation hopping between the spins

A gate is the evolution under a Hamiltonian for a fixed time, and following the state continuously in time shows how entanglement builds up. We keep only the transverse part of the exchange,

$$\hat{\mathcal{H}}_{XX} = \frac{J}{2}\left(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1\right) = J\left(|01\rangle\langle 10| + |10\rangle\langle 01|\right). \tag{9.3}$$

The second form follows from $\hat{X}\otimes\hat{X}|01\rangle = |10\rangle$ and $\hat{Y}\otimes\hat{Y}|01\rangle = (i|1\rangle)\otimes(-i|0\rangle) = |10\rangle$, together with the same relations for $|10\rangle$. Both terms exchange $|00\rangle$ and $|11\rangle$ with opposite signs and cancel. With the ladder operators (2.16), $|01\rangle\langle 10| = \hat{S}_0^-\hat{S}_1^+$ and $|10\rangle\langle 01| = \hat{S}_0^+\hat{S}_1^-$, so $\hat{\mathcal{H}}_{XX} = J(\hat{S}_0^+\hat{S}_1^- + \hat{S}_0^-\hat{S}_1^+)$ lowers one spin and raises the other, moving one excitation from spin to spin. In the subspace $\{|01\rangle, |10\rangle\}$ it acts as $J\hat{X}$, and the evolution from $|01\rangle$, qubit 0 up and qubit 1 down, is the rotation $\hat{R}_x(2Jt)$ of that subspace,

$$|\psi(t)\rangle = e^{-i\hat{\mathcal{H}}_{XX}t}|01\rangle = \cos(Jt)|01\rangle - i\sin(Jt)|10\rangle. \tag{9.4}$$

The magnetisations oscillate in opposite phase,

$$\langle \hat{Z}_0(t) \rangle = \cos^2(Jt) - \sin^2(Jt) = \cos(2Jt) = -\langle \hat{Z}_1(t) \rangle, \tag{9.5}$$

and the Schmidt coefficients are $|\cos Jt|$ and $|\sin Jt|$. The excitation is shared equally, and the register maximally entangled, at $Jt = \pi/4$ and $3\pi/4$, where the smaller squared Schmidt coefficient reaches 0.500 (Figure 25). We compute the evolution numerically from the spectral form (3.5). Records of $\langle \hat{Z}_0(t) \rangle$ and $\langle \hat{Z}_1(t) \rangle$ at many times are the data from which Hamiltonian learning infers the unknown Hamiltonian of a device.

The three couplings of the two spins differ in their evolution operators and in the times at which they entangle maximally (Table 6). The Ising coupling $\frac{J}{2}\hat{X}_0\hat{X}_1$ produces $\hat{R}_{xx}(\theta)$ with $\theta = Jt$, so its Bell point $\theta = \pi/2$ is reached at $Jt = \pi/2$. The hopping Hamiltonian $\hat{\mathcal{H}}_{XX}$ entangles $|01\rangle$ maximally at $Jt = \pi/4$.

Hamiltonian	evolution after the time t	maximal entanglement
Ising coupling $\frac{J}{2}\hat{X}_0\hat{X}_1$	$\hat{R}_{xx}(\theta)$ with $\theta = Jt$	$\|00\rangle \to (\|00\rangle - i\|11\rangle)/\sqrt{2}$ at $Jt = \pi/2$
hopping $\frac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$	$\hat{R}_x(2Jt)$ in $\{\|01\rangle, \|10\rangle\}$	from $\|01\rangle$ at $Jt = \pi/4$
exchange $J(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1 + \hat{Z}_0\hat{Z}_1)$, equal to $4J\,\hat{\mathbf{S}}_0 \cdot \hat{\mathbf{S}}_1$	phase e^{-iJt} on the symmetric states and e^{3iJt} on the singlet	the singlet ground state for $J > 0$

Table 6: The two-spin couplings, their evolution operators and the times or states of maximal entanglement.

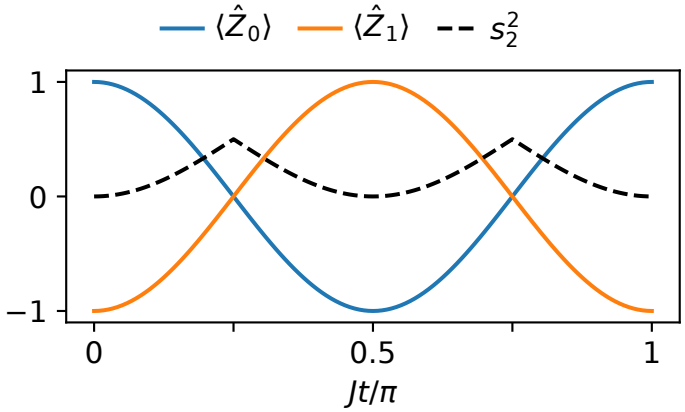


Figure 25: An excitation hopping between the two spins under $\hat{\mathcal{H}}_{XX}$, started in $|01\rangle$. The magnetisations $\langle \hat{Z}_0 \rangle$ and $\langle \hat{Z}_1 \rangle$ oscillate in opposite phase, and the smaller squared Schmidt coefficient (dashed) reaches 1/2 whenever they cross zero.

9.3 Energies from bit strings

Every term of $\hat{\mathcal{H}}$ is a Pauli string whose expectation value we estimate from measurement records as in Section 6. One shot of $\hat{Z}_0$ yields the value $(-1)^{b_0}$, and one shot of $\hat{Z}_0\hat{Z}_1$ the parity $(-1)^{b_0+b_1}$ of equation (6.2). A Pauli string with $\hat{X}$ or $\hat{Y}$ factors is measured in the matching setting of Section 6.2 and evaluated with the same parity.

One setting measures several strings at once, and the setting $\hat{Z}\hat{Z}$ gives $\hat{Z}_0$, $\hat{Z}_1$ and $\hat{Z}_0\hat{Z}_1$ from the same shots. In general a setting measures every Pauli string that carries, on each qubit, either the identity or the Pauli matrix of the setting, and such strings commute with each other. The Hamiltonian (9.1) therefore needs three settings, $\hat{X}\hat{X}$ for $\hat{X}_0\hat{X}_1$, $\hat{Y}\hat{Y}$ for $\hat{Y}_0\hat{Y}_1$, and $\hat{Z}\hat{Z}$ for $\hat{Z}_0\hat{Z}_1$, $\hat{Z}_0$ and $\hat{Z}_1$. Each shot in the setting $\hat{Z}\hat{Z}$ yields one value of $J\hat{Z}_0\hat{Z}_1 + h(\hat{Z}_0 + \hat{Z}_1)$. Because the three parts

of the energy come from independent records, their standard errors add in quadrature,

$$\bar{E} = \overline{\langle J\hat{X}_0\hat{X}_1\rangle} + \overline{\langle J\hat{Y}_0\hat{Y}_1\rangle} + \overline{\langle J\hat{Z}_0\hat{Z}_1 + h(\hat{Z}_0+\hat{Z}_1)\rangle}, \qquad \mathrm{std}(\bar{E}) = \Big(\sum_{\text{settings}} \mathrm{std}^2\Big)^{1/2}. \tag{9.6}$$

A bar denotes the sample mean over the shots of the setting that measures the term. We estimate the energy at $J = 1$, $h = 0.5$ from 1000 simulated shots in each setting (Table 7). In the Bell state the three exchange strings take the definite values $+1$, -1 and $+1$, and the exchange part equals J in every shot. The field part has mean zero but takes the values $\pm 2h$ from shot to shot, because in the setting $\hat{Z}\hat{Z}$ the two bits agree and each is a fair coin, so its standard error is $2h/\sqrt{M}$. The product state $|0\rangle|+\rangle$ has $\langle\hat{X}_0\hat{X}_1\rangle = \langle\hat{Y}_0\hat{Y}_1\rangle = \langle\hat{Z}_0\hat{Z}_1\rangle = 0$ and the exact energy h. Its bits are random in more settings, and its estimate carries the larger standard error. Both estimates agree with the exact energies within their standard errors. Variational quantum algorithms use this estimate of the energy of a prepared state from a few measurement settings as their measurement step.

state	estimate $\bar{E}$	standard error	exact energy
Bell state $\|\Phi^+\rangle$	$+1.030$	$2h/\sqrt{M}$	$+1.000$
product state $\|0\rangle\|+\rangle$	$+0.590$	0.065	$+0.500$

Table 7: Energy of the Hamiltonian (9.1) at $J = 1$, $h = 0.5$ estimated from $M = 1000$ simulated shots in each of the settings $\hat{X}\hat{X}$, $\hat{Y}\hat{Y}$ and $\hat{Z}\hat{Z}$.

10 Density matrices

10.1 The density matrix of a pure state

The correlated coin of Section 8.3 and one qubit of an entangled register have no state vector, and both call for a description of incompletely known states. We build it from the expectation value (2.11) rewritten with the trace. The trace is cyclic for any matrices, $\mathrm{Tr}(\hat{A}\hat{B}) = \mathrm{Tr}(\hat{B}\hat{A})$, and therefore

$$\langle\psi|\hat{O}|\psi\rangle = \mathrm{Tr}(\hat{O}|\psi\rangle\langle\psi|) = \mathrm{Tr}(\hat{O}\hat{\rho}), \qquad \hat{\rho} = |\psi\rangle\langle\psi|. \tag{10.1}$$

The outer product $\hat{\rho}$ is the *density matrix* of the pure state $|\psi\rangle$. For one qubit with $|\psi\rangle = (a,b)^T$,

$$\hat{\rho} = \begin{pmatrix} a \\ b \end{pmatrix}\begin{pmatrix} a^* & b^* \end{pmatrix} = \begin{pmatrix} |a|^2 & ab^* \\ a^*b & |b|^2 \end{pmatrix}. \tag{10.2}$$

The diagonal holds the Born probabilities of the computational basis, and the off-diagonal elements, the *coherences*, carry the relative phase. The global phase cancels in $\hat{\rho}$, since $e^{i\alpha}|\psi\rangle\langle\psi|e^{-i\alpha} = \hat{\rho}$, and the density matrix keeps everything observable about the state. For the two-qubit product state $|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle$ of (5.1), the rule $(\hat{A}\otimes\hat{B})(\hat{C}\otimes\hat{D}) = \hat{A}\hat{C}\otimes\hat{B}\hat{D}$ yields

$$\hat{\rho} = (|\psi^{(0)}\rangle \otimes |\psi^{(1)}\rangle)(\langle\psi^{(0)}| \otimes \langle\psi^{(1)}|) = \hat{\rho}_0 \otimes \hat{\rho}_1 = \begin{pmatrix} |a|^2\hat{\rho}_1 & ab^*\hat{\rho}_1 \\ a^*b\,\hat{\rho}_1 & |b|^2\hat{\rho}_1 \end{pmatrix}, \qquad \hat{\rho}_1 = \begin{pmatrix} |c|^2 & cd^* \\ c^*d & |d|^2 \end{pmatrix}, \tag{10.3}$$

where $\hat{\rho}_0 = |\psi^{(0)}\rangle\langle\psi^{(0)}|$ is the density matrix of qubit 0 and $\hat{\rho}_1 = |\psi^{(1)}\rangle\langle\psi^{(1)}|$ that of qubit 1. The register matrix $\hat{\rho}$ is 4×4, with rows and columns in the order $00, 01, 10, 11$. A pure-state density matrix is a projector of rank one, since $\langle\psi|\psi\rangle = 1$ and

$$\hat{\rho}^2 = |\psi\rangle\langle\psi|\psi\rangle\langle\psi| = \hat{\rho}, \qquad \mathrm{Tr}\hat{\rho}^2 = \mathrm{Tr}\hat{\rho} = 1. \tag{10.4}$$

10.2 Mixtures and superpositions

A state vector describes a system whose preparation we know completely, and often we know less. A source may emit $|\psi_1\rangle$ with probability p_1 and $|\psi_2\rangle$ with probability p_2 without revealing which one it emitted. The expectation value of an observable is then the average of the expectation values of the possible states, and by (10.1)

$$\langle \hat{O} \rangle = \sum_i p_i \langle \psi_i | \hat{O} | \psi_i \rangle = \sum_i p_i \mathrm{Tr}(\hat{O} |\psi_i\rangle\langle\psi_i|) = \mathrm{Tr}(\hat{O}\hat{\rho}), \qquad \hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|. \tag{10.5}$$

One matrix, the *density matrix* or density operator $\hat{\rho}$ of the mixture, contains all predictions of the source. The probability of the outcome k of a measurement in an orthonormal basis $\{|k\rangle\}$ is $p_k = \langle k|\hat{\rho}|k\rangle$, the diagonal of $\hat{\rho}$ in that basis.

To see how a mixture differs from a superposition, we compare the superposition $|+\rangle$ with a source that emits $|0\rangle$ or $|1\rangle$ with probability one half each,

$$\hat{\rho}_+ = |+\rangle\langle+| = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{mix}} = \frac{1}{2}|0\rangle\langle 0| + \frac{1}{2}|1\rangle\langle 1| = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{10.6}$$

Sharing their diagonal, the two matrices give 0 and 1 with probability one half in the $\hat{Z}$ basis. In the $\hat{X}$ basis the superposition always gives the outcome $+$, while the mixture still gives $+$ and $-$ with probability one half. The coherences of $\hat{\rho}_+$ carry this difference, and simulated records of 2000 shots in each basis show it. The $\hat{X}$ basis thus distinguishes these two particular states. One fixed basis cannot determine an arbitrary density matrix, because its outcome probabilities fix only the diagonal of $\hat{\rho}$ in that basis.

10.3 Properties, purity and the Bloch ball

The mixture $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ gives every density matrix three properties.

- It is Hermitian, $\hat{\rho}^\dagger = \hat{\rho}$, because each projector $|\psi_i\rangle\langle\psi_i|$ is Hermitian.
- Its trace is one, $\mathrm{Tr}\hat{\rho} = \sum_i p_i \langle\psi_i|\psi_i\rangle = 1$, so the outcome probabilities sum to one.
- It is positive, $\langle v|\hat{\rho}|v\rangle = \sum_i p_i |\langle v|\psi_i\rangle|^2 \geq 0$ for every vector v, so every outcome probability is non-negative.

Its eigenvalues λ_k are therefore non-negative and sum to one, so they are probabilities. The *purity*

$$\mathrm{Tr}\hat{\rho}^2 = \sum_k \lambda_k^2 \tag{10.7}$$

equals one if and only if one eigenvalue is one and the others vanish, that is, for a pure state. It is smallest, $1/d$ in dimension d, for the *maximally mixed state* $\hat{I}/d$. Table 8 contrasts the two cases.

	pure state	mixed state
form	$\hat{\rho} = \|\psi\rangle\langle\psi\|$	$\hat{\rho} = \sum_i p_i \|\psi_i\rangle\langle\psi_i\|$, at least two $p_i > 0$ with $\|\psi_i\rangle$ not all equal
rank	1	≥ 2
$\hat{\rho}^2$	$= \hat{\rho}$	$\neq \hat{\rho}$
purity $\mathrm{Tr}\hat{\rho}^2$	1	< 1, down to $1/d$ for $\hat{I}/d$
one-qubit Bloch vector	on the sphere, $\|\mathbf{r}\| = 1$	inside the ball, $\|\mathbf{r}\| < 1$
example	$\|+\rangle$, $\|\Phi^+\rangle$	$\hat{\rho}_{\mathrm{mix}} = \hat{I}/2$, the correlated coin

Table 8: Pure and mixed density matrices.

For one qubit the four matrices $\hat{I}, \hat{X}, \hat{Y}, \hat{Z}$ form a basis of the Hermitian 2×2 matrices, and the three properties allow only the form

$$\hat{\rho} = \tfrac{1}{2}(\hat{I} + r_x\hat{X} + r_y\hat{Y} + r_z\hat{Z}), \qquad r_\alpha = \mathrm{Tr}(\hat{\rho}\,\hat{\sigma}_\alpha) = \langle\hat{\sigma}_\alpha\rangle, \qquad |\mathbf{r}| \leq 1, \qquad \mathrm{Tr}\hat{\rho}^2 = \tfrac{1}{2}(1 + |\mathbf{r}|^2). \tag{10.8}$$

The trace condition fixes the coefficient of $\hat{I}$. The coefficients r_α are the expectation values because $\mathrm{Tr}(\hat{\sigma}_\alpha\hat{\sigma}_\beta) = 2\delta_{\alpha\beta}$, and the eigenvalues $\frac{1}{2}(1 \pm |\mathbf{r}|)$ are non-negative only for $|\mathbf{r}| \leq 1$. Pure states lie on the Bloch sphere $|\mathbf{r}| = 1$ of Section 2.7, and mixed states fill its interior, the *Bloch ball*. The maximally mixed state $\hat{I}/2$ sits at the centre, with purity 0.5. A mixture averages Bloch vectors with its weights, so the mixture of two pure states lies on the straight segment between them (Figure 27).

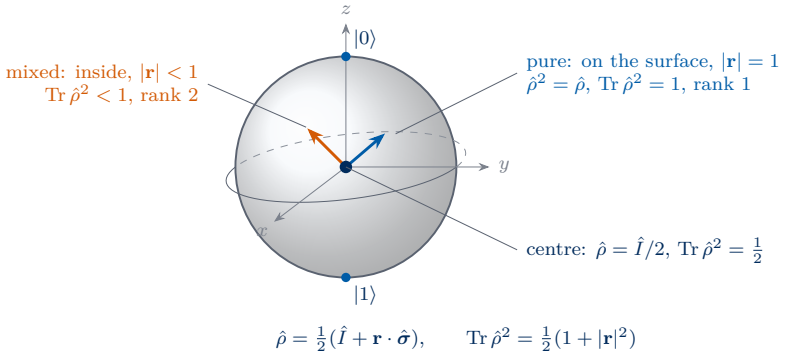


Figure 26: Pure states lie on the Bloch sphere and have purity one, mixed states lie inside the ball with purity $(1 + |\mathbf{r}|^2)/2 < 1$, and the maximally mixed state $\hat{I}/2$ sits at the centre.

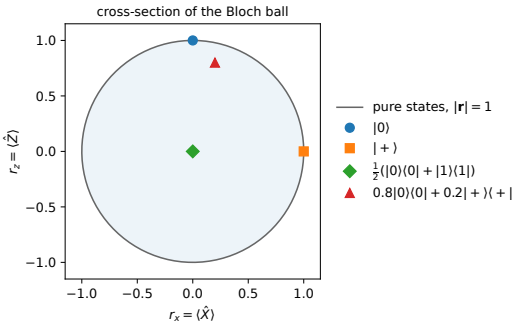


Figure 27: Cross-section of the Bloch ball in the x–z plane. Pure states lie on the circle, mixtures inside it, and the mixture of $|0\rangle$ and $|1\rangle$ with equal weights at the centre.

10.4 The correlated coin as a density matrix

The Bell state and the correlated coin of Section 8.3 now become two density matrices,

$$\hat{\rho}_{\mathrm{Bell}} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 \end{pmatrix}, \qquad \hat{\rho}_{\mathrm{coin}} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{10.9}$$

in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. They share the diagonal and differ only in the coherences between $|00\rangle$ and $|11\rangle$, the entries responsible for $\langle\hat{X}\hat{X}\rangle$ and $\langle\hat{Y}\hat{Y}\rangle$. Evaluating $\langle\hat{O}\rangle = \mathrm{Tr}(\hat{O}\hat{\rho})$ with these two matrices reproduces the correlators of Table 5. The Bell register is pure, with purity one, whereas the coin register is mixed, with rank two and purity $1/2$.

10.5 The partial trace

Qubit 0 of an entangled register has a Bloch vector inside the ball (Section 8), so it is in no pure state. The density matrix that describes one qubit of the register follows from the register state written with one index for each qubit, $|\psi\rangle = \sum_{a,b} \psi_{ab}|a\rangle|b\rangle$. Here a is the bit of qubit 0 and b that of qubit 1.

A measurement of qubit 0 alone corresponds to an observable $\hat{O}_0 \otimes \hat{I}$, and in its expectation value the bra carries its own indices, which we write with primes, a' and b',

$$\langle\psi|\hat{O}_0 \otimes \hat{I}|\psi\rangle = \sum_{a,a',b,b'} \psi^*_{a'b'}\,(\hat{O}_0)_{a'a} \underbrace{\delta_{b'b}}_{\text{identity on qubit 1}} \psi_{ab} = \sum_{a,a'} (\hat{O}_0)_{a'a} \underbrace{\sum_b \psi_{ab}\,\psi^*_{a'b}}_{(\hat{\rho}_0)_{aa'}} = \mathrm{Tr}(\hat{O}_0\hat{\rho}_0). \quad (10.10)$$

The identity on qubit 1 produced the Kronecker delta $\delta_{b'b}$, which sets the index of qubit 1 in ψ and in ψ^* equal before it is summed. The result depends on the state only through the 2×2 matrix

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\,\psi^*_{a'b}, \quad (10.11)$$

the *reduced density matrix* of qubit 0, which contains everything that measurements on qubit 0 alone can reveal. It is produced by the *partial trace* over qubit 1, written $\hat{\rho}_0 = \mathrm{Tr}_1|\psi\rangle\langle\psi|$, a linear map from 4×4 matrices to 2×2 matrices. On the diagonal the partial trace sums the joint probabilities over the bit of qubit 1, as marginalisation does for a joint distribution in probability theory. The reduced state of qubit 1 follows in the same way, $(\hat{\rho}_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'}$, as the partial trace over qubit 0. Element by element for two qubits,

$$\hat{\rho}_0 = \begin{pmatrix} |\psi_{00}|^2 + |\psi_{01}|^2 & \psi_{00}\psi^*_{10} + \psi_{01}\psi^*_{11} \\ \psi_{10}\psi^*_{00} + \psi_{11}\psi^*_{01} & |\psi_{10}|^2 + |\psi_{11}|^2 \end{pmatrix}. \quad (10.12)$$

The diagonal holds the marginals of Section 6, the probabilities that qubit 0 is found in 0 or in 1, summed over both outcomes of qubit 1. Each off-diagonal element pairs amplitudes that agree on qubit 1.

On paper the same matrix follows from the 4×4 density matrix $\hat{\rho} = |\psi\rangle\langle\psi|$ of the register, with rows and columns labelled by the bit strings ab at the index $2a + b$. Grouping them by the bit of qubit 0 divides $\hat{\rho}$ into four 2×2 blocks. The partial trace over qubit 1 replaces each block by its trace,

$$\hat{\rho} = \begin{pmatrix} \rho^{(00)} & \rho^{(01)} \\ \rho^{(10)} & \rho^{(11)} \end{pmatrix}, \qquad \hat{\rho}_0 = \mathrm{Tr}_1\hat{\rho} = \begin{pmatrix} \mathrm{Tr}\rho^{(00)} & \mathrm{Tr}\rho^{(01)} \\ \mathrm{Tr}\rho^{(10)} & \mathrm{Tr}\rho^{(11)} \end{pmatrix}, \quad (10.13)$$

where the block $\rho^{(aa')}$ holds the elements $\psi_{ab}\psi^*_{a'b'}$. Its trace sets $b' = b$ and sums over b, reproducing formula (10.11). The partial trace over qubit 0 instead adds the two diagonal blocks, $\hat{\rho}_1 = \rho^{(00)} + \rho^{(11)}$. For a product state $\hat{\rho} = \hat{\rho}_0 \otimes \hat{\rho}_1$ the blocks are $(\hat{\rho}_0)_{aa'}\hat{\rho}_1$, and the block traces return $\hat{\rho}_0$, since $\mathrm{Tr}\hat{\rho}_1 = 1$. A third form uses the 2×2 matrix $\Psi_{ab} = \psi_{ab}$ of Section 8.2, with rows labelled by qubit 0, in which the sum formula becomes a matrix product,

$$\hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^T. \quad (10.14)$$

Index by index, $(\Psi\Psi^\dagger)_{aa'} = \sum_b \psi_{ab}\psi^*_{a'b}$ is formula (10.11), while $(\Psi^\dagger\Psi)_{b'b} = \sum_a \psi^*_{ab'}\psi_{ab} = (\hat{\rho}_1)_{bb'}$ carries the indices of $\hat{\rho}_1$ in the opposite order, hence the transpose.

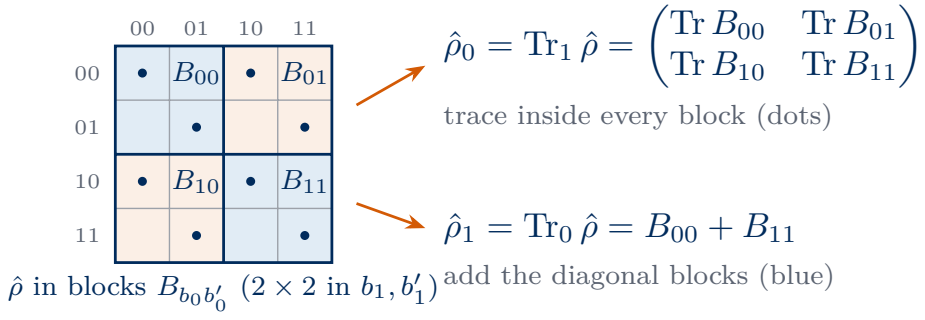


Figure 28: The partial traces of a 4×4 density matrix as block operations. Tracing out qubit 1 replaces each 2×2 block by its trace, and tracing out qubit 0 adds the two diagonal blocks.

On a computer the partial trace contracts the state tensor $\psi[a,b]$ with its conjugate over the index of qubit 1 and leaves both indices of qubit 0 open, `"ab,Ab->aA"` with the operands `(psi, psi.conj())`, where an upper-case letter stands for a primed index. In the first diagram of Figure 29 the legs of qubit 0 stay open and become the two indices of $\hat{\rho}_0$, while the joined legs of qubit 1 form the sum over b. In the second the density matrix, reshaped to a tensor with four legs, has its ket leg b joined to its bra leg b', which is the block trace (10.13). The contraction never builds the $2^N \times 2^N$ matrix and is the form used for many qubits.

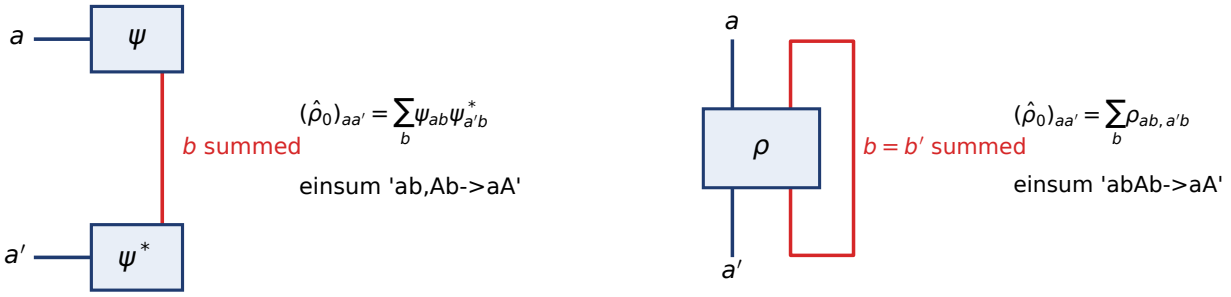


Figure 29: The partial trace over qubit 1 as tensor networks. Left, the state tensor ψ and its conjugate with the legs of qubit 1 joined. Right, the density matrix as a tensor with the ket and bra legs of qubit 1 joined.

We apply formula (10.11) to three register states.

- The running product state $|0\rangle \otimes |+\rangle$ has the amplitudes $\psi_{00} = \psi_{01} = 1/\sqrt{2}$ and $\psi_{10} = \psi_{11} = 0$. Only $a = a' = 0$ gives a nonzero term, and the two reduced states are

$$\hat{\rho}_0 = \sum_b \psi_{ab}\psi^*_{a'b} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = |0\rangle\langle 0|, \qquad (\hat{\rho}_1)_{bb'} = \sum_a \psi_{ab}\psi^*_{ab'} = \psi_{0b}\psi^*_{0b'} = \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = |+\rangle\langle +|, \tag{10.15}$$

  the pure states of the two qubits themselves.

- The Bell state has $\psi_{00} = \psi_{11} = 1/\sqrt{2}$, so $\Psi = \hat{I}/\sqrt{2}$ and $\hat{\rho}_0 = \Psi\Psi^\dagger = \hat{I}/2$. The register is in a completely known pure state, while each qubit alone is the maximally mixed state of rank two. The local state $\hat{I}/2$ fixes every prediction for one qubit, a fair coin in every basis, and the correlations of Section 8.3 appear only in joint measurements. A pure state of the whole with maximally mixed parts has no classical counterpart and is the signature of entanglement.

- The family $|\psi_\theta\rangle = \cos\theta|00\rangle + \sin\theta|11\rangle$ has $\Psi = \mathrm{diag}(\cos\theta, \sin\theta)$ and $\hat{\rho}_0 = \mathrm{diag}(\cos^2\theta, \sin^2\theta)$. It interpolates between the pure $|0\rangle\langle 0|$ at $\theta = 0$ and $\hat{I}/2$ at $\theta = \pi/4$.

By (10.14) the rank of the reduced state equals the rank of Ψ, the Schmidt rank. The reduced state of an entangled pure state is therefore always mixed, with rank two for two qubits.

10.6 Purity of the reduced state

The purity $\mathrm{Tr}\hat{\rho}_0^2$ measures how mixed qubit 0 looks on its own. For the three cases it is 1 for the product state, $\mathrm{Tr}(\hat{I}/4) = 1/2$ for the Bell state, and $\cos^4\theta + \sin^4\theta$ for the family. The last falls from 1 at $\theta = 0$ to 1/2 at $\theta = \pi/4$ and equals 0.75 at $\theta = \pi/8$. In terms of the optional concurrence of Section 8.2, a general pure state of two qubits has $\mathrm{Tr}\hat{\rho}_0^2 = 1 - C^2/2$. The register is pure in all three cases, so a reduced purity below one comes entirely from the correlations between the qubits.

10.7 Entanglement entropy

The eigenvalues λ_k of $\hat{\rho}_0$ are probabilities, and their Shannon entropy in bits is the *von Neumann entropy*

$$S(\hat{\rho}_0) = -\mathrm{Tr}\,\hat{\rho}_0 \log_2 \hat{\rho}_0 = -\sum_k \lambda_k \log_2 \lambda_k, \tag{10.16}$$

with the convention $0 \log 0 = 0$. For a pure register it is called the *entanglement entropy*, zero for a product state and one bit for the Bell state. For the family $|\psi_\theta\rangle$ it is the binary entropy $S_{\mathrm{bin}}(\cos^2\theta)$, with $S_{\mathrm{bin}}(p) = -p\log_2 p - (1-p)\log_2(1-p)$, which equals 0.60 bit at $\theta = \pi/8$ (Figure 30).

The entropy follows without forming $\hat{\rho}_0$ from the SVD (8.5) of the reshaped state, $\Psi = U\Sigma V^\dagger$, which gives with $V^\dagger V = I$

$$\hat{\rho}_0 = \Psi\Psi^\dagger = U\Sigma V^\dagger V \Sigma U^\dagger = U\Sigma^2 U^\dagger = U\,\mathrm{diag}(s_1^2, s_2^2)\,U^\dagger, \tag{10.17}$$

so the eigenvalues of $\hat{\rho}_0$ are the squared Schmidt coefficients, $\lambda_k = s_k^2$, and

$$S = -\sum_k s_k^2 \log_2 s_k^2. \tag{10.18}$$

The matrix $\Psi^\dagger\Psi$ has the same nonzero eigenvalues as $\Psi\Psi^\dagger$, so the two qubits of a pure register always have equal entropies, $S(\hat{\rho}_0) = S(\hat{\rho}_1)$. A random state, whose four amplitudes are drawn as independent complex Gaussian numbers and then normalised, has for example 0.36 bit on each side. Table 9 collects the criteria for the three cases. The singular-value route extends to a cut of N qubits into two blocks, where Ψ becomes a $2^{N_A} \times 2^{N-N_A}$ matrix, and underlies matrix product states.

pure register state	Schmidt coefficients	Schmidt rank	$\hat{\rho}_0$	$\mathrm{Tr}\hat{\rho}_0^2$	$S(\hat{\rho}_0)$ [bit]
product $\|0\rangle\|+\rangle$	$1,\ 0$	1	$\|0\rangle\langle 0\|$	1	0
$\|\psi_\theta\rangle$, $0 < \theta < \frac{\pi}{2}$	$\|\cos\theta\|,\ \|\sin\theta\|$	2	$\mathrm{diag}(\cos^2\theta, \sin^2\theta)$	$\cos^4\theta + \sin^4\theta$	$S_{\mathrm{bin}}(\cos^2\theta)$
Bell $\|\Phi^+\rangle$	$\frac{1}{\sqrt{2}},\ \frac{1}{\sqrt{2}}$	2	$\hat{I}/2$	1/2	1

Table 9: Entanglement criteria for pure two-qubit states. All follow from the singular values of the reshaped amplitudes, with $\lambda_k = s_k^2$.

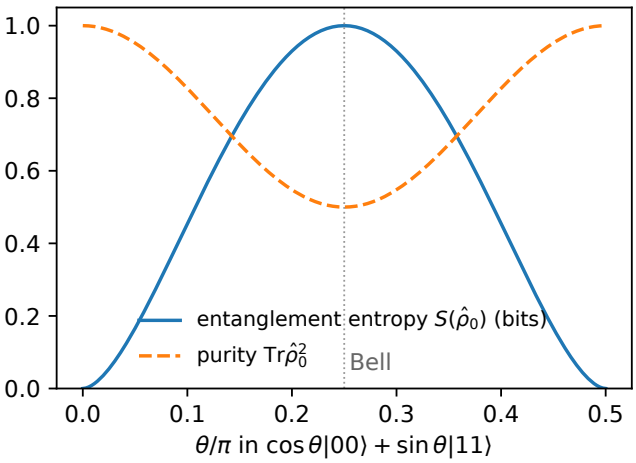


Figure 30: Entanglement entropy and purity of qubit 0 for $\cos\theta|00\rangle + \sin\theta|11\rangle$. The entropy rises from zero for $|00\rangle$ to one bit at the Bell state, $\theta = \pi/4$, and returns to zero at $|11\rangle$. The purity mirrors it between one and one half.

10.8 Mixed registers

The entropy of a reduced state measures entanglement only when the whole register is pure. In the correlated coin, a mixed register, qubit 0 is also a fair coin, $\hat{\rho}_0 = \hat{I}/2$, with one bit of entropy as for the Bell state. That bit is ordinary ignorance about which of $|00\rangle$ and $|11\rangle$ was emitted, and the coin contains no entanglement. The two registers look identical on one qubit. They differ in the correlations $\langle \hat{X}\hat{X}\rangle$ and $\langle \hat{Y}\hat{Y}\rangle$ and in the purity of the register, one for the Bell state and one half for the coin. For mixed registers the reduced entropy combines classical uncertainty with entanglement, and other measures are needed to quantify entanglement. Table 10 lists rank, purity and von Neumann entropy of the density matrices met so far. The reduced state of the Bell state shares all three numbers with the coin register.

density matrix	rank	purity $\mathrm{Tr}\hat{\rho}^2$	entropy $S(\hat{\rho})$ [bit]
product register $\|0\rangle\|+\rangle$	1	1	0
Bell register $\|\Phi^+\rangle\langle\Phi^+\|$	1	1	0
qubit 0 of the Bell register, $\hat{I}/2$	2	1/2	1
correlated coin register	2	1/2	1
maximally mixed register $\hat{I}/4$	4	0.25	2

Table 10: Rank, purity and entropy of pure and mixed density matrices. A pure state has rank one, purity one and zero entropy.

11 Measurement in general

11.1 Projective measurement and the Lüders rule

So far we have read the qubits in a basis and recorded bit strings. The general description of a measurement assigns a matrix $\hat{P}_k$ to every outcome k. For a *projective measurement* these matrices satisfy three conditions.

- Hermiticity, $\hat{P}_k^\dagger = \hat{P}_k$.
- Orthogonality, $\hat{P}_k\hat{P}_l = \delta_{kl}\hat{P}_k$. For $k = l$ this says that each $\hat{P}_k$ is a projector, and for $k \neq l$ that different outcomes exclude each other.
- Completeness, $\sum_k \hat{P}_k = \hat{I}$, so that one of the outcomes always occurs.

By the Born rule each outcome occurs with the probability $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$, or $\langle\psi|\hat{P}_k|\psi\rangle$ for a pure state. Completeness makes the probabilities sum to $\mathrm{Tr}\hat{\rho} = 1$. After the outcome k the *Lüders rule* updates the state,

$$\hat{\rho} \;\longrightarrow\; \hat{\rho}_k = \frac{\hat{P}_k\,\hat{\rho}\,\hat{P}_k}{p_k}, \tag{11.1}$$

where the denominator restores the unit trace. A repeated measurement with the same projectors returns the same outcome, since $\hat{P}_k\hat{P}_k = \hat{P}_k$. For one qubit in the state (2.4), measured in the $\hat{Z}$ basis, the projector $\hat{P}_0 = |0\rangle\langle 0|$ yields $p_0 = \langle\psi_{\mathrm{ex}}|\hat{P}_0|\psi_{\mathrm{ex}}\rangle = 3/4$ and the updated state $\hat{P}_0|\psi_{\mathrm{ex}}\rangle\langle\psi_{\mathrm{ex}}|\hat{P}_0/p_0 = |0\rangle\langle 0|$. On the register, a $\hat{Z}$-basis measurement of qubit 0 uses $\hat{P}_0 = |0\rangle\langle 0|\otimes\hat{I}$ and $\hat{P}_1 = |1\rangle\langle 1|\otimes\hat{I}$. The $\hat{X}$ basis has the projectors $|\pm\rangle\langle\pm|\otimes\hat{I}$, with outcome bit 0 for $+$, and the $\hat{Y}$ basis has $|\pm i\rangle\langle\pm i|\otimes\hat{I}$.

11.2 Measuring an observable and measuring a Pauli string

An observable $\hat{O}$ with distinct eigenvalues o has the spectral decomposition $\hat{O} = \sum_o o\,\hat{P}_o$, where $\hat{P}_o$ projects onto all eigenvectors with the eigenvalue o. Measuring the observable is the projective measurement $\{\hat{P}_o\}$, and its mean is $\sum_o o\,\mathrm{Tr}(\hat{P}_o\hat{\rho}) = \mathrm{Tr}(\hat{O}\hat{\rho})$, the rule (10.1). Measuring in a basis is the special case in which every projector has rank one. When an eigenvalue is degenerate, measuring the observable reveals less than measuring in a basis and disturbs the state less.

A Pauli string $\hat{P}$ has only the eigenvalues ± 1. Its measurement has two outcomes, with the projectors

$$\hat{P}_{\pm} = \frac{\hat{I} \pm \hat{P}}{2}, \qquad \hat{P} = \hat{P}_{+} - \hat{P}_{-}. \tag{11.2}$$

For $\hat{Z}_0\hat{Z}_1$ the projector $\hat{P}_{+}$ keeps the bit strings 00 and 11 of even parity and $\hat{P}_{-}$ keeps 01 and 10. We obtain the same outcome statistics by measuring both qubits in the $\hat{Z}$ basis and multiplying the two values $(-1)^{b_0}(-1)^{b_1}$, but the post-measurement states differ. The measurement of $\hat{Z}_0\hat{Z}_1$ returns $+1$ with certainty on $0.6|00\rangle + 0.8|11\rangle$ and leaves this state unchanged, coherence included, whereas reading both bits leaves $|00\rangle$ or $|11\rangle$. For $\hat{X}_0\hat{X}_1$ we apply Hadamard gates to both qubits before the same projectors, because $(\hat{H}\otimes\hat{H})(\hat{Z}\otimes\hat{Z})(\hat{H}\otimes\hat{H}) = \hat{X}\otimes\hat{X}$, the basis rotation of Section 6.2.

11.3 A parity measurement with an ancilla

A device can measure a Pauli string without reading out the qubits themselves. It copies the parity onto an extra qubit prepared in $|0\rangle$, an *ancilla*, and reads only the ancilla (Figure 31). On a basis state the two CNOT gates add both bits to the ancilla modulo two, $|b_0b_1\rangle|0\rangle \to |b_0b_1\rangle|b_0 \oplus b_1\rangle$. A general register state $\sum \psi_{b_0b_1}|b_0b_1\rangle$ becomes $\hat{P}_{+}|\psi\rangle|0\rangle + \hat{P}_{-}|\psi\rangle|1\rangle$. The ancilla reads 0 with the probability $\langle\psi|\hat{P}_{+}|\psi\rangle$, after which the register is in $\hat{P}_{+}|\psi\rangle$, normalised. The circuit performs the projective measurement (11.2) of $\hat{Z}_0\hat{Z}_1$ without reading either qubit. Quantum error correction is built on such *mid-circuit measurements*, followed by further gates that may depend on the outcome.

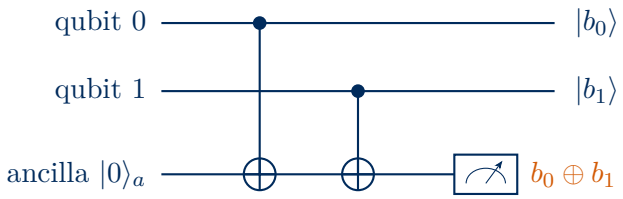


Figure 31: Measuring the Pauli string $\hat{Z}_0\hat{Z}_1$ with an ancilla. Two CNOT gates add the bits of qubits 0 and 1 to the ancilla, $|b_0b_1\rangle|0\rangle_a \to |b_0b_1\rangle|b_0 \oplus b_1\rangle_a$, whose readout gives the parity $b_0 \oplus b_1$ while the two qubits are never read. The outcome projects the qubits on the even or the odd subspace and keeps superpositions within it, so a Bell state $(|00\rangle + |11\rangle)/\sqrt{2}$ is left unchanged. For $\hat{X}_0\hat{X}_1$ Hadamard gates act on both qubits before and after the circuit.

11.4 Back-action in the register

In an entangled register a measurement on one qubit also changes the other. Measuring qubit 0 of the Bell state in the $\hat{Z}$ basis gives the outcome 0 with probability 1/2 and leaves the register in $|00\rangle$, with qubit 1 in $|0\rangle$. The outcome $+$ in the $\hat{X}$ basis leaves qubit 1 in $|+\rangle$, as the Bell state written in the $\hat{X}$ basis shows,

$$|\Phi^{+}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{|++\rangle + |--\rangle}{\sqrt{2}}. \tag{11.3}$$

Before the measurement the partner qubit is a fair coin. Afterwards it is in a definite pure state, the eigenstate of the chosen basis that is correlated with the observed outcome. We simulate records by measuring qubit 0 first and qubit 1 afterwards in the same basis, sampling each outcome from the updated state. The two outcomes of the Bell state agree in every shot in $\hat{Z}$ and $\hat{X}$ and disagree in every shot in $\hat{Y}$, as $\langle\hat{Y}\hat{Y}\rangle = -1$ predicts. For the correlated coin they agree in every shot only in $\hat{Z}$, whereas in $\hat{X}$ they agree in a fraction 0.48 of the shots, like two independent coins. Only the entangled register shows back-action in every basis.

11.5 No signalling

The back-action transmits no signal. An observer at qubit 1 who does not know the outcome on qubit 0 describes qubit 1 by the average of the post-measurement states, weighted with the outcome

probabilities. The average is independent of the basis chosen at qubit 0,

$$\sum_k p_k \,\mathrm{Tr}_0 \hat{\rho}_k = \sum_k \mathrm{Tr}_0 [(\hat{\Pi}_k \otimes \hat{I}) \hat{\rho} (\hat{\Pi}_k \otimes \hat{I})] = \mathrm{Tr}_0 \Big[\Big(\sum_k \hat{\Pi}_k \Big) \otimes \hat{I}\, \hat{\rho} \Big] = \mathrm{Tr}_0 \hat{\rho} = \hat{\rho}_1, \tag{11.4}$$

where $\hat{\Pi}_k$ is the one-qubit projector and $\hat{P}_k = \hat{\Pi}_k \otimes \hat{I}$. In the middle step the right factor $\hat{\Pi}_k \otimes \hat{I}$ moves cyclically to the left inside the partial trace over qubit 0. The cyclic move is allowed for any operator $\hat{A}$ acting on the traced qubit alone. In index form, for any register matrix $\hat{R}$,

$$\big(\mathrm{Tr}_0 [\hat{R}\, (\hat{A} \otimes \hat{I})]\big)_{bb'} = \sum_{a,a''} R_{ab,a''b'} A_{a''a} = \sum_{a,a''} A_{a''a} R_{ab,a''b'} = \big(\mathrm{Tr}_0 [(\hat{A} \otimes \hat{I})\, \hat{R}]\big)_{bb'}, \tag{11.5}$$

where $R_{ab,a''b'}$ is the element of $\hat{R}$ in the row of the bit string ab and the column of $a''b'$. The partial trace sets the row and column bits of qubit 0 equal and sums over them, the second equality only reorders two numbers, and the last one renames the summed indices $a \leftrightarrow a''$. With $\hat{A} = \hat{\Pi}_k$ the two projectors meet in $\hat{\Pi}_k^2 = \hat{\Pi}_k$, and the sum over k collapses by $\sum_k \hat{\Pi}_k = \hat{I}$. Qubit 1 of the Bell state remains $\hat{I}/2$ whichever basis is measured at qubit 0. Measurements at one place cannot detect the choice of measurement at another place. The correlations appear only when the two records are compared.

11.6 Unread measurements and dephasing

When a measurement happens but its outcome is not recorded, the state is the average of the post-measurement states weighted with their probabilities,

$$\hat{\rho} \;\longrightarrow\; \sum_k p_k \hat{\rho}_k = \sum_k \hat{P}_k\, \hat{\rho}\, \hat{P}_k. \tag{11.6}$$

The map removes the coherences between different outcome sectors, the ranges of different projectors, and keeps those inside each sector. An unread measurement in a basis keeps the diagonal of $\hat{\rho}$ in that basis and removes all off-diagonal elements. The parity measurement of $\hat{Z}_0\hat{Z}_1$ keeps the coherence between $|00\rangle$ and $|11\rangle$. An unread $\hat{Z}$ measurement turns $|+\rangle$ into the mixture $\hat{\rho}_{\mathrm{mix}}$ and lowers the purity from 1 to 1/2. On qubit 0 of the Bell state an unread $\hat{Z}$ measurement produces the correlated coin, in which $\langle \hat{X}\hat{X} \rangle$ drops from 1 to 0 and $\langle \hat{Z}\hat{Z} \rangle$ stays at 1. Loss of coherence in a fixed basis is called *dephasing*.

11.7 General measurements

Projective measurements describe ideal detectors. A real $\hat{Z}$-basis detector may report the wrong bit, say with the probability ϵ for either input, and then reads 0 with the probability $(1-\epsilon)\langle\psi|0\rangle\langle 0|\psi\rangle + \epsilon\langle\psi|1\rangle\langle 1|\psi\rangle = \langle\psi|\hat{\pi}_0|\psi\rangle$ with the matrices

$$\hat{\pi}_0 = (1-\epsilon)|0\rangle\langle 0| + \epsilon|1\rangle\langle 1|, \qquad \hat{\pi}_1 = \hat{I} - \hat{\pi}_0 = \epsilon|0\rangle\langle 0| + (1-\epsilon)|1\rangle\langle 1|. \tag{11.7}$$

With $\epsilon = 0.1$ the detector yields $p(0) = 0.9$ for the state $|0\rangle$ and $p(0) = 0.5$ for $|+\rangle$. Although $\hat{\pi}_0$ and $\hat{\pi}_1$ are no projectors, since $\hat{\pi}_0^2 \neq \hat{\pi}_0$ for $0 < \epsilon < 1$, they reproduce all outcome probabilities through $p_k = \mathrm{Tr}(\hat{\pi}_k \hat{\rho})$. Matrices of this kind form a *positive operator-valued measure* (POVM), a set of matrices $\hat{\pi}_k$, the POVM elements, with two conditions.

- Positivity, $\langle v|\hat{\pi}_k|v\rangle \geq 0$ for every vector v, written $\hat{\pi}_k \geq 0$, so that every probability is non-negative.
- Completeness, $\sum_k \hat{\pi}_k = \hat{I}$, so that the probabilities sum to one.

The elements need be neither orthogonal nor projectors, and a POVM may have more outcomes than the dimension of the system. The noisy detector describes the readout error of a device.

A POVM fixes only the statistics. The state after the outcome k is described by a *measurement instrument*, a set of matrices $\hat{M}_{k,\alpha}$ with

$$\hat{\pi}_k = \sum_\alpha \hat{M}_{k,\alpha}^\dagger \hat{M}_{k,\alpha}, \qquad \hat{\rho} \;\longrightarrow\; \hat{\rho}^{(k)} = \frac{1}{p_k} \sum_\alpha \hat{M}_{k,\alpha}\, \hat{\rho}\, \hat{M}_{k,\alpha}^\dagger. \tag{11.8}$$

We use the *square-root instrument*, with one matrix $\hat{M}_k = \sqrt{\hat{\pi}_k}$ per outcome, computed in the eigenbasis of $\hat{\pi}_k$. For a projective measurement $\sqrt{\hat{P}_k} = \hat{P}_k$ and the update reduces to the Lüders rule (11.1). Other instruments realise the same POVM. With any unitary $\hat{V}_k$ the matrices $\hat{M}'_k = \hat{V}_k \sqrt{\hat{\pi}_k}$ satisfy $\hat{M}_k'^\dagger \hat{M}'_k = \hat{\pi}_k$ and yield the same probabilities with a different post-measurement state. The state update therefore depends on how the device is built.

Every POVM can be realised as a projective measurement on a larger system (optional, Naimark's theorem). The map $|\psi\rangle \to \sum_k \hat{M}_k|\psi\rangle|k\rangle_A$ with an ancilla A preserves the norm, because $\sum_k \hat{M}_k^\dagger \hat{M}_k = \hat{I}$. A $\hat{Z}$-basis measurement of the ancilla then gives the outcome k with the probability $\langle\psi|\hat{M}_k^\dagger \hat{M}_k|\psi\rangle = \langle\psi|\hat{\pi}_k|\psi\rangle$.

11.8 Informationally complete measurements

A measurement is *informationally complete* when its outcome probabilities determine the density matrix. Its elements must then span the real space of Hermitian $d \times d$ matrices, of dimension d^2. An informationally complete POVM therefore has at least d^2 outcomes and a minimal one has d^2. The state follows from the probabilities as a linear function. Two such families are the starting point of quantum state tomography.

The Pauli strings of Section 7.5 form a basis of the Hermitian matrices, and their expectation values determine $\hat{\rho} = \frac{1}{2^N} \sum_{\hat{P}} \langle \hat{P} \rangle \hat{P}$. A setting of Section 6.2 assigns $\hat{X}$, $\hat{Y}$ or $\hat{Z}$ to every qubit. It measures every string that carries, on each qubit, the identity or the Pauli matrix of that setting (Section 9.3), and the 3^N settings together yield all 4^N expectation values. The Pauli strings themselves are not positive and are therefore no POVM elements. A POVM appears when we choose the setting at random with the probability 3^{-N}. The setting and the bit string then label the outcome together, and the projectors of all settings, each weighted with 3^{-N}, form the POVM.

The second family is the *symmetric informationally complete* POVM (SIC-POVM) of one qubit, built on the four Bloch vectors of a regular tetrahedron,

$$\mathbf{s}_1 = (0, 0, 1), \quad \mathbf{s}_2 = (\tfrac{2\sqrt{2}}{3}, 0, -\tfrac{1}{3}), \quad \mathbf{s}_3 = (-\tfrac{\sqrt{2}}{3}, \sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}), \quad \mathbf{s}_4 = (-\tfrac{\sqrt{2}}{3}, -\sqrt{\tfrac{2}{3}}, -\tfrac{1}{3}), \tag{11.9}$$

and the elements $\hat{\pi}_a = \frac{1}{4}(\hat{I} + \mathbf{s}_a \cdot \hat{\boldsymbol{\sigma}})$. Each $\hat{\pi}_a$ is half of a pure-state projector and hence positive, and $\sum_a \mathbf{s}_a = 0$ ensures $\sum_a \hat{\pi}_a = \hat{I}$. A state with Bloch vector $\mathbf{r}$ yields the probabilities $p_a = \frac{1}{4}(1 + \mathbf{s}_a \cdot \mathbf{r})$, which are non-negative and sum to one. Since $\mathbf{s}_a \cdot \mathbf{s}_b = -1/3$ for $a \neq b$, we recover the Bloch vector as $\mathbf{r} = 3 \sum_a p_a \mathbf{s}_a$. The four elements are linearly independent, and their number is $d^2 = 4$ for one qubit. For N qubits the Kronecker products $\hat{\pi}_{a_0} \otimes \cdots \otimes \hat{\pi}_{a_{N-1}}$ provide 4^N outcomes in one fixed setting, so the SIC-POVM allows state tomography without changing the measurement setting from shot to shot.

12 Qubit errors and noise channels

12.1 An environment that records the qubit

A qubit in a laboratory is never completely isolated. It interacts with its surroundings, the *environment* E, which we do not observe, and the qubit alone then evolves from a pure state into a mixed one. The smallest example is an environment that keeps a record of the qubit's value. A perfect record acts like the unread measurement of Section 11, and a partial record produces partial dephasing. As the environment E we take a second qubit, prepared in $|0\rangle_E$. It couples to the system qubit through

the Hamiltonian $\hat{\mathcal{H}} = \frac{\Omega}{2}|1\rangle\langle 1| \otimes \hat{Y}$, which rotates the environment about y only when the system is in $|1\rangle$. Switched on for a time t, the coupling produces the controlled rotation

$$\hat{U} = e^{-i\hat{\mathcal{H}}t} = |0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{R}_y(g), \qquad g = \Omega t. \tag{12.1}$$

A system state $\alpha|0\rangle + \beta|1\rangle$ becomes

$$\hat{U}(\alpha|0\rangle + \beta|1\rangle)|0\rangle_E = \alpha|0\rangle|e_0\rangle + \beta|1\rangle|e_1\rangle, \qquad |e_0\rangle = |0\rangle, \quad |e_1\rangle = \cos\tfrac{g}{2}|0\rangle + \sin\tfrac{g}{2}|1\rangle. \tag{12.2}$$

The partial trace (10.11) over the environment, written Tr_E, sums over the bit of the environment qubit and gives the state of the system,

$$\hat{\rho}_S = \begin{pmatrix} |\alpha|^2 & \alpha\beta^*\langle e_1|e_0\rangle \\ \alpha^*\beta\langle e_0|e_1\rangle & |\beta|^2 \end{pmatrix}, \qquad \langle e_0|e_1\rangle = \cos\tfrac{g}{2}. \tag{12.3}$$

The populations are unchanged, while the coherence is multiplied by the overlap of the two environment states. We take $0 \le g \le \pi$, where the overlap is non-negative. At $g = \pi$ the environment states are orthogonal, the environment holds a perfect record of the system's bit, and the result is the unread measurement (11.6). For $\pi < g \le 2\pi$ the negative overlap adds a phase flip $\hat{Z}$ to the dephasing. At $g = 2\pi$ the environment is back in $|0\rangle_E$ and the system has acquired the gate $\hat{Z}$. A finite environment such as one qubit can therefore give the information back, and irreversible decay requires an environment with many degrees of freedom.

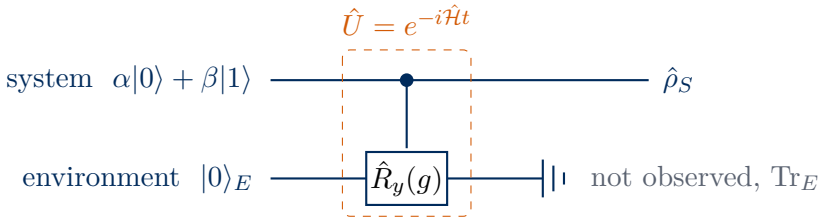


Figure 32: A qubit recorded by an environment qubit. A controlled rotation correlates the environment with the system, and tracing out the environment leaves a mixed state of the system. The partial matrix elements of the joint unitary are the Kraus operators.

12.2 Kraus operators from an unobserved environment

The general rule follows the same steps. The environment starts in a state $|0\rangle_E$ and evolves jointly with the system by a unitary $\hat{U}$. Afterwards the partial trace Tr_E over the environment describes the system, and we take it in an orthonormal basis $\{|k\rangle_E\}$ of the environment. The *partial matrix element* ${}_E\langle k|\hat{U}|0\rangle_E$ takes the matrix element of $\hat{U}$ between the environment states $|0\rangle_E$ and $|k\rangle_E$ only. It is a matrix that still acts on the system, with the entries $\big({}_E\langle k|\hat{U}|0\rangle_E\big)_{ss'} = \langle s|\langle k|_E\,\hat{U}\,|s'\rangle|0\rangle_E$. With it

$$\hat{\rho} \;\longrightarrow\; \mathcal{E}(\hat{\rho}) = \mathrm{Tr}_E\big[\hat{U}(\hat{\rho}\otimes|0\rangle\langle 0|_E)\hat{U}^\dagger\big] = \sum_k {}_E\langle k|\hat{U}|0\rangle_E\,\hat{\rho}\,{}_E\langle 0|\hat{U}^\dagger|k\rangle_E = \sum_k \hat{K}_k\,\hat{\rho}\,\hat{K}_k^\dagger, \tag{12.4}$$

with the *Kraus operators* $\hat{K}_k = {}_E\langle k|\hat{U}|0\rangle_E$, one for each basis state of the environment. Unitarity of $\hat{U}$ gives the completeness relation

$$\sum_k \hat{K}_k^\dagger\hat{K}_k = \sum_k {}_E\langle 0|\hat{U}^\dagger|k\rangle_E\,{}_E\langle k|\hat{U}|0\rangle_E = {}_E\langle 0|\hat{U}^\dagger\hat{U}|0\rangle_E = \hat{I}, \tag{12.5}$$

which keeps the trace of $\hat{\rho}$ equal to one. A map of the form (12.4) with (12.5) is a *quantum channel*, a completely positive and trace-preserving (CPTP) map [21]. Trace preservation, $\mathrm{Tr}\mathcal{E}(\hat{\rho}) = \mathrm{Tr}\hat{\rho}$, is the completeness relation. Complete positivity requires that $\mathcal{E}$ map density matrices to density matrices also when it acts on one part of a larger system. The condition reads $(\mathcal{E}\otimes\mathrm{id})(\hat{\rho}_{SA}) \ge 0$ for every state of the system and an ancilla A. Every map of the Kraus form has this property, and conversely

every completely positive map has a Kraus form. Positivity alone is weaker. The transpose $\hat{\rho} \to \hat{\rho}^T$ maps every one-qubit density matrix to a density matrix. Applied to one qubit of the Bell state, however, it produces a matrix with the eigenvalue $-1/2$ and is therefore no physical operation.

Every physical noise process has the Kraus form. One Kraus operator, $\hat{K}_0 = \hat{U}_S$, is ordinary unitary evolution. Several Kraus operators describe a mixture of things that may have happened to the qubit, the branch k occurring with the probability $\mathrm{Tr}(\hat{K}_k \hat{\rho} \hat{K}_k^\dagger)$. The Kraus operators of a channel are not unique. Any unitary $r \times r$ matrix u produces a new set $\hat{\tilde{K}}_j = \sum_l u_{jl} \hat{K}_l$ with the same $\mathcal{E}(\hat{\rho})$, because the sum $\sum_j \hat{\tilde{K}}_j \hat{\rho} \hat{\tilde{K}}_j^\dagger$ contains $\sum_j u_{jl} u^*_{jl'} = \delta_{ll'}$. Different physical environments can thus produce identical noise on the qubit. For the environment of Section 12.1 with $0 \le g \le \pi$, the controlled rotation yields, with $\kappa = \sin^2 \frac{g}{2}$,

$$\hat{K}_0 = {}_E\langle 0|\hat{U}|0\rangle_E = \begin{pmatrix} 1 & 0 \\ 0 & \cos\frac{g}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\kappa} \end{pmatrix}, \qquad \hat{K}_1 = {}_E\langle 1|\hat{U}|0\rangle_E = \begin{pmatrix} 0 & 0 \\ 0 & \sin\frac{g}{2} \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\kappa} \end{pmatrix}, \tag{12.6}$$

which reproduce (12.3), and $\hat{K}_0^\dagger \hat{K}_0 + \hat{K}_1^\dagger \hat{K}_1 = \hat{I}$.

12.3 Six error models

Six standard error models describe the common imperfections of qubits. For each model we give the Kraus operators, the action on the density matrix and on the Bloch vector $\mathbf{r}$ of equation (10.8), and an example. Every Kraus set below satisfies (12.5). Table 11 summarises the six models, and Figures 33 and 34 show them as deformations of the Bloch ball.

Bit flip

With probability p the gate $\hat{X}$ acts by accident, and otherwise nothing happens,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_1 = \sqrt{p}\,\hat{X}, \qquad \mathcal{E}(\hat{\rho}) = (1-p)\hat{\rho} + p\hat{X}\hat{\rho}\hat{X}. \tag{12.7}$$

Since the conjugation $\hat{X}\hat{\rho}\hat{X}$ keeps r_x and flips the signs of r_y and r_z, the Bloch vector maps to $(r_x, (1-2p)r_y, (1-2p)r_z)$. The state $|1\rangle$ becomes $\mathrm{diag}(p, 1-p)$, a qubit that reads 0 with probability p.

Phase flip (dephasing)

With probability p the gate $\hat{Z}$ acts,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_1 = \sqrt{p}\,\hat{Z}, \qquad \mathcal{E}(\hat{\rho}) = (1-p)\hat{\rho} + p\hat{Z}\hat{\rho}\hat{Z}, \tag{12.8}$$

and $\mathbf{r} \to ((1-2p)r_x, (1-2p)r_y, r_z)$. The populations in the $\hat{Z}$ basis survive and the coherences shrink by $1-2p$. The state $|+\rangle$ becomes

$$\mathcal{E}(|+\rangle\langle+|) = \frac{1}{2}\begin{pmatrix} 1 & 1-2p \\ 1-2p & 1 \end{pmatrix}, \tag{12.9}$$

and at $p = 1/2$ the channel is the unread $\hat{Z}$ measurement (11.6). Dephasing models a qubit whose energy splitting fluctuates, as for a spin in a fluctuating magnetic field, and the fluctuations randomise the relative phase.

Bit-phase flip

With probability p the gate $\hat{Y} = i\hat{X}\hat{Z}$ acts, a bit flip and a phase flip together,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_1 = \sqrt{p}\,\hat{Y}, \qquad \mathbf{r} \to ((1-2p)r_x,\ r_y,\ (1-2p)r_z). \tag{12.10}$$

The state $|1\rangle$ becomes $\mathrm{diag}(p, 1-p)$, as under the bit flip, and $|+\rangle$ loses its coherence as under the phase flip.

Depolarising noise

With probability p one of the three Pauli errors acts, each with probability $p/3$,

$$\hat{K}_0 = \sqrt{1-p}\,\hat{I}, \quad \hat{K}_{1,2,3} = \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}, \qquad \mathbf{r} \to \Big(1-\frac{4}{3}p\Big)\mathbf{r}. \tag{12.11}$$

Each component keeps its sign under one of the three errors and flips it under the other two, and the factor becomes $1-p+\frac{p}{3}-\frac{2p}{3} = 1-\frac{4}{3}p$. Written as $\mathcal{E}(\hat{\rho}) = (1-\frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\frac{\hat{I}}{2}$, the channel mixes the state with the maximally mixed state and shrinks the Bloch ball uniformly towards its centre. Depolarising noise is the basic model of errors without a preferred axis.

Amplitude damping

A qubit encoded in two energy levels loses energy to its surroundings by decaying from the excited state $|e\rangle = |1\rangle$ to the ground state $|g\rangle = |0\rangle$. The joint evolution of the qubit and an environment initially in $|0\rangle_E$ moves the excitation to the environment with probability γ, $|1\rangle|0\rangle_E \to \sqrt{1-\gamma}|1\rangle|0\rangle_E + \sqrt{\gamma}|0\rangle|1\rangle_E$, and leaves $|0\rangle|0\rangle_E$ unchanged. The Kraus operators ${}_E\langle k|\hat{U}|0\rangle_E$ are

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\gamma} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & \sqrt{\gamma} \\ 0 & 0 \end{pmatrix}, \qquad \mathbf{r} \to (\sqrt{1-\gamma}\,r_x,\ \sqrt{1-\gamma}\,r_y,\ \gamma + (1-\gamma)r_z). \tag{12.12}$$

The jump $\hat{K}_1 = \sqrt{\gamma}\,\hat{S}^+$ contains the ladder operator (2.16), which with the ground state $|g\rangle = |0\rangle$ and the excited state $|e\rangle = |1\rangle$ describes the decay, while $\hat{S}^-$ describes the excitation,

$$\text{decay} \quad |g\rangle\langle e| = |0\rangle\langle 1| = \hat{S}^+, \qquad \text{excitation} \quad |e\rangle\langle g| = |1\rangle\langle 0| = \hat{S}^-. \tag{12.13}$$

The operator $\hat{K}_0$ is the branch without a jump, in which the amplitude of $|1\rangle$ shrinks. Under the channel $|1\rangle$ becomes $\mathrm{diag}(\gamma, 1-\gamma)$ and the ball is pushed towards the north pole $|0\rangle$, which stays fixed. For $\gamma \to 1$ every state relaxes to $|0\rangle$. For a decay with the energy relaxation time T_1 the parameter grows as $\gamma = 1 - e^{-t/T_1}$.

Phase damping

The environment of Section 12.1 learns whether the qubit is excited without taking its energy. The channel has the parameter κ, the probability that the environment registers the excitation,

$$\hat{K}_0 = \begin{pmatrix} 1 & 0 \\ 0 & \sqrt{1-\kappa} \end{pmatrix}, \quad \hat{K}_1 = \begin{pmatrix} 0 & 0 \\ 0 & \sqrt{\kappa} \end{pmatrix}, \qquad \mathbf{r} \to (\sqrt{1-\kappa}\,r_x,\ \sqrt{1-\kappa}\,r_y,\ r_z), \tag{12.14}$$

with $\kappa = \sin^2\frac{g}{2}$ for the controlled rotation. The populations stay fixed and the coherences shrink. For $1-2p = \sqrt{1-\kappa}$, phase damping and the phase flip are one channel written with two different sets of Kraus operators. All three descriptions of dephasing in these notes, the unread measurement, the environment record and the phase flip, are therefore one channel with different parameters.

12.4 Pauli channels, unital and non-unital noise

Each of the first four error models applies a Pauli matrix $\hat{P}$ with a probability p_P,

$$\mathcal{E}(\hat{\rho}) = \sum_{\hat{P}\in\{\hat{I},\hat{X},\hat{Y},\hat{Z}\}} p_P\,\hat{P}\hat{\rho}\hat{P}, \qquad \sum_{\hat{P}} p_P = 1, \tag{12.15}$$

which defines a *Pauli channel*. On N qubits Pauli strings take the place of the Pauli matrices. A Pauli channel multiplies each Bloch-vector component by a factor, and the Bloch ball shrinks along the three axes with its centre fixed, $\mathcal{E}(\hat{I}/2) = \hat{I}/2$. A channel with $\mathcal{E}(\hat{I}) = \hat{I}$ is called *unital*. Every mixture of unitaries $\sum_j p_j \hat{U}_j\hat{\rho}\hat{U}_j^\dagger$ is unital, and so is phase damping. Amplitude damping is *non-unital*. It maps $\hat{I}/2$ to $\mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2})$, moves the centre of the ball towards the north pole and has the ground state $|0\rangle$ as its fixed point. Noise that releases energy into the environment always has a preferred final state and is non-unital.

channel	Kraus operators	Bloch vector $(r_x, r_y, r_z) \to$
bit flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{X}$	$(r_x,\ (1-2p)r_y,\ (1-2p)r_z)$
phase flip (dephasing)	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Z}$	$((1-2p)r_x,\ (1-2p)r_y,\ r_z)$
bit-phase flip	$\sqrt{1-p}\,\hat{I},\ \sqrt{p}\,\hat{Y}$	$((1-2p)r_x,\ r_y,\ (1-2p)r_z)$
depolarising	$\sqrt{1-p}\,\hat{I},\ \sqrt{p/3}\,\hat{X},\ \sqrt{p/3}\,\hat{Y},\ \sqrt{p/3}\,\hat{Z}$	$(1-\frac{4}{3}p)\,(r_x, r_y, r_z)$
amplitude damping	$\begin{pmatrix}1 & 0\\ 0 & \sqrt{1-\gamma}\end{pmatrix},\ \begin{pmatrix}0 & \sqrt{\gamma}\\ 0 & 0\end{pmatrix}$	$(\sqrt{1-\gamma}\,r_x,\ \sqrt{1-\gamma}\,r_y,\ \gamma+(1-\gamma)r_z)$
phase damping	$\begin{pmatrix}1 & 0\\ 0 & \sqrt{1-\kappa}\end{pmatrix},\ \begin{pmatrix}0 & 0\\ 0 & \sqrt{\kappa}\end{pmatrix}$	$(\sqrt{1-\kappa}\,r_x,\ \sqrt{1-\kappa}\,r_y,\ r_z)$

Table 11: Single-qubit error models. Each Kraus set satisfies $\sum_k \hat{K}_k^\dagger \hat{K}_k = \hat{I}$, and each Bloch-vector map was checked numerically on random states.

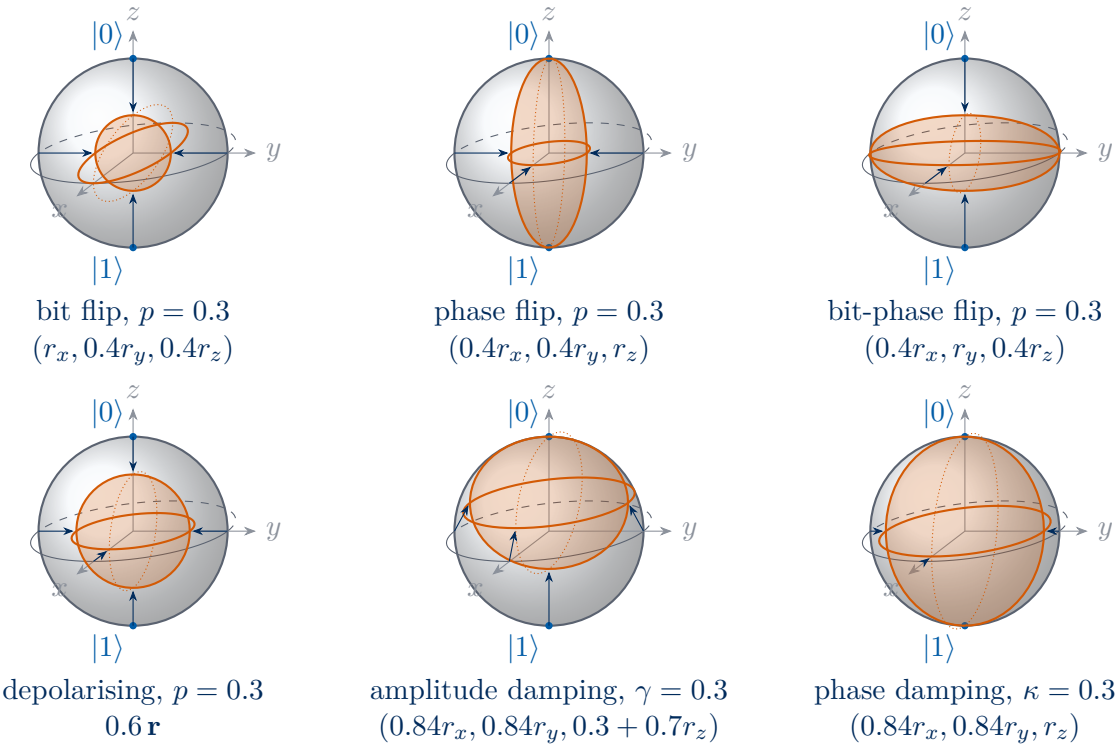


Figure 33: The six error models as deformations of the Bloch ball at the strength $p = \gamma = \kappa = 0.3$. The grey sphere holds the pure states, and the orange surface is its image under the channel. The orange curves are the images of the equator and of a great circle through the poles. Arrows run from pure states to their images.

12.5 The Choi state of a channel (optional)

A channel can be measured in the laboratory by turning it into a state. We prepare the Bell state $|\Phi^+\rangle$, send qubit 0 through the channel and leave qubit 1 alone (Figure 35). The resulting two-qubit density matrix contains the whole channel, so measuring it with the informationally complete settings of Section 11.8 determines the channel. Process tomography thereby reduces to state tomography, a correspondence called channel–state duality and used in Lecture 13. The same matrix also turns the conditions for a channel, which refer to all states of the system and of any ancilla, into properties of one matrix, and it removes the freedom in the choice of Kraus operators. The *Choi state* [42,46] of the channel $\mathcal{E}$ is

$$\hat{\rho}_{\mathcal{E}} = (\mathcal{E} \otimes \mathrm{id})\big(|\Phi^+\rangle\langle\Phi^+|\big) = \frac{1}{2}\sum_{i,j} \mathcal{E}\big(|i\rangle\langle j|\big) \otimes |i\rangle\langle j|, \tag{12.16}$$

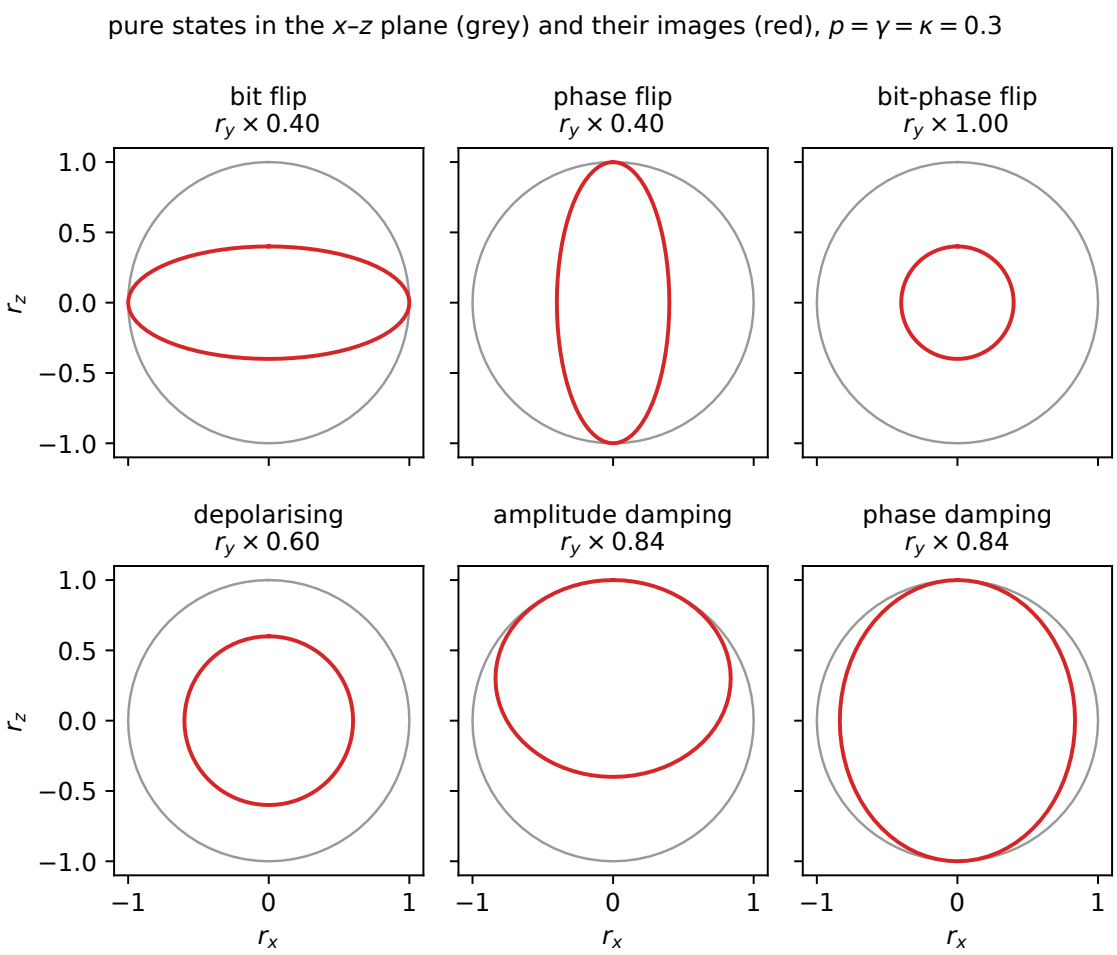


Figure 34: Images of the circle of pure states in the x–z plane (grey) under the six error models (red), with strength $p = \gamma = \kappa = 0.3$. The bit flip keeps the x axis and shrinks z by $1 - 2p = 0.4$. The phase flip and phase damping keep the poles and shrink the coherences, and the bit-phase flip shrinks both axes. Depolarising noise shrinks the whole circle by $1 - \frac{4}{3}p = 0.6$, and amplitude damping shrinks the horizontal axis by $\sqrt{1-\gamma} \approx 0.84$ and shifts the ellipse towards $|0\rangle$.

with the channel on qubit 0 and the identity on qubit 1. Its blocks are the images of the four matrices $|i\rangle\langle j|$, which form a basis, so $\hat{\rho}_{\mathcal{E}}$ determines the channel. Since $\mathrm{Tr}(|i\rangle\langle j|\,\hat{\rho}^T) = \rho_{ij}$, the action on any state is

$$\mathcal{E}(\hat{\rho}) = \sum_{i,j} \rho_{ij}\,\mathcal{E}(|i\rangle\langle j|) = 2\,\mathrm{Tr}_1\big[\hat{\rho}_{\mathcal{E}}\,(\hat{I} \otimes \hat{\rho}^T)\big]. \tag{12.17}$$

- The map preserves the trace if and only if $\mathrm{Tr}\,\mathcal{E}(|i\rangle\langle j|) = \delta_{ij}$, that is, $\mathrm{Tr}_0 \hat{\rho}_{\mathcal{E}} = \hat{I}/2$, where Tr_0 traces out the output qubit 0.
- The map is completely positive if and only if $\hat{\rho}_{\mathcal{E}} \geq 0$ [42]. A completely positive map applied to qubit 0 of the Bell state gives a positive matrix, which shows one direction. The other follows from the next item.
- A positive Choi state has the eigendecomposition $\hat{\rho}_{\mathcal{E}} = \sum_k \lambda_k |u_k\rangle\langle u_k|$ with $\lambda_k \geq 0$. A Kraus operator acts on the Bell state as $(\hat{K} \otimes \hat{I})|\Phi^+\rangle = \frac{1}{\sqrt{2}} \sum_{a,i} K_{ai}|a\rangle|i\rangle$, the matrix K read row by row as a vector of four entries and divided by $\sqrt{2}$. Reading each vector $\sqrt{2\lambda_k}|u_k\rangle$ back as a 2×2 matrix, where the factor $\sqrt{2}$ undoes the $1/\sqrt{2}$ of the Bell state, with qubit 0 labelling the rows as for Ψ in Section 8.2, therefore gives Kraus operators $\hat{K}_k$ with $\sum_k (\hat{K}_k \otimes \hat{I})|\Phi^+\rangle\langle\Phi^+|(\hat{K}_k \otimes \hat{I})^\dagger = \hat{\rho}_{\mathcal{E}}$, and hence, by (12.17), the channel itself. The number of Kraus operators needed is the rank of $\hat{\rho}_{\mathcal{E}}$.

For amplitude damping the images $\mathcal{E}(|0\rangle\langle 0|) = |0\rangle\langle 0|$, $\mathcal{E}(|1\rangle\langle 1|) = \gamma|0\rangle\langle 0| + (1-\gamma)|1\rangle\langle 1|$ and

$\mathcal{E}(|0\rangle\langle 1|) = \sqrt{1-\gamma}\,|0\rangle\langle 1|$ give, in the basis $|00\rangle, |01\rangle, |10\rangle, |11\rangle$,

$$\hat{\rho}_{\mathcal{E}_{\mathrm{AD}}} = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & \sqrt{1-\gamma} \\ 0 & \gamma & 0 & 0 \\ 0 & 0 & 0 & 0 \\ \sqrt{1-\gamma} & 0 & 0 & 1-\gamma \end{pmatrix}, \tag{12.18}$$

with the eigenvalues $1-\gamma/2$, $\gamma/2$, 0 and 0. The normalised eigenvector $(1, 0, 0, \sqrt{1-\gamma})/\sqrt{2-\gamma}$ of the first, multiplied by $\sqrt{2\lambda} = \sqrt{2-\gamma}$ and read as a matrix, is the no-jump operator $\hat{K}_0 = \mathrm{diag}(1, \sqrt{1-\gamma})$. The eigenvector $(0, 1, 0, 0)$, multiplied by $\sqrt{\gamma}$, gives the jump $\hat{K}_1 = \sqrt{\gamma}|0\rangle\langle 1|$. Depolarising noise, $\mathcal{E}(\hat{\rho}) = (1 - \frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\frac{\hat{I}}{2}$, gives

$$\hat{\rho}_{\mathcal{E}_{\mathrm{dep}}} = \Big(1 - \frac{4}{3}p\Big)|\Phi^+\rangle\langle\Phi^+| + \frac{4}{3}p\,\frac{\hat{I}}{4}, \tag{12.19}$$

a mixture of the Bell state and the maximally mixed state of two qubits. Its eigenvalue $1-p$ on $|\Phi^+\rangle$ yields the Kraus operator $\sqrt{1-p}\,\hat{I}$, and the threefold eigenvalue $p/3$ yields the three Pauli errors.

The transpose $\hat{\rho} \to \hat{\rho}^T$ keeps the eigenvalues of every one-qubit state and is therefore positive and trace preserving. Applied to qubit 0 of the Bell state it gives $\frac{1}{2}\sum_{ij}|j\rangle\langle i| \otimes |i\rangle\langle j|$, which maps $|ab\rangle$ to $|ba\rangle/2$ and equals $\widehat{\mathrm{SWAP}}/2$. Its eigenvalue $-1/2$ on the singlet, the value met at the definition of complete positivity, shows that the transpose is not completely positive. Figure 36 compares the three spectra and continues the depolarising formula to $p > 1$. Every output stays a density matrix up to $p = 3/2$, where the Bloch factor $1 - \frac{4}{3}p$ reaches -1. The Choi state has a negative eigenvalue already beyond $p = 1$, so between these values the map is positive without being completely positive.

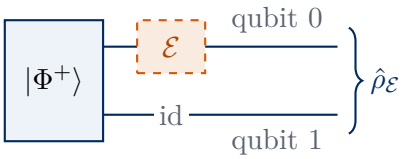


Figure 35: The Choi state (12.16). The channel acts on qubit 0 of the Bell state $|\Phi^+\rangle$ while qubit 1 is left alone, and the two-qubit output is $\hat{\rho}_{\mathcal{E}}$.

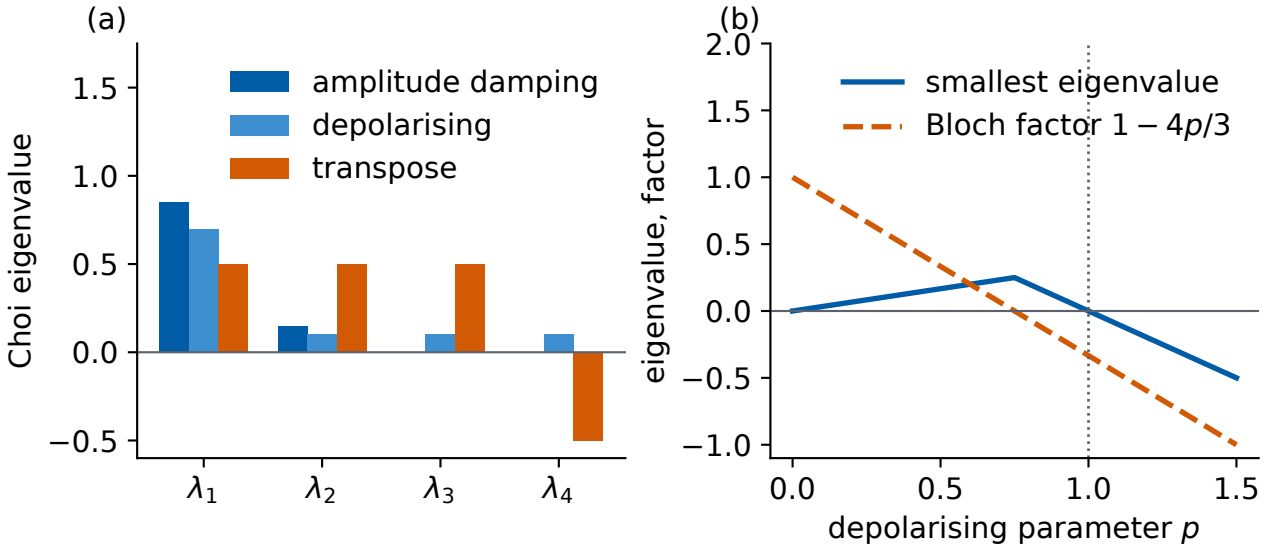


Figure 36: (a) The four eigenvalues of the Choi state for amplitude damping with $\gamma = 0.3$, depolarising noise with $p = 0.3$ and the transpose. (b) Smallest eigenvalue of the Choi state of the map $(1 - \frac{4}{3}p)\hat{\rho} + \frac{4}{3}p\,\hat{I}/2$ and its Bloch factor $1 - \frac{4}{3}p$, against p. The smallest eigenvalue starts at zero, since the identity channel at $p = 0$ has the pure Choi state $|\Phi^+\rangle\langle\Phi^+|$. The map is completely positive up to $p = 1$ (dotted).

12.6 The register under local noise

Let each qubit of the Bell register suffer the same noise independently. The channel on the register has the Kraus operators $\hat{K}_k \otimes \hat{K}_l$,

$$\hat{\rho} \;\longrightarrow\; \sum_{k,l} (\hat{K}_k \otimes \hat{K}_l)\, \hat{\rho}\, (\hat{K}_k \otimes \hat{K}_l)^\dagger . \tag{12.20}$$

On a computer a channel on qubit q acts on the ket and bra legs of that qubit only, as a gate does.

Local dephasing does not touch the populations in the $\hat{Z}$ basis and leaves $\langle \hat{Z}\hat{Z} \rangle = 1$, whereas it multiplies $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ by $(1-2p)^2$, one factor per qubit (Figure 37, left). At $p = 1/2$ the coherences are gone and the register has become the correlated coin. Depolarising noise reduces all three correlators by the common factor $(1 - \frac{4}{3}p)^2$ (Figure 37, right). Under both channels the register purity falls.

The entropy of qubit 0 stays at one bit throughout, since $\hat{\rho}_0 = \hat{I}/2$ for every p under both channels. As in Section 10.8, it misses the transition from the Bell state to the classical coin. Amplitude damping has a preferred final state and changes the reduced states as well. With a common γ on both qubits it turns the reduced state of each qubit into $\mathrm{diag}(\frac{1+\gamma}{2}, \frac{1-\gamma}{2})$ and the correlator into $\langle \hat{Z}\hat{Z} \rangle = 1 - 2\gamma + 2\gamma^2$.

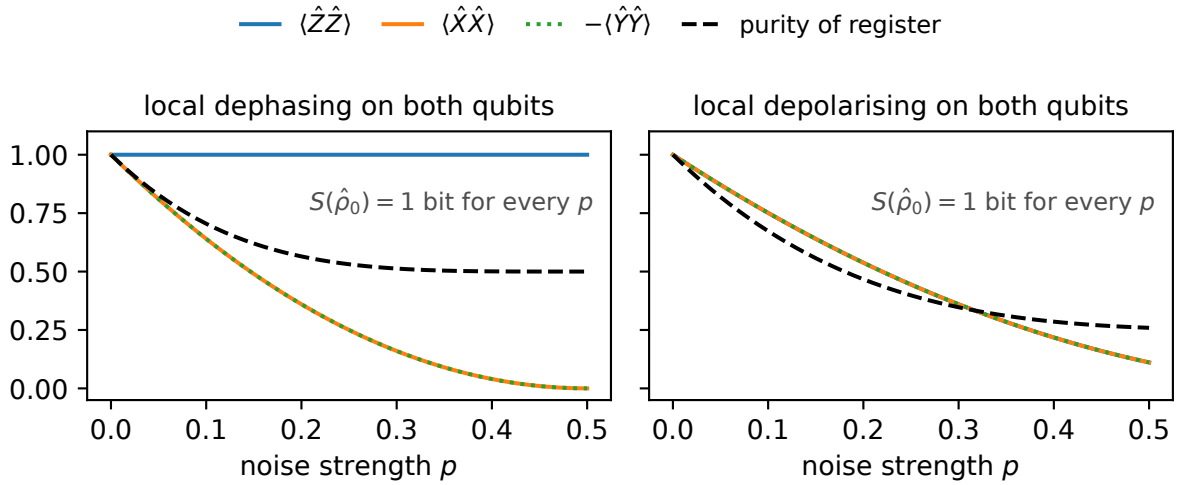


Figure 37: Correlators, register purity and entropy of qubit 0 for the Bell state under local dephasing (left) and local depolarising noise (right), against the noise strength p. Under dephasing $\langle \hat{Z}\hat{Z} \rangle$ stays at one while $\langle \hat{X}\hat{X} \rangle$ and $\langle \hat{Y}\hat{Y} \rangle$ decay as $\pm(1-2p)^2$, and at $p = 1/2$ only the classical correlation of the coin remains. Under depolarising noise all three correlators decay together as $(1 - \frac{4}{3}p)^2$. In both panels the register purity falls while the entropy of qubit 0 stays at one bit, so the reduced state of one qubit cannot see the loss of entanglement.

12.7 The Lindblad equation

A channel describes the noise accumulated over a fixed time. In a laboratory the noise acts continuously, and the density matrix of a system coupled to an environment, an *open system*, obeys a differential equation that takes the place of the Schrödinger equation (3.1). For a closed system $\hat{\rho}(t) = \hat{U}(t)\hat{\rho}(0)\hat{U}(t)^\dagger$ with $\hat{U} = e^{-it\hat{\mathcal{H}}}$, and differentiation yields the von Neumann equation $\frac{d\hat{\rho}}{dt} = -i[\hat{\mathcal{H}}, \hat{\rho}]$, with the commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. An environment that carries information away without returning it, the memoryless or Markovian case, adds extra terms. The equation then takes the *Lindblad form* [22,23],

$$\frac{d\hat{\rho}}{dt} = \mathcal{L}(\hat{\rho}) = -i[\hat{\mathcal{H}}, \hat{\rho}] + \sum_k \Big(\hat{L}_k \hat{\rho} \hat{L}_k^\dagger - \frac{1}{2}\{\hat{L}_k^\dagger \hat{L}_k, \hat{\rho}\} \Big), \tag{12.21}$$

with the anticommutator $\{\hat{A}, \hat{B}\} = \hat{A}\hat{B} + \hat{B}\hat{A}$ and the *jump operators* $\hat{L}_k$, one for each noise process. The first term in the bracket describes the jumps. The anticommutator removes the corresponding

probability from the no-jump evolution and keeps the trace at one. The solution $\hat{\rho}(t) = \mathcal{E}_t(\hat{\rho}(0))$ with $\mathcal{E}_t = e^{t\mathcal{L}}$ is the open-system analogue of $\hat{U}(t) = e^{-it\hat{\mathcal{H}}}$. Every $\mathcal{E}_t$ is a channel, and unlike $\hat{U}(t)$ it has in general no physical inverse, since the information lost to the environment cannot be recovered. On a computer $\hat{\rho}$ is flattened into a vector of d^2 entries, so that $\mathcal{L}$ becomes a $d^2 \times d^2$ matrix and $e^{t\mathcal{L}}$ a matrix exponential.

Two jump operators describe the decoherence of a qubit. *Energy relaxation* with the rate Γ has the jump operator $\hat{L} = \sqrt{\Gamma}\,|g\rangle\langle e| = \sqrt{\Gamma}\,\hat{S}^+$, the decay operator (12.13), which takes the excited state $|1\rangle$ to the ground state $|0\rangle$. With $\hat{L}\hat{\rho}\hat{L}^\dagger = \Gamma\rho_{11}|0\rangle\langle 0|$ and $\hat{L}^\dagger\hat{L} = \Gamma|1\rangle\langle 1|$, the elements of (12.21) with $\hat{\mathcal{H}} = 0$ read

$$\frac{d\rho_{11}}{dt} = -\Gamma\rho_{11}, \qquad \frac{d\rho_{00}}{dt} = +\Gamma\rho_{11}, \qquad \frac{d\rho_{01}}{dt} = -\frac{\Gamma}{2}\,\rho_{01}, \tag{12.22}$$

so $\rho_{11}(t) = e^{-\Gamma t}\rho_{11}(0)$ and $\rho_{01}(t) = e^{-\Gamma t/2}\rho_{01}(0)$. The decay is amplitude damping with $\gamma = 1 - e^{-\Gamma t}$, whose coherence factor $\sqrt{1-\gamma}$ equals $e^{-\Gamma t/2}$, and the *relaxation time* is $T_1 = 1/\Gamma$. *Pure dephasing* with the rate γ_ϕ has $\hat{L} = \sqrt{\gamma_\phi/2}\,\hat{Z}$. With $\hat{L}^\dagger\hat{L} = \frac{\gamma_\phi}{2}\hat{I}$ the equation becomes $\frac{d\hat{\rho}}{dt} = \frac{\gamma_\phi}{2}(\hat{Z}\hat{\rho}\hat{Z} - \hat{\rho})$, and the populations stay constant. Since $\hat{Z}\hat{\rho}\hat{Z}$ flips the sign of the coherences, they decay as $\rho_{01}(t) = e^{-\gamma_\phi t}\rho_{01}(0)$, and the channel is phase damping with $\sqrt{1-\kappa} = e^{-\gamma_\phi t}$. With both processes the populations relax with T_1 and the coherences decay with the *coherence time* T_2,

$$\frac{1}{T_2} = \frac{1}{2T_1} + \gamma_\phi \qquad \Longrightarrow \qquad T_2 \le 2T_1. \tag{12.23}$$

Under a Hamiltonian $\frac{\omega}{2}\hat{Z}$ together with both processes the Bloch vector spirals towards the north pole $|0\rangle$, its z component relaxing with T_1 and its length in the x–y plane shrinking with T_2 (Figure 38).

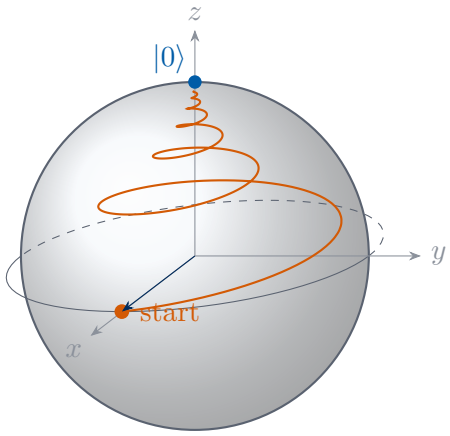


Figure 38: The Bloch vector under the Lindblad equation with a Hamiltonian $\frac{\omega}{2}\hat{Z}$, energy relaxation and pure dephasing, started on the equator. It precesses about the z axis while its z component relaxes towards $|0\rangle$ with the time T_1 and its component in the x–y plane shrinks with the time T_2.

12.8 Noise in quantum circuits

We describe a noisy quantum computer by attaching channels to the circuit. The standard model has four ingredients, all built from the channels above (Figure 39).

Gate errors

Each gate is applied ideally and followed by a channel on the qubits it touches. After a one-qubit gate this is the depolarising channel with the probability p_1, which equals $(1 - \frac{4}{3}p_1)\hat{\rho} + \frac{4}{3}p_1\frac{\hat{I}}{2}$. After a two-qubit gate it is the two-qubit depolarising channel with the error probability p_2, uniform over the

15 Pauli strings of the pair other than $\hat{I} \otimes \hat{I}$,

$$\mathcal{E}_2(\hat{\rho}) = (1 - p_2)\hat{\rho} + \frac{p_2}{15} \sum_{\hat{P} \neq \hat{I} \otimes \hat{I}} \hat{P}\hat{\rho}\hat{P} = \Big(1 - \frac{16}{15} p_2\Big)\hat{\rho} + \frac{16}{15} p_2 \, \frac{\hat{I}}{4} \otimes \mathrm{Tr}_{\mathrm{pair}} \hat{\rho}. \tag{12.24}$$

The second form uses the identity $\sum_{\hat{P}} \hat{P}\hat{\rho}\hat{P} = 4\,\hat{I} \otimes \mathrm{Tr}_{\mathrm{pair}}\hat{\rho}$ for the sum over all 16 strings of the pair. The identity holds because the average over all Pauli strings removes every coherence and population difference of the pair. In this form the channel mixes the pair with its maximally mixed state with the weight $\frac{16}{15}p_2$. It multiplies the expectation value of every non-identity Pauli string on the pair by $1 - \frac{16}{15}p_2$. Some texts quote the mixing weight $\frac{16}{15}p_2$ as the error probability, and error rates from different sources must be compared with this convention in mind. Two-qubit gates are usually the noisiest operations of a device, with p_2 often about ten times larger than p_1.

Idle decoherence

A layer of the circuit lasts t_{layer}. During each layer the qubits that wait while gates act on others undergo amplitude damping with $\gamma = 1 - e^{-t_{\mathrm{layer}}/T_1}$ and phase damping with $\sqrt{1-\kappa} = e^{-t_{\mathrm{layer}}/T_\phi}$, where $1/T_\phi = 1/T_2 - 1/(2T_1)$ by (12.23). Both are finite-time channels of the Lindblad equation of Section 12.7. In our model a qubit acted on by a gate receives only the gate channel, which is taken to include its decoherence during the gate. A deep circuit is limited by the ratio of its total duration to T_1 and T_2. A gate lasts as long as its angle divided by the coupling that produces it, θ/ω for a single-qubit rotation generated by $\frac{\omega}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ (Section 3) and θ/J for a two-qubit rotation (7.14), so larger angles and weaker couplings expose the qubits longer to these channels.

Readout errors

The detector reports the wrong bit with a probability that may depend on the true bit. A *confusion matrix* $A_{mb} = p(\text{read } m \,|\, \text{true } b)$, with columns that sum to one, maps the true probabilities of the bits to the recorded ones,

$$\mathbf{p}_{\mathrm{meas}} = A\,\mathbf{p}_{\mathrm{true}}, \qquad A = \begin{pmatrix} 1 - \epsilon_{0\to1} & \epsilon_{1\to0} \\ \epsilon_{0\to1} & 1 - \epsilon_{1\to0} \end{pmatrix}. \tag{12.25}$$

The confusion matrix corresponds to the POVM $\hat{\pi}_m = \sum_b A_{mb} |b\rangle\langle b|$, which for $\epsilon_{0\to1} = \epsilon_{1\to0} = \epsilon$ is the noisy detector (11.7). With independent readout errors on N qubits the register has the confusion matrix $A_0 \otimes \cdots \otimes A_{N-1}$, acting on the vector of 2^N bit-string probabilities in the order of Table 3. *Readout mitigation* measures A with known basis states as inputs and estimates $\mathbf{p}_{\mathrm{true}} \approx A^{-1}\mathbf{p}_{\mathrm{meas}}$. From a finite record the inverted vector can have small negative entries, which must be handled before the vector serves as a probability distribution.

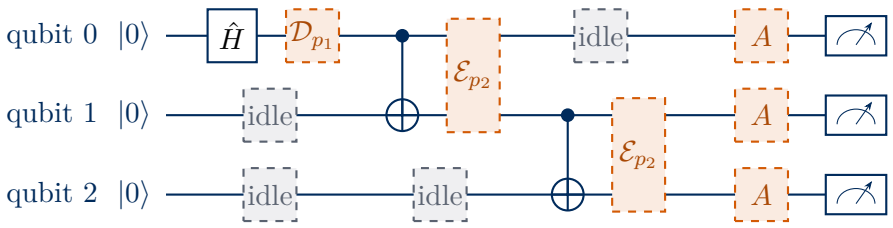


Figure 39: The circuit noise model for the GHZ circuit. Each gate is followed by a depolarising channel on the qubits it touches, and waiting qubits decay with T_1 and T_2. The readout reports wrong bits with the probabilities of the confusion matrix. White boxes are ideal gates, orange boxes the one-qubit depolarising channel $\mathcal{D}_{p_1}$ and the two-qubit channel $\mathcal{E}_{p_2}$ that follow them, grey boxes the decay of waiting qubits and A the readout error. The ideal output is $(|000\rangle + |111\rangle)/\sqrt{2}$.

Coherent and stochastic errors (optional)

A *coherent* error is a small unwanted unitary, for example an over-rotation $\hat{R}_x(\epsilon)$ in place of the identity. Over n repetitions the angles add, $\hat{R}_x(\epsilon)^n = \hat{R}_x(n\epsilon)$, and a qubit started in $|0\rangle$ is found in $|1\rangle$ with the probability $\sin^2(n\epsilon/2) \approx (n\epsilon)^2/4$. Because amplitudes add, the error grows quadratically with the number of gates. A *stochastic* error is a channel, for example a bit flip with probability p at each gate. After n gates the flip probability is $\frac{1}{2}(1-(1-2p)^n) \approx np$, so probabilities add and the error grows linearly. At equal single-gate error, $p = \epsilon^2/4$, the coherent error after n gates is larger by the factor n (Figure 40b). Depolarising models describe stochastic errors only, and coherent errors are usually the first to be calibrated away.

Simulating a noisy circuit

We simulate a noisy circuit with the density matrix. Each gate acts as $\hat{\rho} \to \hat{U}\hat{\rho}\hat{U}^\dagger$, with $\hat{U}$ a Kronecker product with identities on the untouched qubits as in Section 7.5. Each channel acts as $\hat{\rho} \to \sum_k \hat{K}_k \hat{\rho} \hat{K}_k^\dagger$, where a Kraus operator on qubit q is again $\hat{I} \otimes \cdots \otimes \hat{K}_k \otimes \cdots \otimes \hat{I}$. On a computer these operations are contractions with the ket and bra legs of the touched qubits. With 4^N entries, $16 \cdot 4^N$ bytes, the density matrix limits this simulation to about half the number of qubits of a state-vector simulation. The *quantum trajectory* method (optional) keeps a state vector, chooses at each channel the branch k with the probability $\|\hat{K}_k|\psi\rangle\|^2$ and continues with $\hat{K}_k|\psi\rangle / \|\hat{K}_k|\psi\rangle\|$. Averaging $|\psi\rangle\langle\psi|$ over many trajectories reproduces $\hat{\rho}$, with a statistical error that falls as one over the square root of the number of trajectories.

A noisy GHZ circuit

The GHZ circuit (7.22) with one Hadamard gate and $N-1$ CNOT gates prepares a density matrix $\hat{\rho}$. We measure its quality by the fidelity with the target, $F = \langle \mathrm{GHZ}_N | \hat{\rho} | \mathrm{GHZ}_N \rangle$, the probability that $\hat{\rho}$ passes a test projecting on the target state. We add depolarising noise p_1 after the Hadamard gate and p_2 after each CNOT. Each channel does nothing with the weight $1 - \frac{4}{3}p_1$ or $1 - \frac{16}{15}p_2$ and replaces the touched qubits by a maximally mixed state with the remaining weight. These weights are non-negative for $p_1 \le 3/4$ and $p_2 \le 15/16$. In the expanded product of all channels the branch without replacement prepares the ideal GHZ state. Every other branch adds a positive matrix, whose overlap with the GHZ state is non-negative. The fidelity therefore obeys the lower bound

$$F \;\ge\; \Big(1 - \frac{4}{3}p_1\Big)\Big(1 - \frac{16}{15}p_2\Big)^{N-1}, \tag{12.26}$$

which falls exponentially with the number of CNOT gates. The bound overestimates the loss, since some Pauli errors leave the GHZ state unchanged. To first order in the error probabilities, an $\hat{X}$ error after the Hadamard gate leaves $|+\rangle$ as it is. After the first CNOT, three of the 15 errors are harmless, because $\hat{X}\otimes\hat{X}$, $\hat{Y}\otimes\hat{Y}$ and $\hat{Z}\otimes\hat{Z}$ leave the pair state $(|00\rangle + |11\rangle)/\sqrt{2}$ unchanged up to a global phase. After each later CNOT only $\hat{Z}\otimes\hat{Z}$ is harmless, and counting the harmful errors yields

$$1 - F = \frac{2}{3}p_1 + \frac{12}{15}p_2 + (N-2)\,\frac{14}{15}\,p_2 + O(p^2). \tag{12.27}$$

The full model adds idle noise on the waiting qubits of each layer and readout errors, with parameters of the order found on current devices (Figure 40). The simulated fidelity falls with the number of qubits (Figure 40a) and stays close to the cruder estimate $(1-p_1)(1-p_2)^{N-1}$, in which every gate fails with its error probability. The readout errors add recorded bit strings other than $0\ldots0$ and $1\ldots1$, and the parity $\langle \hat{Z}_0 \hat{Z}_{N-1} \rangle$ of the outer qubits, which equals one for the ideal state, falls below one (Figure 40c).

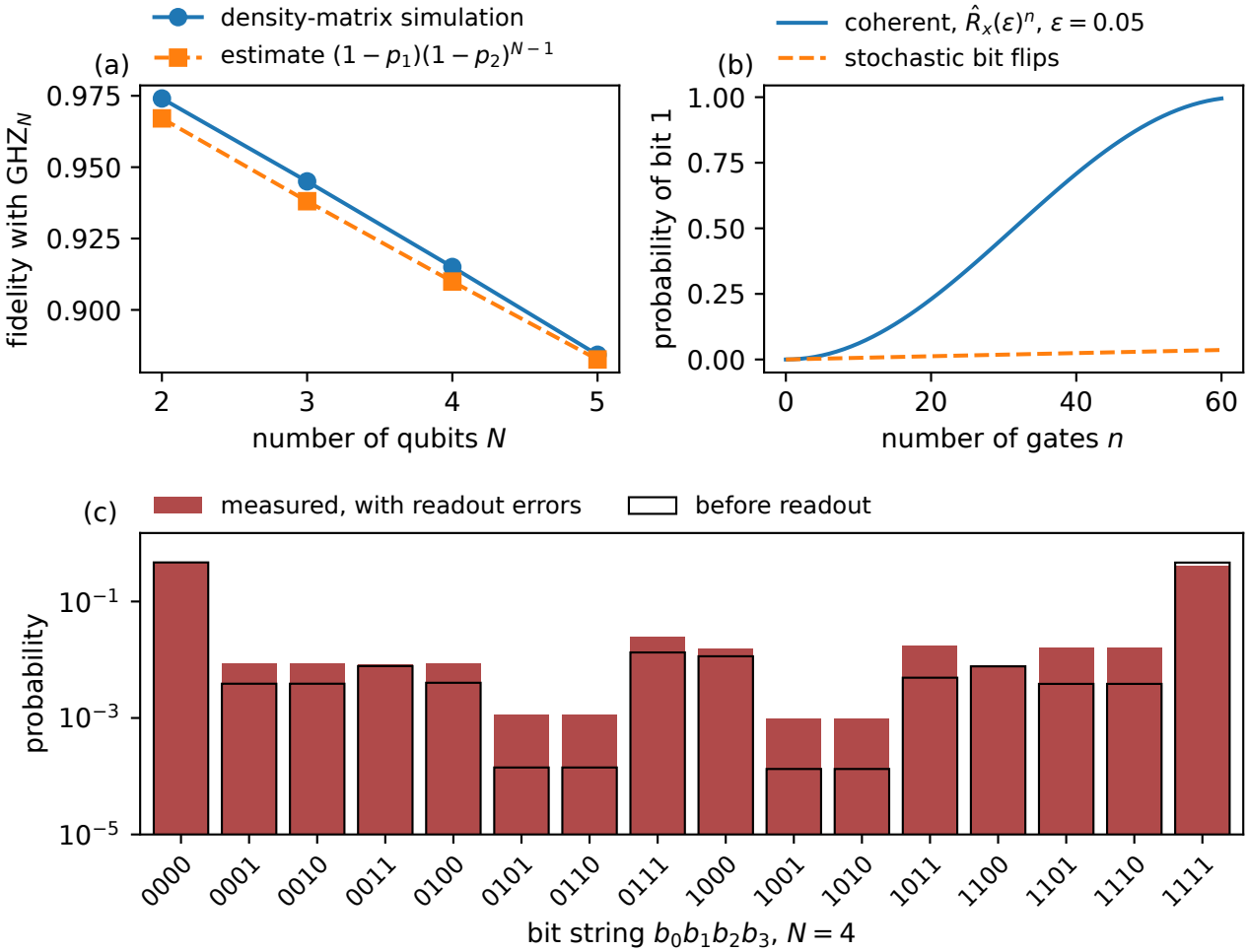


Figure 40: Noisy GHZ preparation with the gate errors $p_1 = 0.003$ and $p_2 = 0.03$, idle noise $\gamma = 0.002$ and $\kappa = 0.004$ per layer on the waiting qubits, and the readout errors $\epsilon_{0\to1} = 0.01$ and $\epsilon_{1\to0} = 0.03$. (a) The fidelity against the number of qubits, with the estimate in which every gate fails with its error probability. (b) The error after n coherent over-rotations, growing quadratically, and after n stochastic bit flips of the same single-gate probability, growing linearly. (c) The bit-string probabilities for four qubits before and after the readout errors, on a logarithmic scale.

13 Comparing states

The *fidelity* between two pure states is their overlap probability

$$F = |\langle\varphi|\psi\rangle|^2, \tag{13.1}$$

which is one for equal states, up to a global phase, and zero for orthogonal ones. It is the probability that $|\psi\rangle$ passes a test that projects on $|\varphi\rangle$. When the target $|\psi\rangle$ is pure and the prepared state is a density matrix $\hat{\sigma}$, the same test gives

$$F = \langle\psi|\hat{\sigma}|\psi\rangle, \tag{13.2}$$

the quantity used for the noisy GHZ circuit of Section 12.8.

For two mixed states (optional) the Uhlmann fidelity

$$F(\hat{\rho},\hat{\sigma}) = \left(\mathrm{Tr}\sqrt{\sqrt{\hat{\rho}}\,\hat{\sigma}\,\sqrt{\hat{\rho}}}\right)^2 \tag{13.3}$$

generalises it. The square root of a positive matrix is computed in its eigenbasis. For $\hat{\rho} = \sum_k \lambda_k |k\rangle\langle k|$ it is $\sqrt{\hat{\rho}} = \sum_k \sqrt{\lambda_k}|k\rangle\langle k|$, the matrix whose square is $\hat{\rho}$. When $\hat{\rho} = |\psi\rangle\langle\psi|$ is pure, $\sqrt{\hat{\rho}} = \hat{\rho}$ and the formula reduces to (13.2).

The *trace distance*

$$D(\hat{\rho},\hat{\sigma}) = \tfrac{1}{2}\|\hat{\rho}-\hat{\sigma}\|_1 = \tfrac{1}{2}\sum_k |\mu_k|, \tag{13.4}$$

with μ_k the eigenvalues of the Hermitian matrix $\hat{\rho}-\hat{\sigma}$, uses the *trace norm* $\|\hat{A}\|_1$ of a Hermitian matrix, the sum of the moduli of its eigenvalues. Since the difference of two density matrices is traceless, its positive and negative eigenvalues cancel in the sum, and D equals the sum of the positive ones. Given one copy that is $\hat{\rho}$ or $\hat{\sigma}$ with equal probability, the best possible measurement identifies it correctly with probability $\frac{1}{2}(1+D)$ (derived in Section 15.2). For one qubit D is half the Euclidean distance between the Bloch vectors $\mathbf{r}_\rho$ and $\mathbf{r}_\sigma$ of the two states, since $\hat{\rho}-\hat{\sigma} = \frac{1}{2}(\mathbf{r}_\rho-\mathbf{r}_\sigma)\cdot\hat{\boldsymbol{\sigma}}$ has the eigenvalues $\pm\frac{1}{2}|\mathbf{r}_\rho-\mathbf{r}_\sigma|$.

The state $|+\rangle$ and the mixture $\hat{\rho}_{\text{mix}}$ have fidelity and trace distance one half, since their Bloch vectors are $\hat{x}$ and 0. The Bell state and the correlated coin also have $F = 0.50$ and $D = 1/2$, so one copy is identified correctly with probability 3/4 at best, for example by measuring $\hat{X}\hat{X}$. For the Bell state after local dephasing with $p = 0.3$, the decomposition $|\Phi^+\rangle\langle\Phi^+| = \frac{1}{4}(\hat{I}+\hat{X}\hat{X}-\hat{Y}\hat{Y}+\hat{Z}\hat{Z})$ gives

$$F = \langle\Phi^+|\hat{\rho}|\Phi^+\rangle = \tfrac{1}{4}(1+\langle\hat{X}\hat{X}\rangle-\langle\hat{Y}\hat{Y}\rangle+\langle\hat{Z}\hat{Z}\rangle) = \tfrac{1}{2}(1+(1-2p)^2) = 0.58, \tag{13.5}$$

with the trace distance $D = 0.42$. Fidelities with a target state and distances between estimated and true states measure success in variational state preparation and in quantum state tomography.

14 Many qubits

14.1 The same rules for N qubits

Everything introduced on the two-qubit register holds for N qubits without change. The state is a vector of 2^N complex amplitudes in the computational basis, ordered by equation (5.5) with qubit 0 as the most significant bit. Reshaping it yields a state tensor with N indices. Gates on one qubit or on a pair of qubits are Kronecker products with identities on the other qubits, as Section 7.5 works out. Observables are combinations of Pauli strings, whose expectation values are estimated from bit strings in measurement settings. The reduced state of any subset of qubits is the partial trace (10.11) with more indices, the contraction of ψ with ψ^* over the indices of the other qubits. The state alone occupies $16\cdot 2^N$ bytes of memory (Section 5.2), which restricts exact simulation to a few tens of qubits.

14.2 GHZ and W states

Two states of three or more qubits show that multipartite entanglement comes in different kinds. The GHZ state (7.22) of Section 7.5 extends the Bell state. It has only two nonzero amplitudes, at the indices 0 and 2^N-1, but a state-vector simulation still stores all 2^N of them. The *W state* of three qubits

$$|W_3\rangle = \frac{|001\rangle+|010\rangle+|100\rangle}{\sqrt{3}} \tag{14.1}$$

shares one excitation among the three qubits. The reduced state of one qubit is $\hat{I}/2$ for the GHZ state and $\mathrm{diag}(2/3,1/3)$ for the W state, as the partial trace with a three-qubit state tensor shows. A measurement of one qubit of the GHZ state in the $\hat{Z}$ basis leaves the others in a product state. In the W state, the outcome 0 on one qubit leaves the other two in a Bell-type state $(|01\rangle+|10\rangle)/\sqrt{2}$. The two kinds of entanglement respond differently to the loss of a qubit and cannot be converted into each other by local operations [24].

14.3 Circuits and the Toffoli gate

A circuit on N qubits is a list of gates, each acting on one or two qubits. The gates are applied in order to $|0\ldots0\rangle$, and a measurement in a chosen setting follows. Its *depth* is the number of layers

of gates that act on disjoint qubits. Its gate counts, in particular the number of two-qubit gates, measure its cost on a device. Among the gates on three qubits, the *Toffoli* gate flips qubit 2 when qubits 0 and 1 are both 1. It is the evolution generated by the projector on $|1\rangle|1\rangle|-\rangle$,

$$\widehat{\text{Toffoli}} = \exp\Big[-\frac{i\pi}{8}(\hat{I}-\hat{Z})\otimes(\hat{I}-\hat{Z})\otimes(\hat{I}-\hat{X})\Big], \tag{14.2}$$

an 8×8 matrix in the binary order of Table 3. The exponent is $-i\pi\hat{P}$ with the projector $\hat{P}$ on $|1\rangle|1\rangle|-\rangle$, so by the projector argument used for CZ in Section 7.2 the gate is $\hat{I}-2\hat{P}$. It flips the sign of $|1\rangle|1\rangle|-\rangle$ and therefore exchanges the amplitudes of 110 and 111. On basis states it maps $|b_0b_1b_2\rangle$ to $|b_0,\, b_1,\, b_2\oplus(b_0\wedge b_1)\rangle$, with the AND operation $\wedge$. For $b_2=0$ it computes the AND of the first two bits and keeps them, so the Toffoli gate is a reversible AND. Since AND and NOT suffice for every classical computation, a quantum computer can run any classical program reversibly. Parametrised circuits combine rotations $\hat{R}_\alpha(\theta)$ with trainable angles and fixed entangling gates or interactions $\hat{R}_{\alpha\alpha}(\theta)$.

15 What quantum information allows and forbids

A register of N qubits holds 2^N complex amplitudes, and it is tempting to treat it as a memory of 2^N numbers that can be copied, read out and processed at will. The four rules of Section 1 set limits to this picture that no device can overcome, and they also give resources that no classical memory has, both of which the methods of the later lectures rely on.

15.1 The no-cloning theorem

A classical bit can be copied, and the CNOT gate (7.3) copies the bit of its control qubit into a target qubit prepared in $|0\rangle$, $\widehat{\text{CNOT}}|b\rangle|0\rangle = |b\rangle|b\rangle$ for $b=0,1$. A device that copies every state of a qubit would be a unitary $\hat{U}$ acting on the state and on a blank qubit with

$$\hat{U}\,|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle \qquad \text{for every } |\psi\rangle. \tag{15.1}$$

If $\hat{U}$ copies two states $|\psi\rangle$ and $|\varphi\rangle$, the inner product of the two inputs equals that of the two outputs, because a unitary preserves inner products,

$$\langle\varphi|\psi\rangle\,\langle 0|0\rangle = \big(\langle\varphi|\otimes\langle\varphi|\big)\big(|\psi\rangle\otimes|\psi\rangle\big) = \langle\varphi|\psi\rangle^2. \tag{15.2}$$

Taking the modulus gives $|\langle\varphi|\psi\rangle| = |\langle\varphi|\psi\rangle|^2$, so $|\langle\varphi|\psi\rangle|$ is 0 or 1, and $\hat{U}$ copies two states only if they are orthogonal or equal up to a global phase. The same conclusion holds when the copies in (15.1) may carry a phase, and together these statements form the *no-cloning theorem* [25,26]. By linearity, a gate that copies $|0\rangle$ and $|1\rangle$ maps the superposition $|+\rangle|0\rangle = (|0\rangle|0\rangle+|1\rangle|0\rangle)/\sqrt{2}$ to $(|00\rangle+|11\rangle)/\sqrt{2}$, the Bell state, whereas the copy would be $|+\rangle|+\rangle = \frac{1}{2}(|00\rangle+|01\rangle+|10\rangle+|11\rangle)$. For the CNOT gate used as a copier, the fidelity of the output with the desired copy falls from one at the poles of the Bloch sphere to 0.50 at $|+\rangle$ (Figure 41), and it vanishes for $|-\rangle$.

The theorem constrains every attempt to learn a state, because a measurement in one basis changes the state and no copy can be kept for a measurement in another basis. Determining an unknown state therefore requires many independent preparations, each measured once, for example in the informationally complete settings of Section 11.8. State tomography and classical shadows, the subject of Lecture 13, are organised around this budget of fresh preparations.

15.2 Telling two states apart with one copy

A source emits one copy of either $\hat{\rho}$ or $\hat{\sigma}$, each with probability one half, and we guess which one it emitted. A guessing rule is a POVM with two elements, $\hat{\Pi}$ and $\hat{I}-\hat{\Pi}$, both positive, which means

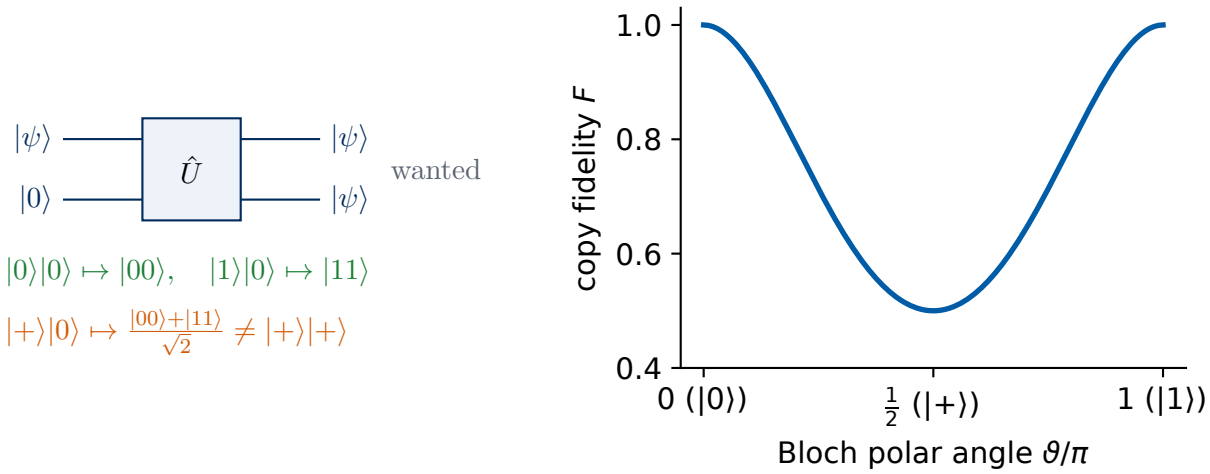


Figure 41: Left, a hypothetical copier $\hat{U}$ of equation (15.1). The CNOT gate copies the basis states and turns $|+\rangle|0\rangle$ into the Bell state. Right, the fidelity $F = |\langle\psi|\langle\psi|\widehat{\mathrm{CNOT}}|\psi\rangle|0\rangle|^2 = \left(\cos^3\frac{\vartheta}{2} + \sin^3\frac{\vartheta}{2}\right)^2$ of the CNOT copier for the states $\cos\frac{\vartheta}{2}|0\rangle + \sin\frac{\vartheta}{2}|1\rangle$ on the x–z meridian, against the polar angle ϑ of the Bloch sphere. It equals one only for the two basis states.

$0 \le \hat{\Pi} \le \hat{I}$. We answer $\hat{\rho}$ on the first outcome and $\hat{\sigma}$ on the second. By the Born rule for POVMs the probability of a correct guess is

$$P_{\mathrm{succ}} = \tfrac{1}{2}\mathrm{Tr}(\hat{\Pi}\hat{\rho}) + \tfrac{1}{2}\mathrm{Tr}\big[(\hat{I}-\hat{\Pi})\hat{\sigma}\big] = \tfrac{1}{2} + \tfrac{1}{2}\mathrm{Tr}\big[\hat{\Pi}\,(\hat{\rho}-\hat{\sigma})\big]. \tag{15.3}$$

We expand the Hermitian difference in its eigenvectors, $\hat{\rho}-\hat{\sigma} = \sum_k \mu_k |k\rangle\langle k|$, as for the trace distance of Section 13. The trace becomes $\sum_k \mu_k \langle k|\hat{\Pi}|k\rangle$, where every weight $\langle k|\hat{\Pi}|k\rangle$ lies between 0 and 1. The sum is largest when the weight is one for every positive μ_k and zero for every negative one, that is, when $\hat{\Pi}$ is the projector on the eigenvectors with positive eigenvalues. The sum of the positive eigenvalues equals the trace distance D, which gives the *Helstrom bound* [27]

$$P_{\mathrm{succ}}^{\max} = \tfrac{1}{2}\big[1 + D(\hat{\rho},\hat{\sigma})\big], \qquad \hat{\Pi}_{\mathrm{opt}} = \sum_{k:\,\mu_k>0} |k\rangle\langle k|. \tag{15.4}$$

Equation (15.4) gives the probability quoted in Section 13 and holds for every measurement, projective or general.

For two pure states $\hat{\rho} = |\psi\rangle\langle\psi|$ and $\hat{\sigma} = |\varphi\rangle\langle\varphi|$ the difference has rank two. Its two nonzero eigenvalues sum to $\mathrm{Tr}(\hat{\rho}-\hat{\sigma}) = 0$, and their squares sum to $\mathrm{Tr}(\hat{\rho}-\hat{\sigma})^2 = 2 - 2|\langle\varphi|\psi\rangle|^2$, so they are $\pm\sqrt{1-|\langle\varphi|\psi\rangle|^2}$ and

$$P_{\mathrm{succ}}^{\max} = \tfrac{1}{2}\Big(1 + \sqrt{1-|\langle\varphi|\psi\rangle|^2}\Big). \tag{15.5}$$

Only orthogonal states are told apart with certainty, which is the measurement counterpart of the no-cloning theorem. For one qubit $\hat{\rho}-\hat{\sigma} = \frac{1}{2}(\mathbf{r}_\rho - \mathbf{r}_\sigma)\cdot\hat{\boldsymbol{\sigma}}$ with the Bloch vectors $\mathbf{r}_\rho$ and $\mathbf{r}_\sigma$, and the optimal projector measures the spin along $\mathbf{r}_\rho - \mathbf{r}_\sigma$. For $|0\rangle$ and $|+\rangle$ this direction is $(\mathbf{e}_z - \mathbf{e}_x)/\sqrt{2}$, with the unit vectors $\mathbf{e}_x$ and $\mathbf{e}_z$ of the Bloch sphere (Figure 42). The overlap is $1/2$, and the optimal measurement succeeds with probability $\frac{1}{2}(1+1/\sqrt{2}) = 0.85$, whereas a measurement in the $\hat{Z}$ basis succeeds with probability $3/4$ (Figure 43). A classifier of quantum data that assigns a label to one copy of a state, as in Lecture 14, cannot do better than equation (15.4).

15.3 How many bits N qubits carry

A sender encodes a classical message x, chosen with probability p_x, in a state $\hat{\rho}_x$ of N qubits, and a receiver measures the state. The receiver, who does not know x, holds the average state $\hat{\bar{\rho}} = \sum_x p_x \hat{\rho}_x$. The information gained is the *mutual information* between x and the measurement outcome, defined as the reduction in bits of the Shannon entropy of x upon learning the outcome. The *Holevo quantity*

$$\chi = S(\hat{\bar{\rho}}) - \sum_x p_x\, S(\hat{\rho}_x), \tag{15.6}$$

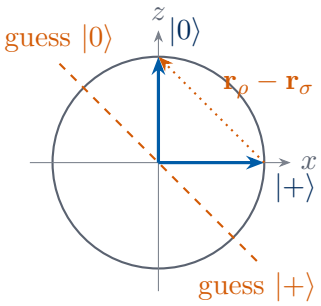


Figure 42: The states $|0\rangle$ and $|+\rangle$ in the x–z plane of the Bloch sphere, with the Bloch vector $\mathbf{r}_\rho$ of $|0\rangle$ and $\mathbf{r}_\sigma$ of $|+\rangle$. The optimal measurement for one copy is the spin component along $\mathbf{r}_\rho - \mathbf{r}_\sigma$ (dashed axis), and each of its two outcomes is read as a guess of the nearer state.

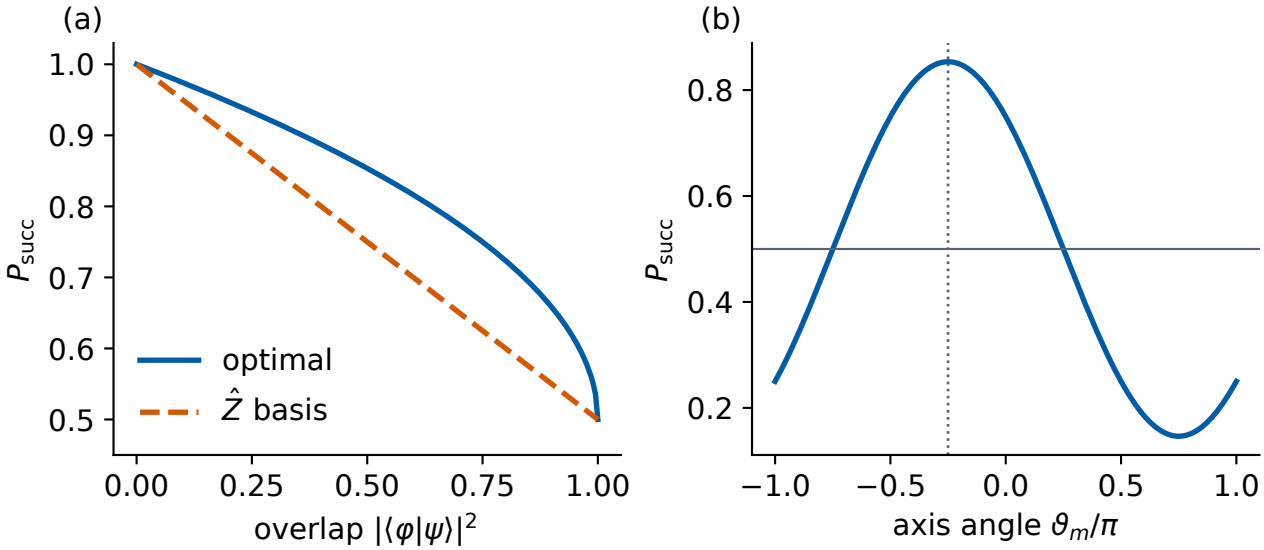


Figure 43: (a) Best success probability (15.5) of telling two pure states apart from one copy, against their overlap, and the success probability $1-|\langle\varphi|\psi\rangle|^2/2$ of a $\hat{Z}$ measurement for $|0\rangle$ against a second state. (b) Success probability for $|0\rangle$ against $|+\rangle$ when the spin is measured along the axis $(\sin\vartheta_m, 0, \cos\vartheta_m)$. The maximum lies at $\vartheta_m = -\pi/4$ (dotted), the direction of $\mathbf{e}_z - \mathbf{e}_x$, and the thin line marks the random guess 1/2.

built from the von Neumann entropy (10.16), bounds this mutual information for every measurement [28]. The entropy of a state of N qubits is at most $\log_2 2^N = N$ bits, reached by $\hat{I}/2^N$, and the entropies $S(\hat{\rho}_x)$ are non-negative, so

$$I(x\text{:outcome}) \;\le\; \chi \;\le\; N \text{ bits}. \tag{15.7}$$

The proof of the first inequality uses properties of the entropy beyond these notes [28,1]. By the second inequality, a receiver who shares no entanglement with the sender learns at most N bits from a state of N qubits, however many different states the sender uses, although 2^N continuous amplitudes fix the state. The amplitudes determine probabilities, and only one sample of these probabilities is read out per register.

For two pure states with equal probabilities, $|0\rangle$ and $\cos\frac{\alpha}{2}|0\rangle + \sin\frac{\alpha}{2}|1\rangle$ at the angle α on the Bloch sphere, the average state has the Bloch vector of length $\cos\frac{\alpha}{2}$ and the eigenvalues $\frac{1}{2}(1 \pm \cos\frac{\alpha}{2})$. With $S(\hat{\rho}_x) = 0$ the Holevo quantity is the binary entropy

$$\chi = S_{\text{bin}}\Big(\tfrac{1}{2}(1+\cos\tfrac{\alpha}{2})\Big). \tag{15.8}$$

The Helstrom measurement guesses wrongly with the probability $p_{\text{err}} = \frac{1}{2}(1-\sin\frac{\alpha}{2})$ by equation (15.5), the same for either message. Its outcome therefore leaves $S_{\text{bin}}(p_{\text{err}})$ of the one bit of uncertainty about x, and it carries $1 - S_{\text{bin}}(p_{\text{err}})$ bits. For $|0\rangle$ and $|+\rangle$, at $\alpha = \pi/2$, these are 0.40 bit against $\chi = 0.60$ bit (Figure 44a). When the sender uses K random pure states of N qubits, χ grows with K and saturates at N bits (Figure 44b).

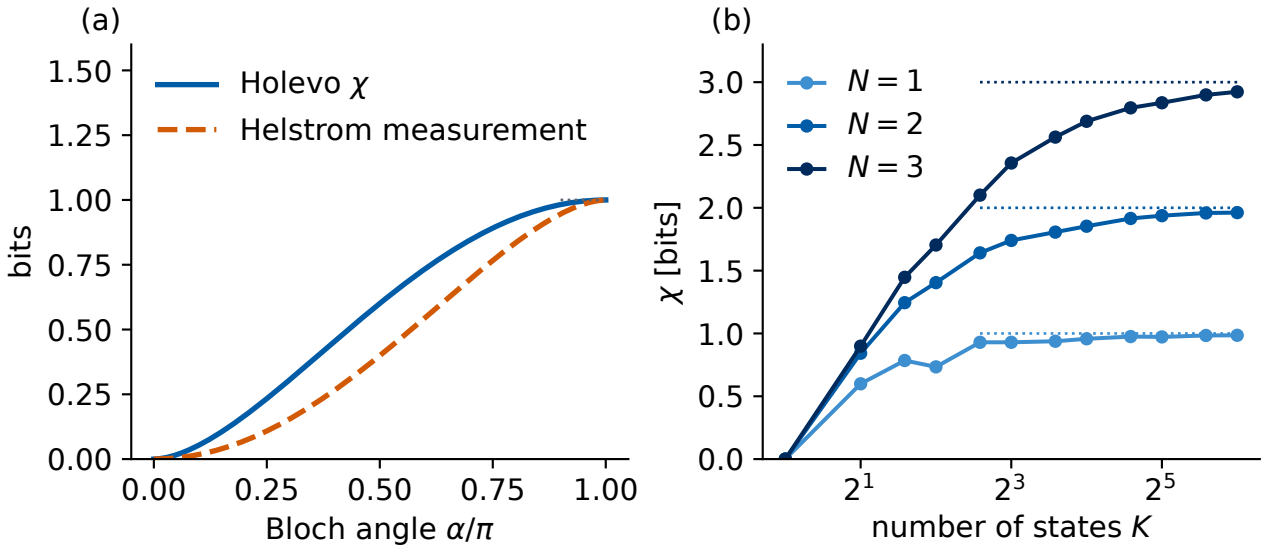


Figure 44: (a) Holevo quantity (15.8) of two equally likely pure states at the Bloch angle α, and the mutual information of the Helstrom measurement. Both reach one bit for orthogonal states. (b) Holevo quantity of K random pure states of N qubits with equal probabilities, averaged over five draws. It saturates at N bits (dotted).

15.4 Universality and the cost of a general unitary

Section 7.3 stated that single-qubit rotations and an entangling two-qubit gate, for example CNOT, are universal [29]. Every unitary on 2^N basis states is a product of *two-level unitaries*, each acting on only two basis states. They are found by removing the off-diagonal entries of the matrix one at a time, in the manner of Gaussian elimination. A two-level unitary acting on two bit strings that differ in one bit is a single-qubit gate on that bit, controlled by the values of all other bits. Two bit strings that differ in several bits are first brought to differ in one bit by CNOT gates, which permute basis states. Controlled gates, in turn, are made from CNOT gates and rotations. Gates with several controls are reduced to singly controlled ones and CNOT gates in the same way, as the Toffoli gate of Section 14.3 illustrates [29].

Error-correcting codes protect only a finite set of gates, for example $\hat{H}$, $\hat{T}$ and CNOT, and continuous angles must then be approximated. The *Solovay–Kitaev theorem* [43] guarantees that any single-qubit gate is approximated to an accuracy ε by a sequence of such gates whose length grows only as a power of $\log(1/\varepsilon)$ [30].

Universality says nothing about cost, and counting parameters shows that most unitaries are expensive. A unitary $2^N \times 2^N$ matrix has 4^N real parameters, one of them an unobservable global phase. A two-qubit gate has at most $4^2 - 1 = 15$ parameters, and single-qubit gates can be absorbed into the neighbouring two-qubit gates. A circuit of G two-qubit gates can therefore reach a generic unitary only if

$$15\,G \;\geq\; 4^N - 1. \tag{15.9}$$

For ten qubits this requires at least 69905 two-qubit gates. Useful circuits are therefore special, short circuits that reach a tiny part of all unitaries.

Controlled gates from CNOT and rotations (optional)

For a single-qubit gate written in the Euler form of Section 3.5, $\hat{U} = e^{i\eta}\hat{R}_z(\theta_1)\hat{R}_y(\theta_2)\hat{R}_z(\theta_3)$ with the global phase η, we define the three rotations

$$\hat{A} = \hat{R}_z(\theta_1)\hat{R}_y(\tfrac{\theta_2}{2}), \qquad \hat{B} = \hat{R}_y(-\tfrac{\theta_2}{2})\hat{R}_z(-\tfrac{\theta_3+\theta_1}{2}), \qquad \hat{C} = \hat{R}_z(\tfrac{\theta_3-\theta_1}{2}). \tag{15.10}$$

Their product is $\hat{A}\hat{B}\hat{C} = \hat{I}$, since the inner rotations about the same axis cancel. With an $\hat{X}$ on each side of $\hat{B}$, the relations $\hat{X}\hat{R}_y(\theta)\hat{X} = \hat{R}_y(-\theta)$ and $\hat{X}\hat{R}_z(\theta)\hat{X} = \hat{R}_z(-\theta)$ reverse the angles of $\hat{B}$,

and $\hat{A}\hat{X}\hat{B}\hat{X}\hat{C} = e^{-i\eta}\hat{U}$. In the circuit of Figure 45 the CNOT gates act as $\hat{X}$ on qubit 1 only when qubit 0 is $|1\rangle$, so qubit 1 receives $\hat{A}\hat{B}\hat{C} = \hat{I}$ in one branch and $e^{-i\eta}\hat{U}$ in the other. The phase gate $\mathrm{diag}(1, e^{i\eta}) = e^{i\eta/2}\hat{R}_z(\eta)$ on qubit 0 restores the phase,

$$|0\rangle\langle 0| \otimes \hat{I} + |1\rangle\langle 1| \otimes \hat{U} = \big[\mathrm{diag}(1, e^{i\eta}) \otimes \hat{I}\big]\, (\hat{I} \otimes \hat{A})\, \widehat{\mathrm{CNOT}}\, (\hat{I} \otimes \hat{B})\, \widehat{\mathrm{CNOT}}\, (\hat{I} \otimes \hat{C}). \tag{15.11}$$

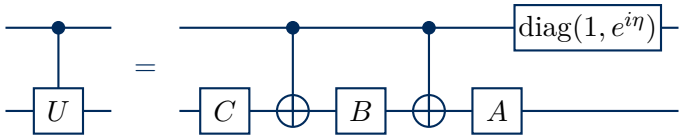


Figure 45: A controlled single-qubit gate $\hat{U} = e^{i\eta}\hat{R}_z(\theta_1)\hat{R}_y(\theta_2)\hat{R}_z(\theta_3)$ from two CNOT gates, the rotations $\hat{A}$, $\hat{B}$, $\hat{C}$ of equation (15.10) and the phase gate $\mathrm{diag}(1, e^{i\eta})$ on the control qubit.

15.5 Clifford gates and the Gottesman–Knill theorem

The Pauli strings of Section 7.5, multiplied by the phases ± 1 and $\pm i$, form the *Pauli group*. A *Clifford gate* is a unitary $\hat{U}$ that maps every Pauli string to a Pauli string under conjugation, $\hat{U}\hat{P}\hat{U}^\dagger = \pm\hat{Q}$. The gates $\hat{H}$, $\hat{S}$ and CNOT have this property,

$$\begin{aligned}
&\hat{H}\hat{X}\hat{H} = \hat{Z}, \quad \hat{H}\hat{Z}\hat{H} = \hat{X}, \quad \hat{H}\hat{Y}\hat{H} = -\hat{Y},\\
&\hat{S}\hat{X}\hat{S}^\dagger = \hat{Y}, \quad \hat{S}\hat{Y}\hat{S}^\dagger = -\hat{X}, \quad \hat{S}\hat{Z}\hat{S}^\dagger = \hat{Z},\\
&\widehat{\mathrm{CNOT}}\, \hat{X}_0\, \widehat{\mathrm{CNOT}} = \hat{X}_0\hat{X}_1, \quad \widehat{\mathrm{CNOT}}\, \hat{Z}_1\, \widehat{\mathrm{CNOT}} = \hat{Z}_0\hat{Z}_1,\\
&\widehat{\mathrm{CNOT}}\, \hat{Z}_0\, \widehat{\mathrm{CNOT}} = \hat{Z}_0, \quad \widehat{\mathrm{CNOT}}\, \hat{X}_1\, \widehat{\mathrm{CNOT}} = \hat{X}_1,
\end{aligned} \tag{15.12}$$

and they generate all Clifford gates [31]. The CNOT rules follow from equation (7.3). An $\hat{X}$ on the control flips the target as well, and a $\hat{Z}$ on the target, a phase that depends on the target bit, becomes after CNOT a phase that depends on the parity $b_0 \oplus b_1$. The rules for $\hat{Y} = i\hat{X}\hat{Z}$ follow from those for $\hat{X}$ and $\hat{Z}$. A Pauli string on N qubits is stored as $2N$ bits and a sign, with $(x_q, z_q) = (0,0), (1,0), (0,1), (1,1)$ for $\hat{I}, \hat{X}, \hat{Z}, \hat{Y}$ on qubit q. A gate changes only the bits of the qubits it acts on, so one Pauli string is followed through a Clifford circuit of G gates with a few bit operations per gate, without any $2^N \times 2^N$ matrix.

A Pauli string $\hat{P}$ *stabilises* a state $|\psi\rangle$ if $\hat{P}|\psi\rangle = |\psi\rangle$. The state $|0\ldots0\rangle$ is stabilised by $\hat{Z}_0, \ldots, \hat{Z}_{N-1}$. A Clifford circuit $\hat{V}$ turns them into the stabilisers $\hat{V}\hat{Z}_q\hat{V}^\dagger$ of the output, since $\hat{V}\hat{Z}_q\hat{V}^\dagger\hat{V}|0\ldots0\rangle = \hat{V}|0\ldots0\rangle$, and such outputs are called *stabiliser states*. For the GHZ circuit (7.22) the rules (15.12) give, column by column, the stabilisers of Figure 46. The three-qubit GHZ state is stabilised by $\hat{X}_0\hat{X}_1\hat{X}_2$, $\hat{Z}_0\hat{Z}_1$ and $\hat{Z}_1\hat{Z}_2$, and the Bell state by $\hat{X}_0\hat{X}_1$ and $\hat{Z}_0\hat{Z}_1$, the two parities that are certain in its records (Section 8). The expectation value of a Pauli string $\hat{P}$ in $\hat{V}|0\ldots0\rangle$ is $\langle 0\ldots0|\hat{V}^\dagger\hat{P}\hat{V}|0\ldots0\rangle$, where $\hat{V}^\dagger\hat{P}\hat{V} = \pm\hat{Q}$ follows from the same rules applied to the inverse gates. It equals ± 1 when $\hat{Q}$ contains only $\hat{Z}$ and $\hat{I}$ factors and zero otherwise, and measurement outcomes are sampled with the same bit representation [32]. Clifford circuits acting on $|0\ldots0\rangle$ and followed by Pauli measurements are therefore simulated efficiently on a classical computer, which is the *Gottesman–Knill theorem* [31,32]. Tracking all N stabilisers costs a number of operations proportional to N per gate [32], and the stabilisers of a thousand-qubit GHZ state follow in less than a second. A circuit can therefore create strong entanglement and still be simulated efficiently.

The $\hat{T}$ gate is not a Clifford gate, since it maps $\hat{T}\hat{X}\hat{T}^\dagger = (\hat{X} + \hat{Y})/\sqrt{2}$, a sum of two Pauli matrices. Adding $\hat{T}$ to $\hat{H}$, $\hat{S}$ and CNOT gives a universal set [33], the finite set of the Solovay–Kitaev theorem.

15.6 Random states, concentration and 2-designs

A uniformly random, or *Haar-random*, state of N qubits is a unit vector drawn so that every unitary leaves its distribution unchanged. The rotated state $\hat{U}|\psi\rangle$ is then as random as $|\psi\rangle$. A computer draws

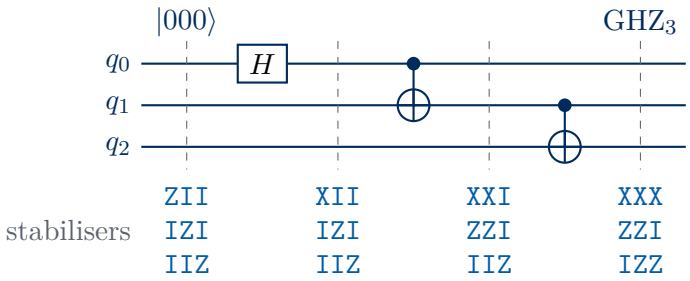


Figure 46: The stabilisers of $|000\rangle$ followed through the GHZ circuit with the rules (15.12). Each column lists the three stabilisers after the gates to its left, written as Pauli strings with qubit 0 first. The last column stabilises $|\mathrm{GHZ}_3\rangle$.

it by normalising a vector of 2^N independent complex Gaussian numbers. Gates such as $\hat{X}_q$ permute the bit strings, so every bit string has the same average probability, and since the probabilities sum to one,

$$\mathbb{E}\,|\langle b|\psi\rangle|^2 = \frac{1}{2^N}, \tag{15.13}$$

where $\mathbb{E}$ denotes the average over the random state.

For Pauli strings we use the expansion of a density matrix in the orthogonal basis of Section 7.5. With $\mathrm{Tr}(\hat{P}\hat{Q}) = 2^N\delta_{PQ}$, every state of N qubits reads $\hat{\rho} = 2^{-N}\sum_{\hat{P}}\langle\hat{P}\rangle\hat{P}$, and its purity is

$$\mathrm{Tr}\hat{\rho}^2 = \frac{1}{2^N}\sum_{\hat{P}}\langle\hat{P}\rangle^2. \tag{15.14}$$

A pure state has purity one, and the identity contributes $\langle\hat{I}\rangle^2 = 1$, so the $4^N - 1$ non-identity strings share $\sum_{\hat{P}\neq\hat{I}}\langle\hat{P}\rangle^2 = 2^N - 1$. For any two non-identity Pauli strings $\hat{P}$ and $\hat{P}'$ there is a Clifford circuit $\hat{V}$ with $\hat{V}\hat{P}'\hat{V}^\dagger = \pm\hat{P}$, since by the rules (15.12) $\hat{H}$ and $\hat{S}$ turn any Pauli matrix into any other on one qubit and CNOT moves factors between qubits [31]. Then $\langle\psi|\hat{P}|\psi\rangle = \pm\langle\phi|\hat{P}'|\phi\rangle$ with $|\phi\rangle = \hat{V}^\dagger|\psi\rangle$, and $|\phi\rangle$ is distributed like $|\psi\rangle$ because the Haar distribution is unchanged by every unitary. Every non-identity string therefore has the same average of $\langle\hat{P}\rangle^2$, namely the shared sum divided by the number of strings. In the same way a Pauli string $\hat{Q}$ that anticommutes with $\hat{P}$ gives $\hat{Q}\hat{P}\hat{Q} = -\hat{P}$, so $\langle\hat{P}\rangle$ and $-\langle\hat{P}\rangle$ are equally likely and the average of $\langle\hat{P}\rangle$ vanishes,

$$\mathbb{E}\,\langle\hat{P}\rangle = 0, \qquad \mathrm{Var}\,\langle\hat{P}\rangle = \mathbb{E}\,\langle\hat{P}\rangle^2 = \frac{2^N - 1}{4^N - 1} = \frac{1}{2^N + 1}, \tag{15.15}$$

where the variance is taken over the random state. The expectation values of a random state concentrate at zero, with a standard deviation that falls as $2^{-N/2}$. By Chebyshev's inequality the probability of $|\langle\hat{P}\rangle| > a$ is at most $1/[a^2(2^N + 1)]$ for any threshold $a > 0$, so a random state of many qubits looks maximally mixed to every Pauli measurement (Figure 47a). Concentration of this kind makes the gradients of randomly initialised parametrised circuits that are deep enough to behave like random unitaries vanish exponentially in N [44], the barren plateaus of Lecture 12.

Unitary 2-designs (optional)

Equations (15.13) and (15.15) use only averages of quantities that are quadratic in $\hat{U}$ and quadratic in $\hat{U}^\dagger$, such as $\langle\hat{P}\rangle^2 = |\langle 0\ldots0|\hat{U}^\dagger\hat{P}\hat{U}|0\ldots0\rangle|^2$ for $|\psi\rangle = \hat{U}|0\ldots0\rangle$. An ensemble of unitaries that reproduces the Haar averages of all such quantities is a *unitary 2-design*. The Clifford gates, drawn uniformly, form a 2-design [34], and even a 3-design [35]. A random stabiliser state has $\langle\hat{P}\rangle \in \{0, \pm1\}$, a distribution unlike the Haar one, yet its variance is again $1/(2^N + 1)$ (Figure 47b). Random Clifford circuits can be followed with the bit update of Section 15.5, and their 2-design property underlies the classical shadows [45] of Lecture 13.

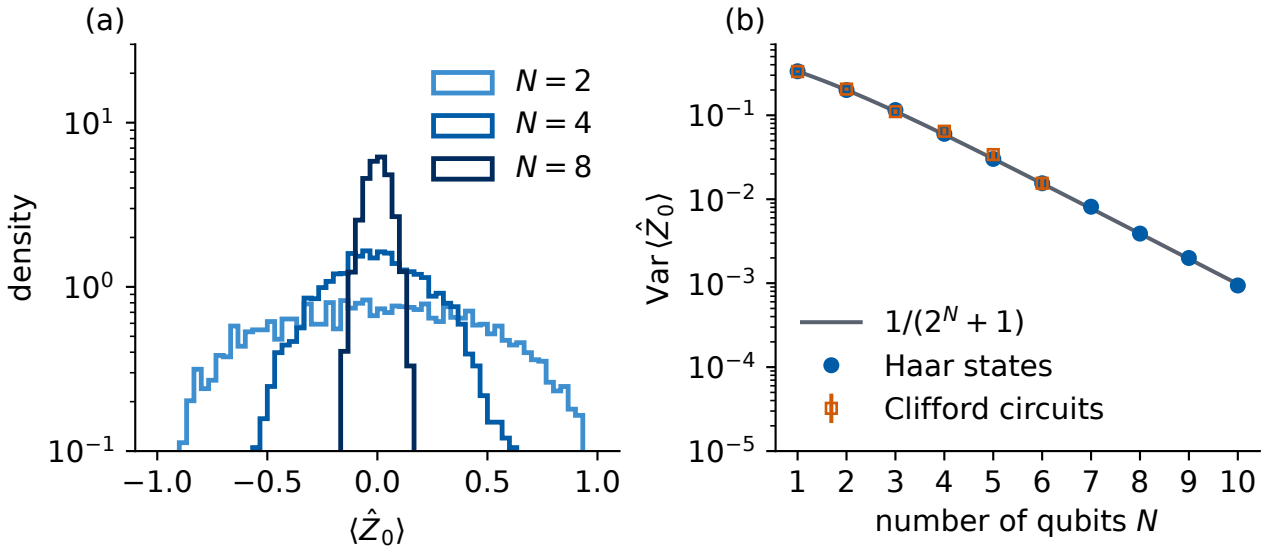


Figure 47: (a) Distribution of $\langle \hat{Z}_0 \rangle$ over 4000 Haar-random states of $N = 2$, 4 and 8 qubits, on a logarithmic density scale. (b) Variance of $\langle \hat{Z}_0 \rangle$ for Haar-random states and for states prepared by random circuits of $6N^2 + 20$ gates drawn from $\hat{H}$, $\hat{S}$, $\hat{S}^\dagger$ and CNOT, with statistical errors, against the curve $1/(2^N + 1)$ of equation (15.15). Such deep random circuits approach uniformly drawn Clifford gates.

15.7 Entanglement of mixed states and the partial transpose

For a pure register the entropy of one qubit measures entanglement, whereas for a mixed register it does not (Section 10.8). A mixed state of two qubits is *separable* if it is a mixture of product states,

$$\hat{\rho} = \sum_k p_k \, \hat{\rho}_0^{(k)} \otimes \hat{\rho}_1^{(k)}, \qquad p_k \geq 0, \quad \sum_k p_k = 1, \tag{15.16}$$

and *entangled* otherwise. The correlated coin is separable. A test follows from the transpose, which maps density matrices to density matrices but is not completely positive (Section 12.5). The *partial transpose* transposes the indices of qubit 1 only,

$$(\hat{\rho}^{T_1})_{ab,a'b'} = \rho_{ab',a'b}, \tag{15.17}$$

and maps the separable state (15.16) to $\sum_k p_k \, \hat{\rho}_0^{(k)} \otimes (\hat{\rho}_1^{(k)})^T$, again a density matrix. A negative eigenvalue of $\hat{\rho}^{T_1}$ therefore proves that $\hat{\rho}$ is entangled, the *Peres criterion* [36]. For two qubits a positive partial transpose also proves separability [37]. The *negativity* [38]

$$\mathcal{N}(\hat{\rho}) = \tfrac{1}{2}(\|\hat{\rho}^{T_1}\|_1 - 1) = \sum_{\lambda<0} |\lambda|, \tag{15.18}$$

with the trace norm of Section 13 and the eigenvalues λ of $\hat{\rho}^{T_1}$, quantifies the entanglement. Its two forms agree because the eigenvalues of $\hat{\rho}^{T_1}$ sum to one. For a pure state with Schmidt coefficients s_1, s_2 the negative eigenvalue is $-s_1 s_2$, so $\mathcal{N} = s_1 s_2 = C/2$ with the concurrence (8.8).

The *Werner state* [39] mixes the Bell state with the maximally mixed state,

$$\hat{\rho}_W = v \, |\Phi^+\rangle\langle\Phi^+| + (1 - v) \, \frac{\hat{I}}{4}, \qquad 0 \leq v \leq 1, \tag{15.19}$$

with the *visibility* v. The Bell matrix of Section 10 has the entries $\frac{1}{2}$ at the positions $(00, 00)$, $(00, 11)$, $(11, 00)$ and $(11, 11)$. The partial transpose moves the two coherences to $(01, 10)$ and $(10, 01)$ and gives $\widehat{\text{SWAP}}/2$, the normalised Choi state of the transpose (Section 12.5). The SWAP gate has the eigenvalue $+1$ on the three symmetric states and -1 on the singlet (Section 7.2), and $\hat{I}^{T_1} = \hat{I}$, so

$$\hat{\rho}_W^{T_1} = \frac{v}{2} \, \widehat{\text{SWAP}} + \frac{1 - v}{4} \, \hat{I} \quad \text{has the eigenvalues} \quad \underbrace{\frac{1 + v}{4}}_{\text{symmetric states, three times}}, \quad \underbrace{\frac{1 - 3v}{4}}_{\text{singlet}}. \tag{15.20}$$

The Werner state is entangled if and only if $v > 1/3$, with $\mathcal{N} = \max\left(0, \frac{3v-1}{4}\right)$ (Figures 48 and 49). At the threshold its fidelity with the Bell state is $\langle\Phi^+|\hat{\rho}_W|\Phi^+\rangle = (1+3v)/4 = 1/2$. The normalised Choi state (12.19) of depolarising noise is a Werner state with $v = 1 - \frac{4}{3}p$.

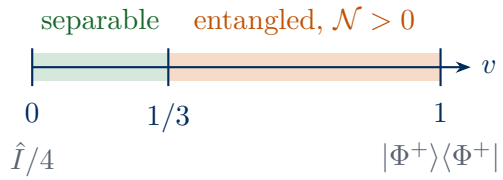


Figure 48: The Werner states (15.19) on a line of visibility v, from the maximally mixed state to the Bell state. They are separable up to $v = 1/3$ and entangled above.

The criterion settles the question left open by the noisy Bell register of Section 12, where the entropy of qubit 0 stayed at one bit for every noise strength. Depolarising noise of strength p on each qubit mixes each qubit with $\hat{I}/2$ with the weight $\frac{4}{3}p$ and turns the Bell state into a Werner state with $v = (1 - \frac{4}{3}p)^2$. It becomes separable at $p = \frac{3}{4}(1 - 1/\sqrt{3}) \approx 0.32$. Local dephasing leaves the coherence $\frac{1}{2}(1-2p)^2$ between $|00\rangle$ and $|11\rangle$. The partial transpose moves it between $|01\rangle$ and $|10\rangle$ with the eigenvalues $\pm\frac{1}{2}(1-2p)^2$, so $\mathcal{N} = \frac{1}{2}(1-2p)^2$ vanishes only at $p = 1/2$, where the register has become the correlated coin (Figure 49b).

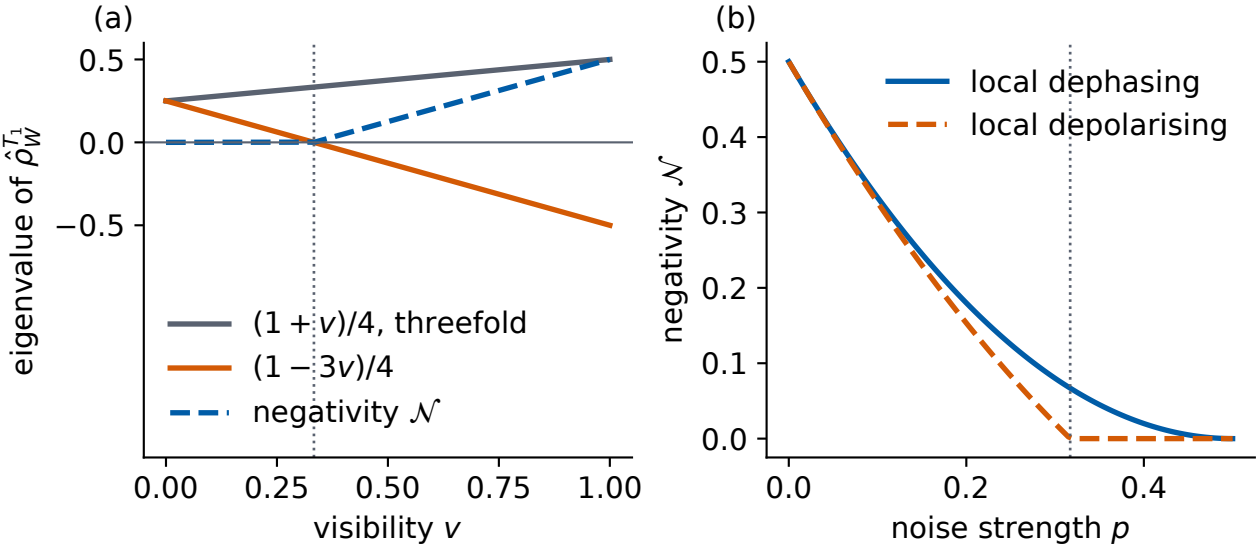


Figure 49: (a) Eigenvalues (15.20) of the partial transpose of the Werner state and its negativity, against the visibility v. The lowest eigenvalue crosses zero at $v = 1/3$ (dotted). (b) Negativity of the Bell state after local dephasing and after local depolarising noise of strength p on both qubits. The depolarised state becomes separable at $p \approx 0.32$ (dotted).

15.8 Quantum Fisher information and the precision of a phase

A phase θ is imprinted on N qubits by the collective rotation $e^{-i\theta\hat{J}_z} = \hat{R}_z(\theta)^{\otimes N}$ with the generator

$$\hat{J}_z = \frac{1}{2}\sum_{j=0}^{N-1} \hat{Z}_j, \tag{15.21}$$

which applies the Ramsey phase of Section 3.4 to every qubit. We prepare the state $|\psi_\theta\rangle = e^{-i\theta\hat{J}_z}|\psi\rangle$ M times, measure each copy and compute an estimate $\tilde{\theta}$ from the outcomes. The estimate is *unbiased* if its average equals the true phase, and its error $\Delta\theta$ is its standard deviation. How well the phases θ and $\theta + \mathrm{d}\theta$ can be told apart follows from the fidelity of the two states. Expanding

$\langle\psi|e^{-i\,\mathrm{d}\theta\,\hat{J}_z}|\psi\rangle = 1 - i\,\mathrm{d}\theta\langle\hat{J}_z\rangle - \frac{1}{2}\mathrm{d}\theta^2\langle\hat{J}_z^2\rangle$ and taking the squared modulus, $\left(1-\frac{1}{2}\mathrm{d}\theta^2\langle\hat{J}_z^2\rangle\right)^2 + \mathrm{d}\theta^2\langle\hat{J}_z\rangle^2 \approx 1 - \mathrm{d}\theta^2(\langle\hat{J}_z^2\rangle - \langle\hat{J}_z\rangle^2)$ to second order, gives

$$\left|\langle\psi_\theta|\psi_{\theta+\mathrm{d}\theta}\rangle\right|^2 = 1 - \frac{F_Q}{4}\,\mathrm{d}\theta^2, \qquad F_Q = 4\,\mathrm{Var}(\hat{J}_z) = 4(\langle\hat{J}_z^2\rangle - \langle\hat{J}_z\rangle^2), \tag{15.22}$$

where F_Q is the *quantum Fisher information* of a pure state. A large variance of the generator makes the state move away quickly as θ changes. For a mixed state F_Q is defined by the same second-order expansion of the fidelity, and it never exceeds $4\,\mathrm{Var}(\hat{J}_z)$ [40]. For any measurement and any unbiased estimate the error obeys the *quantum Cramér–Rao bound* [40]

$$\Delta\theta \;\geq\; \frac{1}{\sqrt{M\,F_Q}}. \tag{15.23}$$

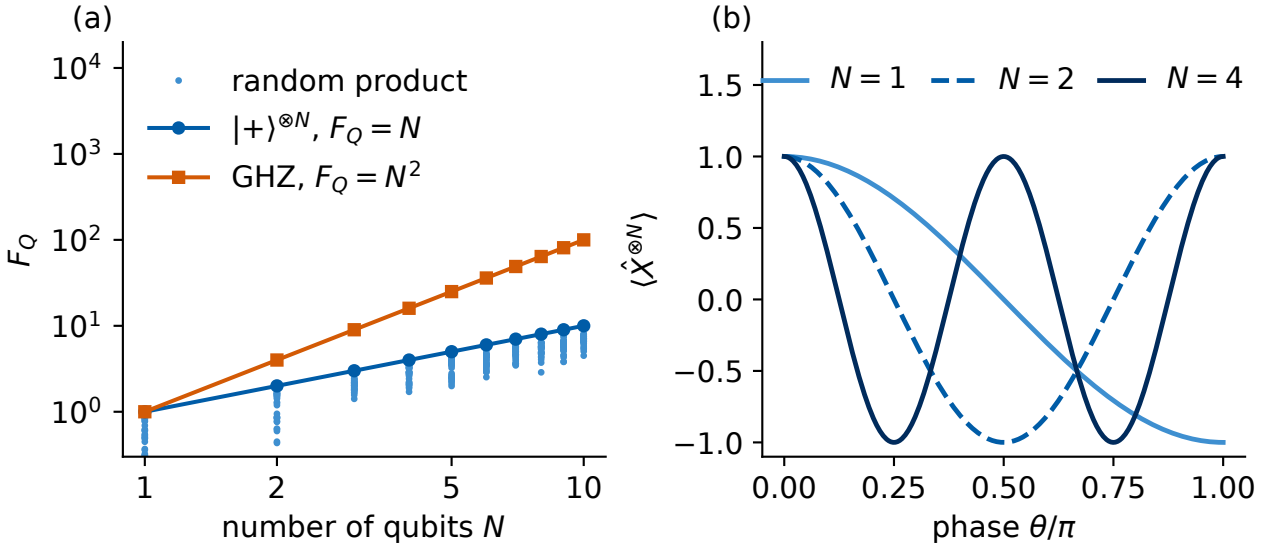


Figure 50: (a) Quantum Fisher information (15.22) against the number of qubits for $|+\rangle^{\otimes N}$, for the GHZ state and for thirty random product states at each N (dots), on logarithmic axes. (b) Parity $\langle\hat{X}^{\otimes N}\rangle = \cos N\theta$ of the GHZ state after the rotation $e^{-i\theta\hat{J}_z}$, for $N = 1$, 2 and 4.

For a product state the variances of the independent qubits add, $\mathrm{Var}(\hat{J}_z) = \frac{1}{4}\sum_j \mathrm{Var}(\hat{Z}_j) \leq N/4$, because $\mathrm{Var}(\hat{Z}_j) = 1 - \langle\hat{Z}_j\rangle^2 \leq 1$. Hence $F_Q \leq N$, with equality for $|+\rangle^{\otimes N}$, and the error falls as $1/\sqrt{MN}$. The GHZ state (7.22) has $\langle\hat{J}_z\rangle = 0$ and $\langle\hat{J}_z^2\rangle = N^2/4$, since its two components have $\hat{J}_z = \pm N/2$, so

$$F_Q^{\text{product}} \leq N, \qquad F_Q^{\text{GHZ}} = N^2. \tag{15.24}$$

Its two components acquire the relative phase $N\theta$, and the parity $\langle\hat{X}^{\otimes N}\rangle = \cos N\theta$ oscillates N times faster than a single-qubit fringe (Figure 50). The error then falls as $1/(N\sqrt{M})$, the *Heisenberg scaling* in N. Every separable state, pure or mixed, obeys $F_Q \leq N$ for this generator [41], so a value $F_Q/N > 1$ certifies entanglement without a full reconstruction of the state. Lecture 13 uses this criterion to certify entanglement from measured data.

15.9 Methods built on these foundations

The limits and resources of this section reappear in the methods of the following lectures. Classical shadows rest on the Clifford 2-design, barren plateaus on the concentration of random states, classifiers of quantum data on the Helstrom bound, and entanglement certification on the quantum Fisher information. Quantum machine-learning methods combine these with the objects of the earlier sections. Variational quantum algorithms prepare states with parametrised rotations and entangling gates. They estimate the energy of a Hamiltonian written as Pauli strings from bit strings in a

few measurement settings, as in Section 9.3, and budget the shot noise of Section 4.2. Quantum state tomography reconstructs density matrices from records in many settings and uses POVMs and randomised measurements. Hamiltonian learning infers the generator of the dynamics from time traces such as those of the hopping excitation. Ground states of spin models can be classified by their phases, and classical data enter quantum states through rotation gates. Tensor networks turn the reshape-and-SVD test of Section 8.2 into matrix product states and the contractions of these notes into algorithms for many qubits. Part II describes the physical devices that supply the drives and couplings used in Part I as given generators of gates.

Part II
Physical realisations of qubits

The second part shows how nature and engineering provide the objects of Part I. A physical qubit is a pair of levels of an atom, of an electrical circuit or of a spin, or the two polarisations of a photon, and its Hamiltonian follows from the physics of that system. We obtain the rotating-frame Hamiltonian of a driven qubit, use it to apply single-qubit gates with pulses of chosen area and phase, and then show how physical interactions between qubits produce the two-qubit gates. The full derivation of the qubit Hamiltonian of an atom from its electrons and nucleus, and the derivation of the Mølmer–Sørensen gate of trapped ions, are collected in Appendices A and B, so that the main text follows the physical storyline. The separation of the two parts marks which statements are general rules and which depend on a particular device, such as its native gates, its connectivity and the lifetime of its superpositions.

16 Single physical qubits and their drives

16.1 Requirements for a physical qubit

The gates of Section 3 are rotations generated by 2×2 Hamiltonians switched on for a chosen time. A physical system provides a qubit when two of its states can be separated from all others, driven into each other by a controllable Hamiltonian of this kind and read out. A single atom illustrates the three operations of Figure 2. *Optical pumping*, a few cycles of laser excitation and spontaneous decay that end in one chosen level, prepares the atom in its ground state. A laser pulse tuned to the energy difference of the two levels drives the atom between them, and the duration and phase of the pulse decide the final superposition. A final laser makes the atom fluoresce only in one of the two levels, and a camera that detects light or no light returns one bit.

Where the two basis states are energy levels, light, microwaves or voltage pulses drive transitions between them. Two-qubit gates come from an interaction between qubits, and a final measurement reads out the basis state of each qubit. A physical qubit is thus an addressable two-level system (Figure 51). One of the systems in the figure, the nitrogen-vacancy (NV) centre, is a defect in diamond, a nitrogen atom next to a missing carbon atom, whose electron spin is driven by microwaves and read out by its fluorescence. It is a spin qubit of the kind treated in Section 16.5 and is not discussed separately.

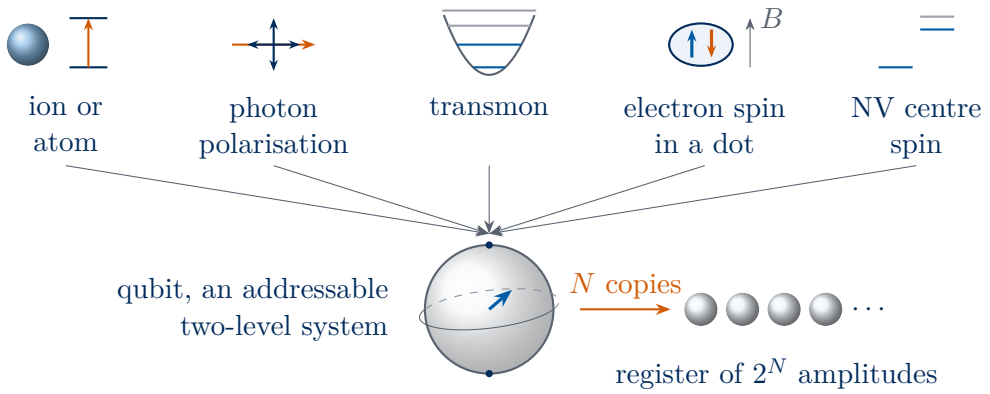


Figure 51: Different physical platforms provide the same abstract object, an addressable two-level system, and a register is N such systems that can be driven, coupled and measured.

In the three systems considered here these conditions hold for different reasons. In an atom, resonance and selection rules, explained in Section 16.2, pick two levels out of an infinite spectrum. In a superconducting circuit the level spacing is engineered until two levels can be addressed alone. A spin-1/2 has only two levels from the start. All three lead to the same Hamiltonian in a frame that

rotates with the drive. A fourth system, photon polarisation, provides a qubit whose two basis states have the same energy.

16.2 An atom in laser light

An atom has discrete energy levels, fixed by the attraction between its electrons and its nucleus. Atoms with one valence electron outside closed shells, the alkali atoms, and singly charged ions such as Ca^+, Sr^+ and Yb^+ have a simple spectrum of this kind and serve as qubits. A laser whose photon energy $\hbar\omega_L$ is close to the energy difference of two levels drives transitions between them. Hydrogen has the simplest spectrum, known in closed form, and we use it to illustrate the numbers. It is not used as a qubit, because its excited level decays within nanoseconds, as discussed at the end of this subsection. Appendix A derives the qubit Hamiltonian of this subsection from the Hamiltonian of the electrons and the nucleus, following the textbook treatments [2, 3].

Light couples two levels through the electric dipole of the atom. In an atom the eigenstates of the energy are either even or odd when all electron positions are inverted, whereas the dipole changes sign, so the light couples only levels of opposite parity. This *selection rule* forbids, for example, the transition from the ground state 1s of hydrogen to its excited state 2s, although 2s has the same energy as the 2p state that the light does excite (Figure 52). The ground state $|g\rangle$ and the excited state $|e\rangle$ are separated by the transition frequency ω_0, and every other allowed transition lies far from the laser frequency ω_L. The atom therefore behaves as a two-level system, a qubit with $|g\rangle = |0\rangle$ and $|e\rangle = |1\rangle$.

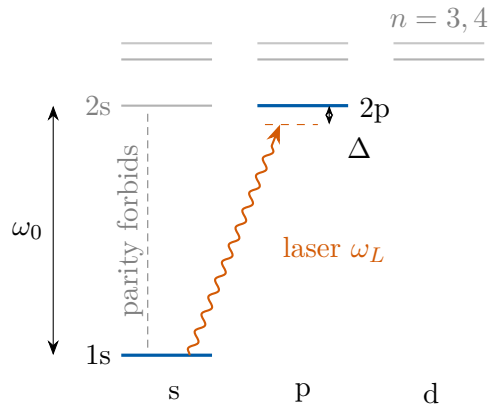


Figure 52: The lowest levels of hydrogen. The laser of angular frequency ω_L is close to the 1s–2p transition, with the gap ω_0 and the detuning Δ. The selection rules forbid the coupling of 1s to 2s, and every other allowed transition is far from resonance, so the atom behaves as a two-level system.

The qubit Hamiltonian in the laboratory

With the zero of energy halfway between the two levels and $\hbar = 1$, the free two-level atom has the Hamiltonian $\hat{\mathcal{H}}_0 = -\frac{\omega_0}{2}\hat{Z}$, so the ground state has the energy $-\omega_0/2$ and the excited state $+\omega_0/2$. A laser field $E_0 \cos(\omega_L t - \phi_L)$ with the amplitude E_0, the angular frequency ω_L and the phase ϕ_L adds a coupling between $|g\rangle$ and $|e\rangle$, and the atom has the Hamiltonian

$$\hat{\mathcal{H}}(t) = -\frac{\omega_0}{2}\hat{Z} + \Omega \cos(\omega_L t - \phi_L)\,\hat{X}, \qquad \Omega = -\frac{d_{eg} E_0}{\hbar}, \tag{16.1}$$

derived in Appendix A. The *Rabi frequency* Ω measures the strength of the coupling. It is proportional to the field amplitude and to the dipole matrix element d_{eg} between the two levels, and its sign depends on the phases chosen for $|g\rangle$ and $|e\rangle$.

Atomic-physics notation

Atomic-physics texts [2, 3] write the two-level atom with $\hat{\sigma}_z^{\text{at}} = |e\rangle\langle e| - |g\rangle\langle g| = -\hat{Z}$ and with the ladder operators $\hat{\sigma}_{\text{at}}^{+} = |e\rangle\langle g|$, which excites the atom, and $\hat{\sigma}_{\text{at}}^{-} = |g\rangle\langle e|$, which lowers it. The free atom is then $\hat{\mathcal{H}}_0 = \frac{\omega_0}{2}\hat{\sigma}_z^{\text{at}}$. These notes keep the Pauli matrices of Part I, in which $|0\rangle = |g\rangle$ has $\hat{Z} = +1$, and the spin ladder operator $\hat{\sigma}^{+} = \hat{X} + i\hat{Y} = 2|g\rangle\langle e|$ lowers the energy of the atom. Formulas taken from atomic-physics texts change sign in their $\hat{\sigma}_z$ terms on translation.

The rotating frame

The Hamiltonian (16.1) oscillates at an optical frequency, whereas the evolution operator $e^{-it\hat{\mathcal{H}}}$ of Section 3 requires a constant Hamiltonian. We remove the fast time dependence by describing the qubit in coordinate axes that turn about z together with the laser. The oscillating coupling $\Omega\cos(\omega_L t)\hat{X}$ along x is the sum of two couplings of half the strength that rotate about z in opposite directions at ω_L, as a linearly polarised field is the sum of two circularly polarised fields. The free Hamiltonian $-\frac{\omega_0}{2}\hat{Z}$ makes the Bloch vector precess clockwise about z at ω_0, as $\hat{R}_z(-\omega_0 t)$. In axes that turn clockwise at ω_L, one rotating component of the drive stands still and the free precession slows down to $\omega_0 - \omega_L$.

To turn the axes clockwise we turn every state counterclockwise, and the state in the *rotating frame* is

$$|\psi_{\text{RF}}(t)\rangle = \hat{R}_z(\omega_L t)\,|\psi(t)\rangle. \tag{16.2}$$

The rotation $\hat{R}_z(\omega_L t)$ changes only the phases of the amplitudes of $|0\rangle$ and $|1\rangle$, so all probabilities of a readout in the computational basis are the same in both frames. Differentiating (16.2) with $\frac{d}{dt}\hat{R}_z(\omega_L t) = -i\frac{\omega_L}{2}\hat{Z}\,\hat{R}_z(\omega_L t)$ and using the Schrödinger equation for $|\psi(t)\rangle$ gives $i\frac{d}{dt}|\psi_{\text{RF}}\rangle = \big[\frac{\omega_L}{2}\hat{Z} + \hat{R}_z(\omega_L t)\,\hat{\mathcal{H}}(t)\,\hat{R}_z(-\omega_L t)\big]|\psi_{\text{RF}}\rangle$. A rotation about z by the angle a turns $\hat{X}$ into

$$\hat{R}_z(a)\,\hat{X}\,\hat{R}_z(-a) = \cos a\,\hat{X} + \sin a\,\hat{Y}, \tag{16.3}$$

as it turns a vector in the x–y plane, which follows from $\hat{X}\hat{Z} = -i\hat{Y}$. With the product formulas for $\cos(\omega_L t - \phi_L)\cos\omega_L t$ and $\cos(\omega_L t - \phi_L)\sin\omega_L t$ the Hamiltonian in the rotating frame becomes

$$\hat{\mathcal{H}}_{\text{RF}}(t) = \frac{\omega_L - \omega_0}{2}\hat{Z} + \underbrace{\frac{\Omega}{2}\big(\cos\phi_L\,\hat{X} + \sin\phi_L\,\hat{Y}\big)}_{\text{co-rotating, constant}} + \underbrace{\frac{\Omega}{2}\big[\cos(2\omega_L t - \phi_L)\,\hat{X} + \sin(2\omega_L t - \phi_L)\,\hat{Y}\big]}_{\text{counter-rotating, turns at } 2\omega_L}. \tag{16.4}$$

The counter-rotating term turns at twice the optical frequency and averages to nearly zero over the much slower evolution caused by Ω and the detuning. Dropping it is the *rotating-wave approximation*, valid for $|\Omega|, |\Delta| \ll \omega_L$, and it leaves the constant Hamiltonian

$$\hat{\mathcal{H}}_{\text{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\big(\cos\phi_L\,\hat{X} + \sin\phi_L\,\hat{Y}\big), \qquad \Delta = \omega_L - \omega_0, \tag{16.5}$$

with the *detuning* Δ. The sign of the $\hat{Z}$ term follows from the slowed precession. In the rotating frame the free precession continues clockwise at $\omega_0 - \omega_L = -\Delta$, and a clockwise precession at this rate is generated by $-\frac{\omega_0-\omega_L}{2}\hat{Z} = +\frac{\Delta}{2}\hat{Z}$. On resonance, $\Delta = 0$, only the static drive remains, and the qubit turns about an axis in the x–y plane fixed by the laser phase.

Limits of the two-level description

The other levels of the atom are only nearly empty. Each far-detuned allowed transition shifts the energies of the qubit levels by an amount proportional to the laser intensity, the *light shift* (A.18) of Appendix A, so the qubit resonance moves when the laser is switched on. The counter-rotating term of (16.4) shifts the resonance as well, by the smaller Bloch–Siegert shift. Both effects are calibrated in an experiment and absorbed into the detuning.

The excited level also decays by spontaneous emission of a photon, an effect of the quantised light field that (16.1) does not contain. The 2p state of hydrogen lives only about 1.60 ns (A.19), far too short for a computation, and this decay is an energy relaxation of the qubit, treated in Section 12. Atomic and ionic qubits therefore use pairs of long-lived levels. One choice is two *hyperfine* levels of the ground state. The electron spin and the nuclear spin each carry a small magnetic moment, and their magnetic interaction splits the ground state into levels that differ in the relative orientation of the two spins, separated by a microwave frequency. Another choice is a *metastable* excited state, which cannot decay by a dipole transition and lives far longer. Microwaves, pairs of laser beams or narrow optical transitions drive these levels with the same Hamiltonian (16.5).

16.3 From the qubit Hamiltonian to gates

To turn the rotating-frame Hamiltonian into gates, we evaluate its evolution operator for pulses of chosen duration, amplitude and phase. The Hamiltonian (16.5) also describes the superconducting circuit and the spin of the following subsections, with a drive frequency ω_d in place of ω_L and a drive phase ϕ in place of ϕ_L, so we write it as

$$\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\left(\cos\phi\,\hat{X} + \sin\phi\,\hat{Y}\right), \qquad \Delta = \omega_d - \omega_0, \tag{16.6}$$

where ω_0 is the transition frequency of the qubit, Δ the detuning, Ω the Rabi frequency and ϕ the phase of the drive. Every result of this subsection uses only (16.6) and therefore holds for any qubit with this Hamiltonian.

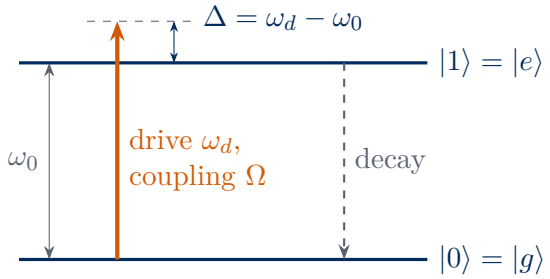


Figure 53: A two-level system with the gap ω_0 driven at the angular frequency ω_d, detuned by $\Delta = \omega_d - \omega_0$. The Rabi coupling Ω connects the ground state $|g\rangle = |0\rangle$ and the excited state $|e\rangle = |1\rangle$, and decay returns $|e\rangle$ to $|g\rangle$.

On resonance, $\Delta = 0$, and for $\phi = 0$ the Hamiltonian is $\frac{\Omega}{2}\hat{X}$, which generates the rotation (3.7) about x,

$$\hat{U}(t) = e^{-i\Omega t\hat{X}/2} = \cos\frac{\Omega t}{2}\,\hat{I} - i\sin\frac{\Omega t}{2}\,\hat{X} = \hat{R}_x(\theta), \qquad \theta = \Omega t. \tag{16.7}$$

Starting in the ground state, the probability of the excited state is

$$P_e(t) = |\langle e|\hat{R}_x(\Omega t)|g\rangle|^2 = \sin^2\frac{\Omega t}{2}, \tag{16.8}$$

the *Rabi oscillation*. A pulse with $\Omega t = \pi$ moves the system from $|g\rangle$ to $|e\rangle$, which is the gate $\hat{X}$ up to the phase $-i$. A pulse with $\Omega t = \pi/2$ produces the equal superposition $(|g\rangle - i|e\rangle)/\sqrt{2}$.

Off resonance and for $\phi = 0$ the Hamiltonian (16.6) is $\frac{W}{2}\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}$ with the *generalised Rabi frequency* $W = \sqrt{\Omega^2 + \Delta^2}$ and the unit vector $\mathbf{n} = (\Omega, 0, \Delta)/W$, an axis tilted from x towards z. By (3.7) it generates the rotation $\hat{R}_{\mathbf{n}}(Wt)$, and its matrix element $\langle 1|\hat{R}_{\mathbf{n}}(Wt)|0\rangle = -i\sin\frac{Wt}{2}\,(n_x + in_y)$ gives

$$P_e(t) = \frac{\Omega^2}{\Omega^2 + \Delta^2}\,\sin^2\frac{\sqrt{\Omega^2 + \Delta^2}\,t}{2}. \tag{16.9}$$

The oscillation is faster than on resonance and reaches the excited state at most with the probability $\Omega^2/(\Omega^2 + \Delta^2)$, because the Bloch vector turns on a cone about the tilted axis that does not reach the south pole. With the drive switched off, $\Omega = 0$, only the detuning remains, and $\frac{\Delta}{2}\hat{Z}$ generates $\hat{R}_z(\Delta t)$ (3.8), a precession about z with the angular frequency Δ in the rotating frame.

Rotation angles set by pulse area

[holds for any qubit] On resonance a drive with the phase ϕ and a slowly varying amplitude produces the Hamiltonian

$$\hat{\mathcal{H}}_{\mathrm{RF}}(t) = \frac{\Omega(t)}{2}\,(\cos\phi\,\hat{X} + \sin\phi\,\hat{Y}), \tag{16.10}$$

where the Rabi frequency $\Omega(t)$ follows the envelope of the field. This Hamiltonian changes in time only through the number $\Omega(t)$ and therefore commutes with itself at different times. The product of the evolutions over short time steps collapses to one exponential of the integrated coupling,

$$\hat{U} = \hat{R}_\phi(\theta) = e^{-i\theta(\cos\phi\,\hat{X}+\sin\phi\,\hat{Y})/2}, \qquad \theta = \int_0^T \Omega(t')\,dt', \tag{16.11}$$

for a pulse of duration T. The rotation angle θ is the *pulse area*, $\theta = \Omega T$ for a rectangular pulse of constant Ω, and (16.7) is the case $\phi = 0$. A smooth envelope, preferred in experiments because its narrower frequency spectrum excites fewer neighbouring transitions, produces the same gate whenever its area is the same. The result requires resonance and a constant phase. With a detuning the Hamiltonian also contains $\hat{Z}$ and no longer commutes with itself at different times, and the gate then depends on the shape of the pulse as well as on its area.

[holds for any qubit] By (16.3) the drive phase turns $\hat{X}$ into $\cos\phi\,\hat{X} + \sin\phi\,\hat{Y}$, and the exponential of a transformed matrix is the transformed exponential, so the drive phase selects the rotation axis in the x–y plane,

$$\hat{R}_\phi(\theta) = \hat{R}_z(\phi)\,\hat{R}_x(\theta)\,\hat{R}_z(-\phi). \tag{16.12}$$

Free evolution with a detuning produces rotations about z, $\theta = \Delta t$, but a rotation about z can also be applied without any time cost. From (16.12) follows $\hat{R}_\phi(\theta)\hat{R}_z(\beta) = \hat{R}_z(\beta)\hat{R}_{\phi-\beta}(\theta)$. A rotation by β about z followed by pulses with the phases $\phi_1, \phi_2, \ldots$ thus equals the same pulses with the phases $\phi_k - \beta$, followed by a final rotation about z. That last rotation changes only the relative phase of $|0\rangle$ and $|1\rangle$ and leaves the probabilities of a readout in the computational basis unchanged. The control electronics therefore apply a rotation about z by shifting the phase of every later pulse, an operation called a *virtual* $\hat{Z}$ gate. It takes no time, so the qubit does not decohere while it is applied, and it carries no calibration error of a pulse area.

Single-qubit gates as pulses

[holds for any qubit] Every gate of Table 2 is a short sequence of resonant pulses and virtual $\hat{Z}$ gates. Table 12 lists the action of each gate on the basis states and on the Bloch sphere together with its implementation, which follows from (16.11) and (16.12). A resonant pulse of area π and phase $\phi = 0$ produces $\hat{R}_x(\pi) = -i\hat{X}$, and the phase gates are rotations about z that cost no time. The Hadamard gate is the product $\hat{H} = \hat{R}_y(\pi/2)\,\hat{Z}$, so in the laboratory it is a virtual $\hat{Z}$ gate followed by a pulse of area $\pi/2$ with the phase $\phi = \pi/2$.

16.4 A superconducting circuit

In a superconducting metal the electrons bind into pairs that move without electrical resistance, so a circuit made of it loses almost no energy, and its oscillations keep their phase for a long time. The circuit behaves as a quantum system with discrete energy levels once it is cooled so far that thermal energy cannot excite it, $k_B T \ll h f_{01}$, with the Boltzmann constant k_B and the qubit frequency f_{01}. A frequency of 5 GHz corresponds to the temperature $h f_{01}/k_B \approx 0.24$ K, so such circuits are operated in dilution refrigerators at about ten millikelvin [11]. The spectrum is set by the design of the circuit, and in the transmon this design turns an electrical oscillator into a qubit [10, 11].

The LC oscillator

An inductance L connected to a capacitance C forms an LC circuit. Its state is described by the charge Q on the capacitor and the magnetic flux Φ through the inductor, and its energy is $Q^2/2C + \Phi^2/2L$.

gate	action on basis states	Bloch sphere	implementation
$\hat{X}$	$\lvert 0\rangle \leftrightarrow \lvert 1\rangle$, $\lvert \pm\rangle \to \pm\lvert \pm\rangle$	half turn about x	pulse of area π, $\phi = 0$
$\hat{Z}$	$\lvert 1\rangle \to -\lvert 1\rangle$, $\lvert +\rangle \leftrightarrow \lvert -\rangle$	half turn about z	virtual $\hat{Z}$ gate
$\hat{H}$	$\lvert 0\rangle \to \lvert +\rangle$, $\lvert 1\rangle \to \lvert -\rangle$, $\lvert \pm\rangle \to \lvert 0\rangle, \lvert 1\rangle$	half turn about $(\hat{x} + \hat{z})/\sqrt{2}$	virtual $\hat{Z}$, then area $\pi/2$ at $\phi = \pi/2$
$\hat{S}$	$\lvert 1\rangle \to i\lvert 1\rangle$, $\lvert +\rangle \to \lvert + i\rangle$	quarter turn about z	virtual $\hat{Z}$ gate
$\hat{T}$	$\lvert 1\rangle \to e^{i\pi/4}\lvert 1\rangle$	eighth of a turn about z	virtual $\hat{Z}$ gate
$\hat{R}_\phi(\theta)$	mixes $\lvert 0\rangle$ and $\lvert 1\rangle$ with amplitudes $\cos\frac{\theta}{2}$, $\sin\frac{\theta}{2}$	turn by θ about $(\cos\phi, \sin\phi, 0)$	pulse of area θ and phase ϕ
$\hat{R}_z(\theta)$	$\lvert 1\rangle \to e^{i\theta}\lvert 1\rangle$ up to a global phase	turn by θ about z	detuning for a time θ/Δ, or virtual $\hat{Z}$

Table 12: Action of the single-qubit gates on basis states and on the Bloch sphere, and their implementation with resonant pulses of phase ϕ and virtual $\hat{Z}$ gates. The table holds for any qubit with the Hamiltonian (16.6).

Quantisation turns the flux and the charge into operators with the commutator $[\hat{\Phi}, \hat{Q}] = i$, as for a position and a momentum, and

$$\hat{\mathcal{H}}_{LC} = \frac{\hat{Q}^2}{2C} + \frac{\hat{\Phi}^2}{2L}. \tag{16.13}$$

It is the harmonic oscillator of the box in Section 16.4 with the position $\hat{\Phi}$, the momentum $\hat{Q}$, the mass C and the angular frequency $\omega_r = 1/\sqrt{LC}$.

The quantum harmonic oscillator

A particle of mass m in the potential $\frac{1}{2}m\omega^2 x^2$ has the Hamiltonian $\hat{\mathcal{H}} = \hat{p}^2/2m + m\omega^2\hat{x}^2/2$, with $[\hat{x}, \hat{p}] = i$ for $\hbar = 1$. Its levels

$$E_k = \omega\Big(k + \frac{1}{2}\Big), \qquad k = 0, 1, 2, \ldots,$$

are equally spaced, and k counts the quanta of oscillation. The *lowering operator* $\hat{b} = \sqrt{m\omega/2}\,\hat{x} + i\hat{p}/\sqrt{2m\omega}$ and its adjoint, the raising operator $\hat{b}^\dagger$, act on the eigenstates $\lvert k\rangle$ as

$$\hat{b}\lvert k\rangle = \sqrt{k}\,\lvert k-1\rangle, \qquad \hat{b}^\dagger\lvert k\rangle = \sqrt{k+1}\,\lvert k+1\rangle, \qquad \hat{b}^\dagger\hat{b}\lvert k\rangle = k\lvert k\rangle, \qquad \hat{\mathcal{H}} = \omega\Big(\hat{b}^\dagger\hat{b} + \frac{1}{2}\Big).$$

The position is $\hat{x} = x_0(\hat{b} + \hat{b}^\dagger)$ with $x_0^2 = 1/(2m\omega)$, the variance of $\hat{x}$ in the ground state, so a coupling proportional to $\hat{x}$ connects only neighbouring levels k and $k \pm 1$. The same box describes the transmon with $\hat{b}$, and the shared motion of trapped ions in Section 17.3, where the lowering operator is written $\hat{a}$ as is customary for vibrations.

The levels of the LC circuit are equally spaced, as in the left part of Figure 54. A microwave drive tuned to the transition between $k = 0$ and $k = 1$ is therefore resonant with the transition from $k = 1$ to $k = 2$ and with every higher one, and it moves the circuit up the whole ladder. Since no pair of levels can be addressed alone, a harmonic oscillator does not provide a qubit.

The Josephson junction and the transmon

A Josephson junction consists of two superconductors separated by an insulating barrier about a nanometre thick. The current in a superconductor is carried by Cooper pairs, bound pairs of electrons with twice the electron charge, which tunnel through the barrier. All Cooper pairs of one superconductor share a single macroscopic wave function, and the complex phase of this wave function is the *superconducting phase*. The energy of the junction depends on the difference ϕ_J of the phases

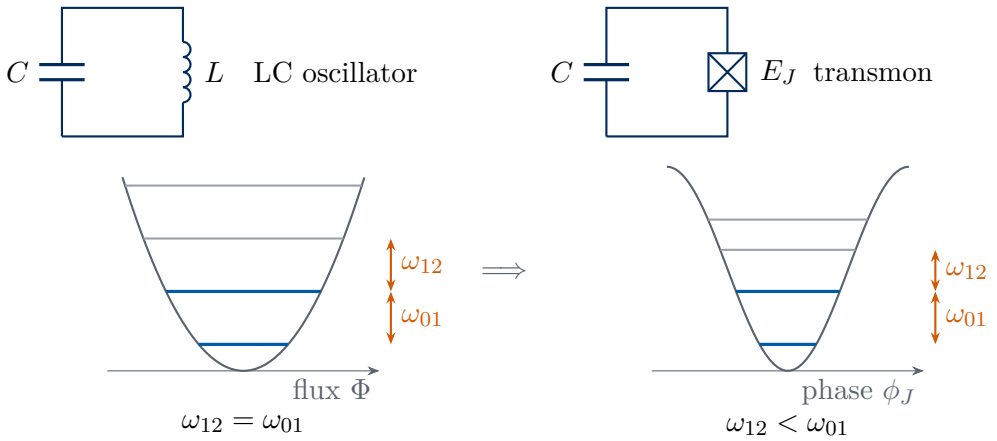


Figure 54: Left, an LC circuit (capacitance C, inductance L) with its parabolic potential in the flux Φ and equally spaced levels. A drive at ω_{01} is also resonant with ω_{12}. Right, the transmon, in which a Josephson junction (boxed cross) with the Josephson energy E_J replaces the inductor. Its cosine potential in the junction phase ϕ_J makes the spacing shrink with energy, so ω_{12} differs from ω_{01} by the anharmonicity α, and the two lowest levels (blue) form the qubit.

of the two superconductors [12] as $-E_J \cos\phi_J$, with a Josephson energy E_J fixed by the area and thickness of the barrier. The subscript J keeps this junction phase apart from the Bloch azimuth φ and from the drive phase ϕ. With $\hbar = 1$ the junction phase is related to the flux by $\hat\phi_J = 2e_0\hat\Phi$, with the elementary charge e_0, and the number of Cooper pairs that have passed through the junction is $\hat n = \hat Q/2e_0$, so that $[\hat\phi_J, \hat n] = [\hat\Phi, \hat Q] = i$. In these variables the capacitive term of (16.13) becomes $\hat Q^2/2C = 4E_C\hat n^2$, with the *charging energy* $E_C = e_0^2/2C$ of one electron on the total capacitance C. The *transmon* is a junction connected in parallel to a large capacitance [10], with the Hamiltonian

$$\hat{\mathcal{H}}_{\rm tr} = 4E_C(\hat n - n_g)^2 - E_J\cos\hat\phi_J, \qquad [\hat\phi_J, \hat n] = i, \tag{16.14}$$

where the offset charge n_g, in units of $2e_0$, is set by nearby electrodes and stray charges. The junction energy depends on $\hat\phi_J$ only through $\cos\hat\phi_J$, so the phase is an angle defined modulo 2π, and $\hat n$ has integer eigenvalues. Its eigenstates $|n\rangle$, with $n = 0, \pm1, \pm2, \ldots$, form a basis. The commutator gives $[\hat n, e^{i\hat\phi_J}] = e^{i\hat\phi_J}$, so $e^{i\hat\phi_J}|n\rangle = |n+1\rangle$ transfers one Cooper pair, and

$$\cos\hat\phi_J = \frac{e^{i\hat\phi_J} + e^{-i\hat\phi_J}}{2} = \frac12\sum_n \big(|n+1\rangle\langle n| + |n\rangle\langle n+1|\big). \tag{16.15}$$

In this basis (16.14) is a tridiagonal matrix with $4E_C(n-n_g)^2$ on the diagonal and $-E_J/2$ beside it, which we truncate at $|n| \le 20$ and diagonalise numerically.

The ratio E_J/E_C is a compromise. Stray charges that change n_g shift the levels by an amount that falls exponentially with $\sqrt{8E_J/E_C}$, whereas the anharmonicity derived below falls relative to the qubit frequency only as $(8E_J/E_C)^{-1/2}$ [10]. A large ratio thus suppresses the charge noise, and a too large ratio leaves the qubit too close to a harmonic oscillator for fast gates. Transmons use E_J/E_C of about 50 or more [10, 11], and for the parameters below changing n_g over its whole range shifts $\omega_{01}/2\pi$ by only about 10 kHz. For such a large ratio the phase stays close to the bottom of the cosine well and never reaches $\pm\pi$, so it behaves as an ordinary coordinate on a line. The unitary $e^{in_g\hat\phi_J}$ shifts $\hat n$ by n_g and removes the offset charge from the Hamiltonian, and with $\cos\phi_J \approx 1 - \phi_J^2/2 + \phi_J^4/24$ we expand

$$\hat{\mathcal{H}}_{\rm tr} \approx -E_J + \underbrace{4E_C\hat n^2 + \frac{E_J}{2}\hat\phi_J^2}_{\text{harmonic oscillator}} \underbrace{- \frac{E_J}{24}\hat\phi_J^4}_{\text{anharmonic}}. \tag{16.16}$$

The harmonic part is the oscillator of the box in Section 16.4 with $\hat x = \hat\phi_J$, $\hat p = \hat n$, the mass $m = 1/8E_C$ and the angular frequency $\omega_p = \sqrt{8E_JE_C}$. Its lowering operator $\hat b$ gives $\hat\phi_J = \phi_0(\hat b + \hat b^\dagger)$ with $\phi_0^2 = 1/(2m\omega_p) = 4E_C/\sqrt{8E_JE_C} = \sqrt{2E_C/E_J}$, the spread of the phase in the ground state,

which is small for $E_J \gg E_C$. First-order perturbation theory adds to the level k the mean of the quartic term in the state $|k\rangle$. Of the sixteen products in $(\hat b + \hat b^\dagger)^4$ only the six with two factors $\hat b$ and two factors $\hat b^\dagger$ return $|k\rangle$ to itself, and with the rules of the box their values add up to $\langle k|(\hat b+\hat b^\dagger)^4|k\rangle = (k+1)(k+2)+(k+1)^2+2k(k+1)+k^2+k(k-1) = 6k^2+6k+3$. With $E_J\phi_0^4 = 2E_C$ the levels are

$$E_k \approx -E_J + \omega_p\Big(k+\frac{1}{2}\Big) - \frac{E_J\phi_0^4}{24}\,(6k^2+6k+3) = -E_J + \omega_p\Big(k+\frac{1}{2}\Big) - \frac{E_C}{12}\,(6k^2+6k+3). \quad (16.17)$$

The transition frequencies follow by subtraction,

$$\omega_{01} = E_1 - E_0 \approx \sqrt{8E_JE_C} - E_C, \qquad \omega_{12} = E_2 - E_1 \approx \sqrt{8E_JE_C} - 2E_C, \qquad \alpha = \omega_{12} - \omega_{01} \approx -E_C. \quad (16.18)$$

The *anharmonicity* α is the difference between the second and the first transition frequency. It is negative because the cosine potential rises more slowly than the parabola, and it is small compared with ω_{01}. Typical transmons have $\omega_{01}/2\pi$ between 3 and 6 GHz and $|\alpha|/2\pi$ between 100 and 300 MHz [11]. The numerical diagonalisation in Figure 55 confirms (16.18) for the qubit frequency, while the first-order value $-E_C$ underestimates $|\alpha|$ by about fifteen per cent because of higher orders in ϕ_0.

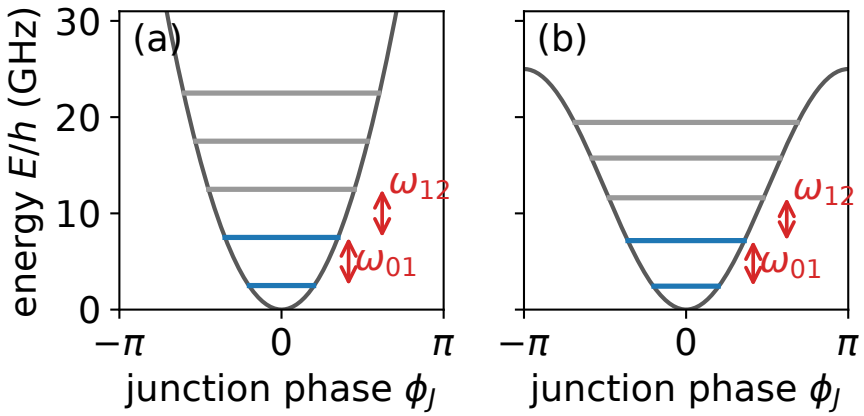


Figure 55: Levels against the junction phase ϕ_J, drawn at their energies above the bottom of the potential, for $E_C/h = 0.25$ GHz and $E_J = 50\,E_C$. (a) The quadratic approximation of the junction energy, an LC oscillator with equally spaced levels. (b) The cosine well of the transmon with the levels obtained by diagonalising (16.14) in the charge basis, which give $\omega_{01}/2\pi = 4.74$ GHz, close to the value 4.75 GHz of (16.18), and $\alpha/2\pi = -287$ MHz, compared with $-E_C/h = -250$ MHz to first order. The two qubit levels are blue. Since energies are plotted as E/h, the arrows have the lengths $\omega_{01}/2\pi$ and $\omega_{12}/2\pi$.

Microwave drive and leakage

A microwave line coupled to the circuit through a small capacitance C_d applies the voltage $V(t) = V_0\cos(\omega_d t-\phi)$ and adds the energy $2e_0\frac{C_d}{C}V(t)\,\hat n$ [11], so the drive couples through the charge operator. We choose the phases of the eigenstates so that the matrix elements $n_{k,k+1} = \langle k|\hat n|k+1\rangle$ are real and positive. At $n_g = 0$ the well is symmetric, its eigenstates have definite parity, and the diagonal elements $\langle k|\hat n|k\rangle$ vanish. Restricted to the two lowest levels, the charge operator therefore becomes $\hat n \to n_{01}\hat X$, and with the zero of energy halfway between the two levels

$$\hat{\mathcal{H}}(t) = -\frac{\omega_{01}}{2}\hat Z + \Omega\cos(\omega_d t-\phi)\,\hat X, \qquad \Omega = 2e_0\frac{C_d}{C}V_0\,n_{01}. \quad (16.19)$$

It has the form (16.1) of the two-level atom in a laser field, with ω_{01} in place of ω_0. The rotating frame and the rotating-wave approximation therefore give (16.6) with $\Delta = \omega_d - \omega_{01}$, and a resonant pulse applies the rotation $\hat R_\phi(\theta)$ of (16.11).

The same drive couples $|1\rangle$ to $|2\rangle$ with the larger matrix element $n_{12} = 1.37\, n_{01}$, close to the value $\sqrt{2}\, n_{01}$ of an oscillator, and this transition is detuned from the drive only by α. In the eigenbasis $\hat{n}$ connects mainly neighbouring levels, as $\hat{b} + \hat{b}^\dagger$ does for the oscillator. For several levels the rotating frame multiplies the amplitude of $|k\rangle$ by $e^{ik\omega_d t}$, the transformation $e^{i\omega_d t \hat{N}}$ with $\hat{N} = \sum_k k|k\rangle\langle k|$, which reduces to (16.2) on two levels up to a global phase. It subtracts $k\omega_d$ from the energy of $|k\rangle$, so in the frame rotating at $\omega_d = \omega_{01}$ the level $|k\rangle$ has the energy $\delta_k = E_k - E_0 - k\,\omega_{01}$, with $\delta_0 = \delta_1 = 0$ and $\delta_2 = \alpha$. The rotating-wave approximation drops the counter-rotating parts of the neighbour couplings,

$$\hat{\mathcal{H}}_{\mathrm{RF}}(t) = \sum_k \delta_k |k\rangle\langle k| + \frac{\Omega(t)}{2} \sum_k \frac{n_{k,k+1}}{n_{01}} \left(|k\rangle\langle k+1| + |k+1\rangle\langle k| \right), \tag{16.20}$$

which reduces to $\frac{\Omega(t)}{2}\hat{X}$ on the two qubit levels. We apply Gaussian pulses of area π and duration T to the five lowest levels. The envelope has the standard deviation $\sigma = T/4$ and is shifted to vanish at $t = 0$ and $t = T$. Its spectrum extends over angular frequencies of about $1/\sigma = 4/T$ around ω_d, and the transition $1 \to 2$ stays unexcited once $\sigma|\alpha| \gtrsim 3$, that is, for

$$T \gtrsim \frac{12}{|\alpha|} \approx 6.6 \text{ ns} \tag{16.21}$$

for the transmon of Figure 55.

Figure 56 confirms this border. A pulse of 4 ns, below the border, leaves about a quarter of the population in $|2\rangle$ and higher levels, the *leakage*, whereas the final leakage falls below 10^{-3} from about 8 ns. Beyond the border it does not fall monotonically, because population that entered $|2\rangle$ during the pulse can return to the qubit, and the spectral overlap of the envelope with the transition $1 \to 2$ oscillates with T. Even a long pulse leaves P_1 slightly below one, because the coupling to $|2\rangle$ also shifts the energy of $|1\rangle$ and detunes the drive. DRAG pulses (derivative removal by adiabatic gate) [13] correct both effects with a *quadrature* component, a second drive component shifted by 90° in phase, which acts along $\hat{Y}$ in the rotating frame and is proportional to the time derivative of the envelope.

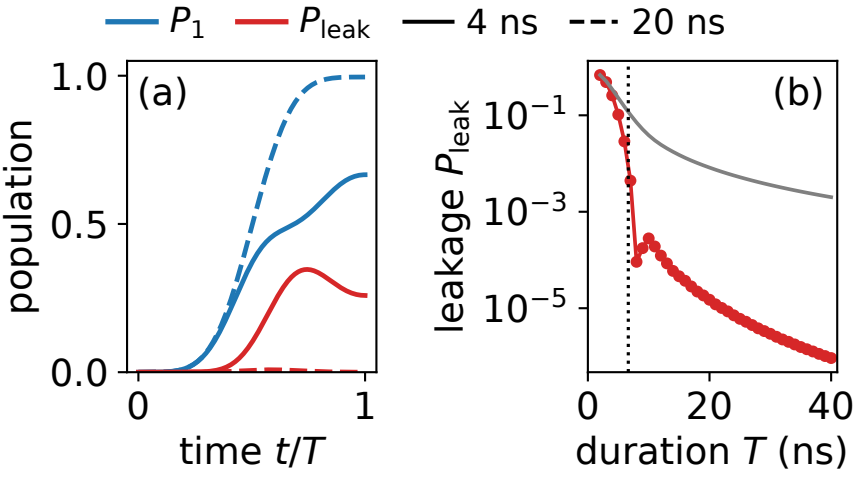


Figure 56: Gaussian π pulses of duration T (standard deviation $T/4$, shifted to vanish at the ends) on the five lowest transmon levels, evolved with (16.20) for the parameters of Figure 55. (a) Population P_1 of $|1\rangle$ and leakage P_{leak}, the total population of $|2\rangle$ and higher levels, during pulses of 4 ns (solid) and 20 ns (dashed). The short pulse ends with $P_{\mathrm{leak}} = 0.26$, the long one with $P_{\mathrm{leak}} = 1.5 \times 10^{-5}$ and $P_1 = 0.9956$. (b) Leakage at the end of the pulse (dots) and its largest value during the pulse (grey line) against the duration. The dotted vertical line marks the border $T = 12/|\alpha|$ of (16.21).

The transmon is read out through a microwave resonator, an electrical circuit with a single oscillation mode, which is coupled to the transmon and detuned from its transition. The resonator frequency is shifted by an amount that depends on whether the transmon is in $|0\rangle$ or $|1\rangle$. A weak microwave tone sent through the resonator returns with a phase that reveals the state of the qubit. This measurement of $\hat{Z}$ is called dispersive readout [14].

16.5 A spin-1/2 in a magnetic field

An electron carries an intrinsic angular momentum, its spin, whose projection on any axis takes only two values, up and down. Such a two-valued spin is called a spin-1/2, and the proton and several other nuclei carry one as well. With $\hbar = 1$ the spin operators are $\hat{\mathbf{S}} = \hat{\boldsymbol{\sigma}}/2$, so a spin is a two-level system from the start and needs no truncation of a larger spectrum. It carries the magnetic moment $\hat{\boldsymbol{\mu}} = \gamma\hat{\mathbf{S}}$, with the *gyromagnetic ratio* γ of the particle, and in a magnetic field $\mathbf{B}$ it has the energy

$$\hat{\mathcal{H}} = -\hat{\boldsymbol{\mu}}\cdot\mathbf{B} = -\gamma\,\mathbf{B}\cdot\hat{\mathbf{S}} = -\frac{\gamma}{2}(B_x\hat{X} + B_y\hat{Y} + B_z\hat{Z}). \tag{16.22}$$

For a static field along z the Hamiltonian is $\hat{\mathcal{H}} = -\frac{\omega_0}{2}\hat{Z}$ with the *Larmor frequency* $\omega_0 = \gamma B_z$, and the two levels are split by ω_0, the *Zeeman splitting*. We orient the z axis so that $\omega_0 > 0$. The state $|0\rangle = |\uparrow\rangle$ is then the ground state, and the spin has the same Hamiltonian as the free two-level atom. The electron has $|\gamma|/2\pi \approx 28$ GHz per tesla and the proton 42.6 MHz per tesla [15], so a field of a fraction of a tesla gives microwave frequencies for an electron and radio frequencies for a nucleus. The electron has $\gamma < 0$, so for $\omega_0 > 0$ its field points along $-z$, and in the ground state its spin points against the field while its magnetic moment points along it.

For a spin the Bloch vector has the components $\langle\hat{\sigma}_a\rangle = 2\langle\hat{S}_a\rangle$, so it points along the mean spin, and spin up along z sits at the north pole. [holds for any qubit] The mean of a time-independent observable $\hat{O}$ changes because the state changes. With $i\frac{d}{dt}|\psi\rangle = \hat{\mathcal{H}}|\psi\rangle$ and its adjoint $-i\frac{d}{dt}\langle\psi| = \langle\psi|\hat{\mathcal{H}}$,

$$\frac{d}{dt}\langle\hat{O}\rangle = \Big(\frac{d}{dt}\langle\psi|\Big)\hat{O}|\psi\rangle + \langle\psi|\hat{O}\Big(\frac{d}{dt}|\psi\rangle\Big) = i\langle\psi|\hat{\mathcal{H}}\hat{O}|\psi\rangle - i\langle\psi|\hat{O}\hat{\mathcal{H}}|\psi\rangle = i\langle[\hat{\mathcal{H}},\hat{O}]\rangle. \tag{16.23}$$

With the commutators $[\hat{S}_a,\hat{S}_b] = i\varepsilon_{abc}\hat{S}_c$, which follow from the Pauli algebra (2.12), the Hamiltonian (16.22) gives

$$\frac{d}{dt}\langle\hat{\mathbf{S}}\rangle = \gamma\,\langle\hat{\mathbf{S}}\rangle\times\mathbf{B}, \tag{16.24}$$

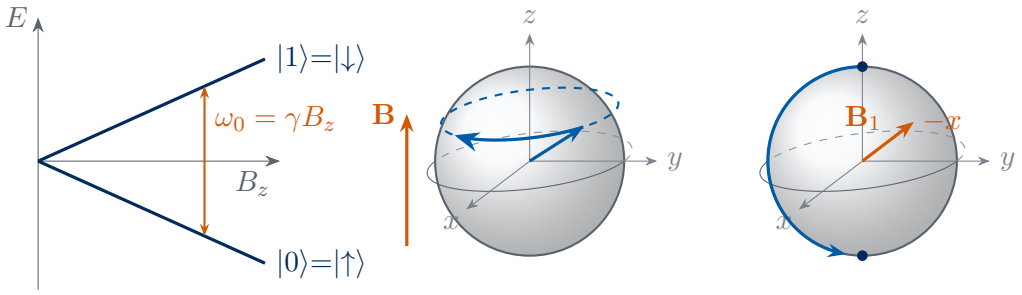
the equation of motion of a magnetic moment turning about a field. In a static field along z the evolution $\hat{U}(t) = e^{i\omega_0 t\hat{Z}/2} = \hat{R}_z(-\omega_0 t)$ turns the Bloch vector about z with the angular frequency ω_0, a motion called *Larmor precession* that runs clockwise seen from the north pole, as in Figure 57(b). It turns in the opposite sense to the precession generated by $+\frac{\omega}{2}\hat{Z}$ in Section 3, because the free spin has the Hamiltonian $-\frac{\omega_0}{2}\hat{Z}$.

Since a static field produces only rotations about its own direction, transitions between the two levels need a transverse field. A field $B_1\cos(\omega_d t)$ along x, oscillating at a drive frequency ω_d close to ω_0, adds to (16.22) and gives

$$\hat{\mathcal{H}}(t) = -\frac{\omega_0}{2}\hat{Z} + \Omega\cos(\omega_d t)\,\hat{X}, \qquad \Omega = -\frac{\gamma B_1}{2}, \tag{16.25}$$

again the Hamiltonian (16.1) of the two-level atom in a laser field. We choose the sign of B_1 so that $\Omega > 0$, and the opposite sign only reverses the sense of the rotation. For $\gamma > 0$ this means $B_1 < 0$, a field oscillating along $-x$. In the frame rotating at ω_d and in the rotating-wave approximation the spin obeys (16.6) with $\phi = 0$ and $\Delta = \omega_d - \omega_0$, and on resonance a pulse of duration t applies $\hat{R}_x(\Omega t)$. As for the atom, the oscillating field is a sum of two fields of amplitude $|B_1|/2$ rotating about z in opposite directions, and the rotating frame turns together with one of them. On resonance the static field disappears in this frame, one rotating component becomes a static field along the $-x$ axis, and the other rotates at $2\omega_d$ and averages out. The spin precesses about the remaining field, $\hat{R}_x(\Omega t)$ carries the Bloch vector from the north pole through $-y$ to the south pole, and the spin flips from up to down with the probability $\sin^2(\Omega t/2)$. This effect is *magnetic resonance* [16], called electron spin resonance for electrons and nuclear magnetic resonance for nuclei. Figure 58 compares the laboratory-frame motion with the rotating-frame rotation.

An electron spin in a semiconductor quantum dot, a nanometre-sized region that holds a single electron, was proposed as a qubit in Ref. [9] and is read out by converting spin into charge. The levels



(a) Zeeman splitting (b) precession, $\hat{R}_z(-\omega_0 t)$ (c) drive, $\hat{R}_x(\Omega t)$

Figure 57: A spin-1/2 in a magnetic field, drawn for $\gamma > 0$ and $B_z > 0$. (a) The energies of $|0\rangle = |\uparrow\rangle$ and $|1\rangle = |\downarrow\rangle$ split linearly with the field B_z, by the Larmor frequency $\omega_0 = \gamma B_z$. (b) In a static field the Bloch vector precesses clockwise on a cone about $\mathbf{B}$. (c) In the frame rotating with a resonant transverse field oriented so that $\Omega > 0$, the field $\mathbf{B}_1$ appears static along $-x$, and the spin turns from the north pole through $-y$ to the south pole, the rotation $\hat{R}_x(\Omega t)$.

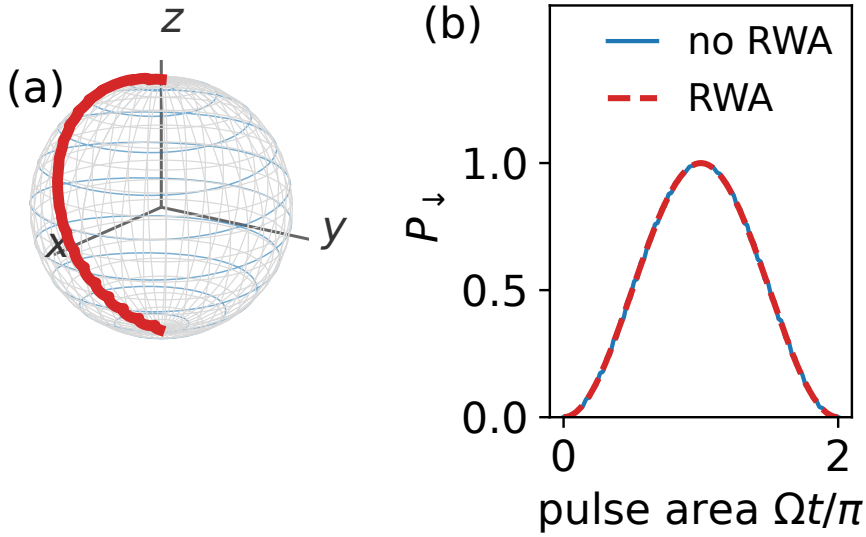


Figure 58: Resonant magnetic drive (16.25) with $\omega_0 = 20\,\Omega$ and $\Omega > 0$, integrated numerically in the laboratory frame without the rotating-wave approximation. (a) Bloch vector during a π pulse. In the laboratory frame (thin blue line) it precesses fast about z and slowly descends to the south pole, and in the rotating frame (thick red line) it makes a half turn about x through $-y$. (b) Probability $P_\downarrow$ of spin down against the pulse area, from the laboratory-frame evolution (no RWA) and from the rotating-frame prediction $\sin^2(\Omega t/2)$ (RWA). The two curves differ by at most 0.013.

of the dot are tuned so that only an electron with spin down can tunnel out to a nearby reservoir. A charge sensor then records whether the electron has left, which is a single-shot measurement of $\hat{Z}$ [17]. In nuclear magnetic resonance a large ensemble of nuclear spins precesses after a pulse, and its oscillating magnetisation induces a voltage in a pick-up coil [18]. This signal measures the ensemble mean of $\hat{X}$ or $\hat{Y}$.

16.6 Photon polarisation

A photon carries a qubit in its polarisation, with $|0\rangle = |H\rangle$ for horizontal and $|1\rangle = |V\rangle$ for vertical polarisation. The two basis states have the same energy, so there is no splitting, no free precession and no resonant drive, and a photon has no drive Hamiltonian of the form (16.6) that could be switched on for a chosen time. Gates are applied instead by passive optical elements. A wave plate is a slab of a crystal whose refractive index depends on the polarisation. It delays one polarisation component relative to the other while the photon passes through it and thereby applies a fixed rotation of the Bloch sphere, set by its thickness and orientation, which play the roles of the pulse area and the

drive phase. A polarising beam splitter transmits H and reflects V towards two detectors and thus measures in the computational basis. A wave plate placed in front of it selects another measurement basis. Photons travel long distances with little disturbance, but two photons barely interact. A two-qubit gate, which acts on two qubits at once and is introduced in Section 7, therefore requires additional photons and measurements [19].

16.7 Comparison of platforms

In the atom, the transmon and the spin, a field oscillating near the qubit transition couples the two levels with a Rabi frequency proportional to the field amplitude and to a matrix element between them, as summarised in Figure 59. The platforms of Figure 51 differ in how the two levels are obtained, driven, coupled and read out, as listed in Table 13. The coupling column names the interaction between two qubits that produces the two-qubit gates of Section 7.

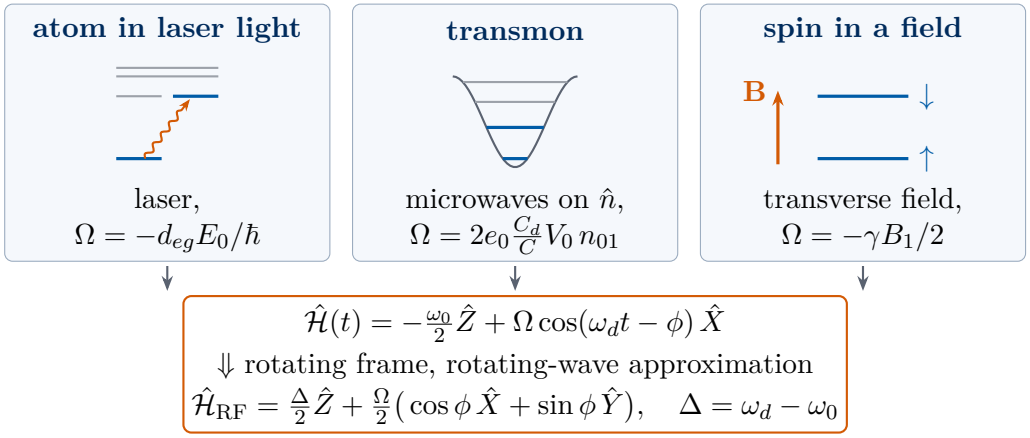


Figure 59: Three systems with the same qubit Hamiltonian, the spin drawn for $\gamma > 0$. In the atom a resonant laser selects two levels, in the transmon the anharmonicity makes two levels addressable, and the spin has only two levels. The drive frequency ω_d and phase ϕ equal ω_L and ϕ_L for the laser, and ω_0 stands for the qubit transition frequency.

platform	the two levels	driven by	coupled by	read out by
trapped ions	two electronic or hyperfine levels of an ion	lasers or microwaves	shared vibrations of the ion chain	laser-induced fluorescence
neutral atoms in optical tweezers	two levels of an atom	lasers	excitation to strongly interacting Rydberg states	fluorescence imaging
superconducting circuits (transmon)	two lowest levels of an anharmonic electrical oscillator	microwave pulses	capacitors or tunable couplers	microwave resonator
spins in semiconductor quantum dots	spin up and down of an electron	microwaves or electric fields	exchange interaction	charge sensing
photons	horizontal and vertical polarisation	wave plates	interference and measurement	photon detectors

Table 13: Qualitative overview of physical platforms for qubits. The same mathematical description applies to all of them. Hyperfine levels are ground-state levels split by the magnetic interaction of the electron spin with the nuclear spin. Optical tweezers are tightly focused laser beams that each hold one atom. Rydberg states are highly excited atomic states with a large orbit, and a resonator is an electrical circuit or cavity that stores one oscillation mode.

The platforms also differ in how long a superposition survives. Atomic and ionic qubits stored in hyperfine or metastable levels (Section 16.2) keep their superpositions for a long time, and since atoms and ions of one isotope are identical, every qubit of a register has the same transition frequency. Superconducting circuits allow fast gates, with durations set by the anharmonicity as in (16.21). Because the circuits are fabricated, every transmon has a slightly different frequency and is calibrated individually, and its superpositions decay sooner than those of the best atomic levels. The interaction between qubits fixes which two-qubit gates a device applies directly, its *native gates*, and which pairs of qubits can be coupled, its *connectivity*. Circuits written for a particular device are built from these native gates.

17 Interacting qubits and two-qubit gates in the laboratory

The two-qubit gates of Section 7 are evolutions under couplings of the form $\frac{J}{2}\hat{\sigma}_\alpha \otimes \hat{\sigma}_\alpha$ and their relatives, switched on for a chosen time. In a device these couplings come from physical interactions between the systems that carry the qubits, and each platform provides some of them directly and the others only after compilation.

17.1 Physical origin of the couplings

The coupling comes from different physics on each platform. Table 14 lists the mechanism, the effective coupling written with Pauli operators and the native gate of the platform (Section 16.7), with the gates $\widehat{\mathrm{iSWAP}}$ and $\sqrt{\widehat{\mathrm{SWAP}}}$ of Section 7.2. Rydberg states have a large principal quantum number n, energies $-\mathrm{Ry}/n^2$ of (A.14) close to the ionisation threshold and large electron orbits. Neutral atoms excited to Rydberg states repel each other with a van der Waals energy V that falls as the inverse sixth power of their distance [5]. Trapped ions share their vibrational modes through the Coulomb repulsion, and a laser with two frequency components converts the shared motion into an effective $\hat{X}\hat{X}$ coupling [6,7]. Superconducting circuits coupled by a capacitor or by a resonator, a circuit that stores one electrical oscillation mode, exchange an excitation. A microwave drive of one qubit at the frequency of the other produces an effective coupling of the form $\hat{Z}\hat{X}$, the cross-resonance coupling, which we quote from Ref. [8] without derivation. Electron spins in neighbouring quantum dots feel the exchange interaction, whose strength $J(t)$ is set by the overlap of the electron orbitals and controlled by gate voltages [9].

platform	physical mechanism	effective coupling	native gate
neutral atoms	van der Waals interaction of two Rydberg atoms	$V\lvert rr\rangle\langle rr\rvert$, blockade of double excitation	$\widehat{\mathrm{CZ}}$ by a laser-pulse sequence that uses the blockade
trapped ions	Coulomb repulsion and shared vibrations	$\hat{X}\hat{X}$	$\hat{R}_{xx}(\theta)$
superconducting circuits	capacitive or resonator coupling	$\hat{X}\hat{X}+\hat{Y}\hat{Y}$, or $\hat{Z}\hat{X}$ with a cross-resonance drive	$\widehat{\mathrm{iSWAP}}$ or its inverse, or CNOT up to one-qubit gates
quantum-dot spins	exchange interaction	$J_{\mathrm{ex}}(t)\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1$	$\sqrt{\widehat{\mathrm{SWAP}}}$

Table 14: Coupling mechanisms of four platforms, their effective two-qubit couplings in Pauli form and the gates they produce most directly.

17.2 Interactions in Pauli language

Writing the projectors on the two levels with Pauli operators, $|0\rangle\langle 0| = (\hat{I}+\hat{Z})/2$ and $|1\rangle\langle 1| = (\hat{I}-\hat{Z})/2$, turns each physical interaction into a combination of Pauli strings. When atoms are used to simulate an Ising model, each atom stores its qubit in the ground state $|g\rangle = |0\rangle$ and a Rydberg state $|r\rangle = |1\rangle$, the *ground–Rydberg encoding*. The interaction acts only on $|rr\rangle = |11\rangle$, and with one projector per atom it becomes

$$V|11\rangle\langle 11| = V\,\frac{\hat{I}-\hat{Z}_0}{2}\,\frac{\hat{I}-\hat{Z}_1}{2} = \frac{V}{4}(\hat{I}-\hat{Z}_0-\hat{Z}_1+\hat{Z}_0\hat{Z}_1). \tag{17.1}$$

The Rydberg interaction is an Ising coupling $\frac{V}{4}\hat{Z}_0\hat{Z}_1$ plus a shift $-\frac{V}{4}\hat{Z}$ of each atom, which acts as a detuning, and a constant. In the same way the exchange of one excitation between two superconducting circuits is the coupling $\frac{J}{2}(\hat{X}_0\hat{X}_1+\hat{Y}_0\hat{Y}_1)$, and the exchange interaction of two quantum-dot spins is the Heisenberg coupling $J_{\mathrm{ex}}\,\hat{\mathbf{S}}_0\cdot\hat{\mathbf{S}}_1 = \frac{J_{\mathrm{ex}}}{4}(\hat{X}_0\hat{X}_1+\hat{Y}_0\hat{Y}_1+\hat{Z}_0\hat{Z}_1)$, both introduced in Section 7.2.

17.3 The Mølmer–Sørensen gate of two trapped ions

Trapped ions are coupled through their shared motion, and the Mølmer–Sørensen scheme [6,7] uses this motion in a way that works also for ions that are not cooled to their motional ground state. It produces an $\hat{X}_0\hat{X}_1$ coupling, and a drive of duration t applies the rotation $\hat{R}_{xx}$ of Section 7.2, which at an angle of magnitude $\pi/2$ turns $|00\rangle$ into a Bell state (Section 8).

Two ions in one trap repel each other, and their joint oscillation about the trap centre is a *motional mode*, a harmonic oscillator of frequency ν shared by both ions, described by the box in Section 16.4 with the lowering operator $\hat{a}$. Its quanta are called *phonons*. A laser tuned to the qubit frequency plus or minus ν flips a qubit and adds or removes one phonon in the same step. The two transitions are the blue and the red *sideband*. Their strength is reduced by the *Lamb–Dicke parameter* η, the spread of the ground-state motion of an ion divided by the reduced wavelength of the light, typically $\eta \approx 0.1$. The gate drives both ions with two tones detuned by $\pm(\nu-\delta)$ from the qubit frequency, close to both sidebands and off them by a small δ. To first order in η and after the rotating-wave approximation the two tones exert a force on the motion whose sign depends on the qubits,

$$\hat{\mathcal{H}}(t) = g\,\hat{S}_x(\hat{a}\,e^{-i\delta t}+\hat{a}^\dagger e^{i\delta t}), \qquad \hat{S}_x = \tfrac{1}{2}(\hat{X}_0+\hat{X}_1), \qquad g = \eta\Omega, \tag{17.2}$$

with the Rabi frequency Ω of each tone. The collective spin $\hat{S}_x$ has the eigenvalues $m = 1, 0, -1$, and for each of them the force pushes the motion around a circle in phase space, the plane of mean position and mean momentum,

$$\langle\hat{a}\rangle(t) = \langle\hat{a}\rangle(0) + m\,\alpha(t), \qquad \alpha(t) = \frac{g}{\delta}(1-e^{i\delta t}). \tag{17.3}$$

The circle closes at the times $t = 2\pi K/\delta$ for every integer K. The motion is then back in its initial state, whatever that state was, and the qubits are no longer entangled with it. The loop leaves only a phase that grows as m^2, which acts on the qubits as the effective coupling $-\frac{J}{2}\hat{X}_0\hat{X}_1$ with $J = g^2/\delta$. Appendix B derives the result after K loops,

$$\hat{U}\Big(\frac{2\pi K}{\delta}\Big) = e^{\,i\Phi\hat{S}_x^2} = e^{i\Phi/2}\,\hat{R}_{xx}(-\Phi), \qquad \Phi = \frac{2\pi K g^2}{\delta^2} = Jt, \tag{17.4}$$

in which the angle grows with the gate time as in (7.14). For $\delta > 0$ the gate rotates by $-\Phi$, the opposite sense to the rotation $\hat{R}_{xx}(+\pi/2)$ used in Part I, and the opposite sign of δ reverses it. One loop with $\delta = 2g$ yields $\Phi = \pi/2$, the maximally entangling gate that takes $|00\rangle$ to $(|00\rangle + i|11\rangle)/\sqrt{2}$.

The simulation of Figure 60 confirms that the gate does not depend on the motional state. It reaches the target state with the same accuracy for a thermal mode as for a cold one, as long as the loop closes. A gate time that misses the closing time leaves the motion displaced and entangled with the qubits, and this error is larger for thermal ions, whose motion is spread over more phonon states.

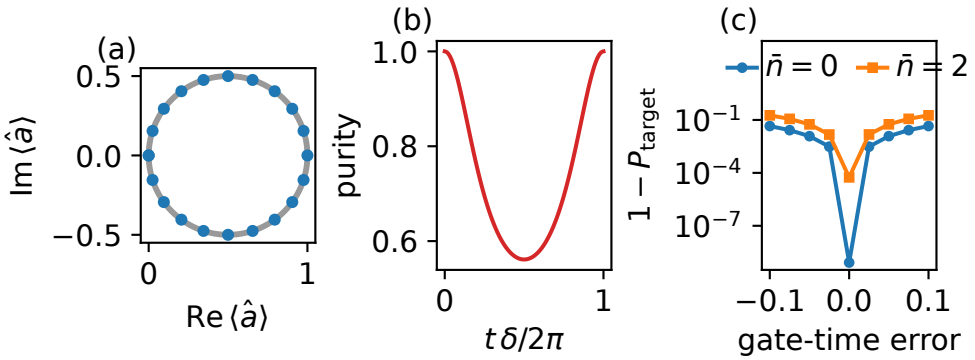


Figure 60: The Mølmer–Sørensen gate simulated with two qubits and one motional mode. (a) The mean motional amplitude for the qubits in $|{+}{+}\rangle$, the eigenstate of $\hat{S}_x$ with $m = 1$ (dots), follows the circle (17.3) (line) and returns to the origin. (b) The purity $\mathrm{Tr}\hat{\rho}^2$ of the two-qubit state starting from $|00\rangle$ drops while the qubits are entangled with the motion, the lower the stronger the entanglement, and returns to one at the gate time. (c) The error $1 - P_{\text{target}}$, with $P_{\text{target}} = |\langle\psi_{\text{target}}|\psi\rangle|^2$ the probability of finding the qubits in $(|00\rangle + i|11\rangle)/\sqrt{2}$, when the gate time misses the closing time by the given relative error, for cold ($\bar{n} = 0$) and thermal ($\bar{n} = 2$) ions. At the closing time $P_{\text{target}} = 0.9999$ also for $\bar{n} = 2$, and a gate time off by ten per cent gives the error 0.045 for $\bar{n} = 0$ and 0.182 for $\bar{n} = 2$.

17.4 Native gates from Ising and exchange couplings

The gates generated by the Ising, exchange and excitation-exchange couplings are those of Section 7.2. New here is which device supplies the coupling. The conditional rotation $\hat{R}_{zz}(\theta)$ is the Ising coupling $\frac{J}{2}\hat{Z}_0\hat{Z}_1$ switched on for the time $t = \theta/J$, provided by the Rydberg interaction in the form (17.1) or by a tunable coupler between superconducting circuits. A device applies CZ by switching the Ising coupling on for the calibrated time $t = \pi/(2J)$ and adding the two z rotations of (7.10) as virtual $\hat{Z}$ gates (Section 16.3). CNOT is then applied as CZ between two Hadamard gates on the target. Superconducting circuits also produce CNOT from the cross-resonance coupling $\hat{Z}\hat{X}$ switched on for a calibrated time, followed by single-qubit corrections [8]. Quantum-dot spins obtain $\sqrt{\widehat{\text{SWAP}}}$ from an exchange pulse of area $\int J_{\text{ex}}\,dt = \pi/2$ [9], and superconducting circuits obtain $\widehat{\text{iSWAP}}$ or its inverse from the excitation exchange $\frac{J}{2}(\hat{X}_0\hat{X}_1 + \hat{Y}_0\hat{Y}_1)$ at $|J|t = \pi/2$.

17.5 Gate parameters set by physical processes

Every angle of a parametrised gate is set by a physical process in the device. For a single-qubit rotation the angle is the pulse area (16.11), and for a two-qubit rotation it is the integrated coupling (7.14). Experiments control $J(t)$ with a tunable coupler between superconducting qubits or with the strength and duration of the laser drive that produces the ion coupling. Between quantum-dot spins the gate voltage sets the exchange. For a fixed gate the duration that gives the required angle is calibrated once and reused, and its local phases are added as virtual $\hat{Z}$ gates. Table 15 connects each gate of these notes with the process that sets its parameter.

Every gate of a circuit is thus a Hamiltonian of the device switched on for a time. A circuit of K steps is a pulse sequence of Section 16.3 acting on the whole register,

$$\hat{U} = e^{-it_K\hat{\mathcal{H}}_K}\cdots e^{-it_2\hat{\mathcal{H}}_2}\,e^{-it_1\hat{\mathcal{H}}_1}, \tag{17.5}$$

where each $\hat{\mathcal{H}}_k$ is a drive of one qubit or a coupling of a pair. Gates on different qubits act simultaneously within one layer, and the circuit lasts for the sum of its layer durations. On many platforms the couplings between qubits are weaker than the drives, so two-qubit gates take longer than single-qubit gates. Each entangling gate of Table 15 is universal together with single-qubit rotations (Section 7.3), so different platforms run the same circuits once each gate is compiled into their native operations.

gate	parameter	physical process	control setting
$\hat{R}_\phi(\theta)$	$\theta = \int \Omega\, dt$, axis angle ϕ	resonant laser or microwave pulse	pulse duration or amplitude, and phase
$\hat{R}_z(\theta)$	$\theta = \Delta t$, or a phase shift	detuning and free evolution, or a virtual $\hat{Z}$ gate	waiting time, or the phase of later pulses
$\hat{S}$, $\hat{T}$	$\hat{R}_z(\pi/2)$, $\hat{R}_z(\pi/4)$ up to a phase	virtual $\hat{Z}$ gates	phase of later pulses
$\hat{H}$, $\hat{X}$	calibrated angles $\pi/2$ and π	resonant pulses and virtual $\hat{Z}$ gates	calibrated durations
$\hat{R}_{zz}(\theta)$	$\theta = \int J\, dt$	Ising coupling, Rydberg interaction or tunable coupler	interaction time
$\hat{R}_{xx}(\theta)$	$\theta = \int J\, dt$	Mølmer–Sørensen drive of trapped ions	drive duration
$\sqrt{\widehat{\text{SWAP}}}$, $\widehat{\text{iSWAP}}$	$\int J_{\text{ex}}\, dt = \pi/2$, $\int J\, dt = \pi/2$	exchange of quantum-dot spins or of superconducting circuits	exchange pulse $J_{\text{ex}}(t)$ or $J(t)$
CZ, CNOT	$\hat{R}_{zz}(\pi/2)$ and fixed phases (7.10)	Ising coupling, Rydberg blockade, or cross-resonance drive	calibrated duration

Table 15: Parametrised and fixed gates and the physical processes that set their parameters.

17.6 A register of interacting qubits

A register of a device consists of N two-level systems driven one at a time and coupled in pairs. Examples are atoms held in an array of optical tweezers, ions in a linear trap and superconducting circuits on a chip. Driving qubit q with the Rabi frequency Ω_q and the detuning Δ_q adds the one-qubit Hamiltonian $\frac{\Omega_q}{2}\hat{X}_q + \frac{\Delta_q}{2}\hat{Z}_q$ of (16.6). Qubits that are close enough, or wired together, interact in pairs, and these pairs are the edges of the *connectivity graph* of the device (Figure 61). With an Ising coupling of strength $J_{qq'}$ on each edge the device Hamiltonian is

$$\hat{\mathcal{H}}_{\text{dev}} = \sum_{q=0}^{N-1} \left(\frac{\Omega_q}{2}\hat{X}_q + \frac{\Delta_q}{2}\hat{Z}_q \right) + \sum_{(q,q')\in\text{edges}} \frac{J_{qq'}}{2}\, \hat{Z}_q \hat{Z}_{q'}, \tag{17.6}$$

a sum of Pauli strings whose coefficients the control electronics switch on and off. The factor $\frac{1}{2}$ matches the two-qubit rotations of Section 7. A quantum circuit is a schedule of such pulses. Lab 068 builds this Hamiltonian as a sparse matrix for a chain and a grid.

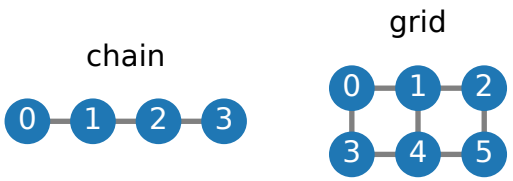


Figure 61: Two connectivity graphs of a register. Each disc is a qubit and each line a pair of qubits that interact directly. In a chain every qubit has at most two neighbours, and in a grid the inner qubits have three or four.

A blockade gate

The blockade gate of Ref. [4] uses the Rydberg interaction without encoding the qubit in the Rydberg state, because Rydberg states live only about a hundred microseconds [5], too short to store a qubit

during a long circuit. Each atom stores its qubit in two long-lived hyperfine ground levels $|0\rangle$ and $|1\rangle$ (Section 16.2), the *hyperfine encoding*, and the Rydberg state is occupied only during the gate. A laser couples only $|1\rangle$ to the Rydberg state $|r\rangle$ with the Rabi frequency Ω. The Hamiltonian of the two three-level atoms is

$$\hat{\mathcal{H}} = \frac{\Omega_0(t)}{2}\left(|1\rangle\langle r| + |r\rangle\langle 1|\right)_0 + \frac{\Omega_1(t)}{2}\left(|1\rangle\langle r| + |r\rangle\langle 1|\right)_1 + V|rr\rangle\langle rr|, \tag{17.7}$$

where the subscript names the atom on which the bracket acts and $\Omega_q(t)$ equals Ω while the laser on atom q is on. The gate consists of a π pulse on the control atom 0, a 2π pulse on the target atom 1 and a second π pulse on the control. A 2π pulse takes $|1\rangle$ to $|r\rangle$ and back and multiplies it by -1. The state $|00\rangle$ is untouched, while $|01\rangle$ and $|10\rangle$ each pass one full cycle and change sign. In $|11\rangle$ the first pulse excites the control, and the interaction V shifts the target transition out of resonance. The target therefore stays in $|1\rangle$, the *blockade*, and the third pulse returns the control with a sign change. The resulting gate is $\mathrm{diag}(1,-1,-1,-1) = (\hat{Z}\otimes\hat{Z})\,\widehat{\mathrm{CZ}}$, a controlled-Z gate up to a $\hat{Z}$ gate on each qubit.

A finite interaction leaves the blockade imperfect. The off-resonant target drive shifts the energy of $|r1\rangle$, the control in $|r\rangle$ and the target in $|1\rangle$, by about $\Omega^2/(4V)$, the light shift (A.18). During the 2π pulse of duration $2\pi/\Omega$ the state $|11\rangle$ acquires the extra phase $\pi\Omega/(2V)$, and the gate error grows as its square, in proportion to $(\Omega/V)^2$. The doubly excited state $|rr\rangle$ is populated only transiently, with a probability of order $(\Omega/V)^2$. We measure the error by the *average gate fidelity* F, the overlap $|\langle\psi|\hat{U}_{\mathrm{ideal}}^{\dagger}\hat{U}|\psi\rangle|^2$ of the simulated and the ideal output averaged over all pure input states $|\psi\rangle$. We allow one $\hat{Z}$ rotation on each qubit before the comparison, applied as a virtual $\hat{Z}$ gate at no cost in time. With this correction the simulated error $1-F$ falls as $(\Omega/V)^2$ over two decades of V/Ω (Figure 62), and an error of 10^{-3} requires an interaction about ten times stronger than the Rabi frequency. This condition fixes how close the atoms must be and which Rydberg level is used.

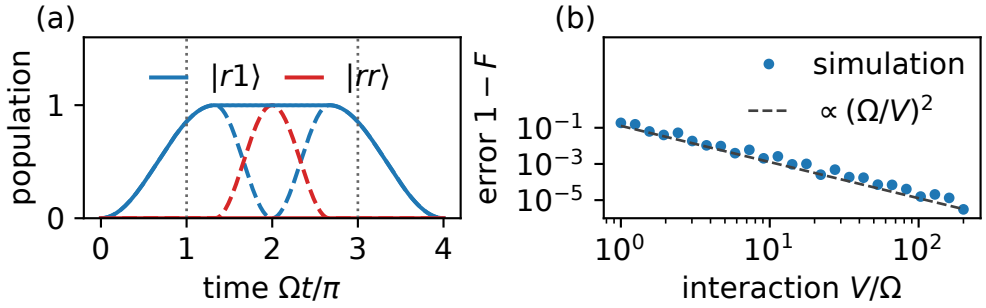


Figure 62: The Rydberg blockade gate. (a) The populations of $|r1\rangle$ (control in $|r\rangle$, target in $|1\rangle$) and $|rr\rangle$ during the three pulses applied to $|11\rangle$, a π pulse on the control, a 2π pulse on the target and a π pulse on the control, separated by the dotted lines. Without interaction (dashed) the target pulse carries the population through $|rr\rangle$, and with $V = 20\,\Omega$ (solid) $|rr\rangle$ never holds more than 2.5×10^{-3}. (b) The gate error $1-F$ after the $\hat{Z}$ corrections against V/Ω. It is 1.3×10^{-3} at $V = 10\,\Omega$ and 7.7×10^{-5} at $V = 40\,\Omega$ and follows the line $\propto(\Omega/V)^2$, with the fitted slope -2.0 on logarithmic axes. An error of 10^{-3} requires $V \approx 11\,\Omega$.

On a chain, where only neighbours interact, a gate between distant qubits is routed with the SWAP chain of Section 7.5. Qubits at distance d need $2(d-1)$ SWAP gates, and each SWAP costs three CNOT gates. A controlled-Z gate between the ends of a four-qubit chain therefore costs 4 SWAP gates, or 12 CNOT gates, in addition to the gate itself. Richer connectivity graphs reduce this overhead. So do qubits that can be moved, such as atoms in tweezers or ions in segmented traps, whose electrodes are divided into zones that shuttle ions between them.

Summary

The results of both parts are collected here in the order of their derivation, each under the number of the section that derives it. The rules of Part I hold for every qubit, and the results of Part II describe how particular physical systems obey them.

Part I Introduction to quantum information theory

1 The four rules

1. *State.* N qubits are described by a unit vector $|\psi\rangle \in \mathbb{C}^{2^N}$, up to a global phase, or by a density matrix $\hat{\rho}$ when the preparation is uncertain.
2. *Evolution.* $|\psi\rangle \to \hat{U}|\psi\rangle$ with a unitary $\hat{U} = e^{-it\hat{\mathcal{H}}}$ for a constant Hamiltonian.
3. *Measurement.* The bit string b appears with the probability $|\langle b|\psi\rangle|^2 = \mathrm{Tr}(|b\rangle\langle b|\hat{\rho})$.
4. *Composition.* Systems combine by the tensor product, computed as the Kronecker product.

2 One qubit and the Pauli algebra

A qubit is a two-level system, with the ground state $|g\rangle = |0\rangle = |\uparrow\rangle$ and the excited state $|e\rangle = |1\rangle = |\downarrow\rangle$. Its state $\alpha|0\rangle + \beta|1\rangle$ with $|\alpha|^2 + |\beta|^2 = 1$ is a point of the Bloch sphere with the Bloch vector $\mathbf{r} = (\langle\hat{X}\rangle, \langle\hat{Y}\rangle, \langle\hat{Z}\rangle)$, and $p(0) = |\alpha|^2 = (1 + r_z)/2$. The Pauli matrices obey (2.12),

$$\hat{\sigma}_a\hat{\sigma}_b = \delta_{ab}\hat{I} + i\sum_c \varepsilon_{abc}\hat{\sigma}_c, \qquad \mathrm{Tr}(\hat{\sigma}_a\hat{\sigma}_b) = 2\delta_{ab}, \qquad \hat{M} = c_0\hat{I} + \mathbf{c}\cdot\hat{\boldsymbol{\sigma}}.$$

The ladder operators $\hat{S}^{\pm} = \hat{S}_x \pm i\hat{S}_y$ (2.16) move between the two basis states. With $|0\rangle = |g\rangle = |\uparrow\rangle$, the decay is $|g\rangle\langle e| = \hat{S}^+$ and the excitation $|e\rangle\langle g| = \hat{S}^-$ (12.13), so $\hat{S}^+$ raises the spin but lowers the energy.

3 Evolution and gates as rotations

The Schrödinger equation $i\frac{d}{dt}|\psi\rangle = \hat{\mathcal{H}}|\psi\rangle$ with a time-independent $\hat{\mathcal{H}}$ is solved by

$$\hat{U}(t) = e^{-it\hat{\mathcal{H}}} = \sum_{k\geq 0}\frac{(-it\hat{\mathcal{H}})^k}{k!} = \sum_n e^{-iE_nt}|E_n\rangle\langle E_n|, \qquad \hat{U}(t_1+t_2) = \hat{U}(t_1)\hat{U}(t_2),$$

where the last form uses the eigenvalues E_n and eigenvectors $|E_n\rangle$ of $\hat{\mathcal{H}}$ (3.5). For any matrix with $\hat{P}^2 = \hat{I}$ (3.6),

$$e^{-ia\hat{P}} = \cos a\,\hat{I} - i\sin a\,\hat{P}, \qquad \hat{R}_{\mathbf{n}}(\theta) = e^{-i\theta\,\mathbf{n}\cdot\hat{\boldsymbol{\sigma}}/2},$$

and every single-qubit gate is a rotation of the Bloch sphere up to a global phase, $\hat{X} = i\hat{R}_x(\pi)$, $\hat{Z} = i\hat{R}_z(\pi)$, $\hat{H} = i\hat{R}_{\mathbf{n}_H}(\pi)$, $\hat{S} = e^{i\pi/4}\hat{R}_z(\pi/2)$, $\hat{T} = e^{i\pi/8}\hat{R}_z(\pi/4)$ (Table 2).

4 Other bases and shots

A measurement in the $\hat{X}$ basis is $\hat{H}$ followed by a $\hat{Z}$ measurement, and in the $\hat{Y}$ basis $\hat{S}^\dagger$, then $\hat{H}$, then $\hat{Z}$. An expectation value estimated from M shots has the standard error $\sqrt{(1-\langle\hat{O}\rangle^2)/M}$ (4.3), which falls as $1/\sqrt{M}$.

5 The register in a computer

The state of N qubits is stored as 2^N complex amplitudes in the computational basis, the common eigenbasis of $\hat{Z}_0, \ldots, \hat{Z}_{N-1}$. The Kronecker product fixes the order,

$$\begin{pmatrix} a \\ b \end{pmatrix} \otimes \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} ac \\ ad \\ bc \\ bd \end{pmatrix}, \qquad i = \sum_{q=0}^{N-1} b_q \, 2^{N-1-q},$$

so qubit 0 is the most significant bit (Table 3). Operators follow the same rule (5.9),

$$(\hat{A}\otimes\hat{B})_{(b_0b_1),(b_0'b_1')} = A_{b_0b_0'}B_{b_1b_1'}, \qquad (\hat{A}\otimes\hat{B})(|u\rangle\otimes|v\rangle) = \hat{A}|u\rangle\otimes\hat{B}|v\rangle, \qquad (\hat{A}\otimes\hat{B})(\hat{C}\otimes\hat{D}) = \hat{A}\hat{C}\otimes\hat{B}\hat{D}.$$

A gate $\hat{G}$ on qubit q of N is $\hat{I}\otimes\cdots\otimes\hat{G}\otimes\cdots\otimes\hat{I}$, and $\widehat{\mathrm{CNOT}}_{i\to j} = |0\rangle\langle 0|_i \otimes \hat{I} + |1\rangle\langle 1|_i \otimes \hat{X}_j$ with identities on all other qubits (Section 7.5). These matrices are sparse, with $2\cdot 2^N$ or 2^N nonzero entries out of 4^N. Pauli strings $\hat{P}_0\otimes\cdots\otimes\hat{P}_{N-1}$ square to the identity and have the eigenvalues ± 1. They form a basis in which every Hamiltonian is a sum $\hat{\mathcal{H}} = \sum_{\hat{P}} c_P \hat{P}$ with $c_P = \mathrm{Tr}(\hat{P}\hat{\mathcal{H}})/2^N$. On a computer every one of these operations is also a tensor contraction of the state tensor $\psi[b_0, \ldots, b_{N-1}]$ (Table 4).

7 Two-qubit gates as rotations

An Ising coupling switched on for a time gives $\hat{R}_{\alpha\alpha}(\theta) = \cos\frac{\theta}{2}\hat{I} - i\sin\frac{\theta}{2}\,\hat{\sigma}_\alpha\otimes\hat{\sigma}_\alpha$ (7.2). $\widehat{\mathrm{CNOT}}$ is the conditional half turn about x, the reversible XOR $(b_0, b_1) \to (b_0, b_0 \oplus b_1)$, and $\widehat{\mathrm{CZ}}$, $\widehat{\mathrm{SWAP}}$ and Toffoli are exponentials of Pauli products in the same way. The angle of a two-qubit rotation is the integrated coupling (7.14), and any entangling two-qubit gate together with single-qubit rotations is universal.

8 Entanglement

Reshaping the four amplitudes into the 2×2 matrix $\Psi_{b_0b_1} = \psi_{b_0b_1}$ and decomposing Ψ by the SVD gives the Schmidt decomposition (8.6),

$$|\psi\rangle = \sum_k s_k |u_k\rangle\otimes|v_k\rangle, \qquad s_1 \geq s_2 \geq 0, \qquad s_1^2 + s_2^2 = 1.$$

Schmidt rank one means a product state. The entanglement entropy is $S = -\sum_k \lambda_k \log_2\lambda_k$ with $\lambda_k = s_k^2$, zero for a product state and one bit for the Bell state $(|00\rangle + |11\rangle)/\sqrt{2} = \widehat{\mathrm{CNOT}}(\hat{H}\otimes\hat{I})|00\rangle$. A gate on one qubit alone leaves the Schmidt coefficients unchanged and cannot create entanglement.

10 Density matrices and the partial trace

A pure state has $\hat{\rho} = |\psi\rangle\langle\psi|$, a rank-one projector with $\hat{\rho}^2 = \hat{\rho}$ and $\mathrm{Tr}\hat{\rho}^2 = 1$. A mixture $\hat{\rho} = \sum_i p_i |\psi_i\rangle\langle\psi_i|$ has rank above one and purity $\mathrm{Tr}\hat{\rho}^2 < 1$. For one qubit $\hat{\rho} = \frac{1}{2}(\hat{I} + \mathbf{r}\cdot\hat{\boldsymbol{\sigma}})$ with $|\mathbf{r}| \leq 1$ and $\mathrm{Tr}\hat{\rho}^2 = (1+|\mathbf{r}|^2)/2$ (10.8). The reduced state of qubit 0 is

$$(\hat{\rho}_0)_{aa'} = \sum_b \psi_{ab}\psi^*_{a'b}, \qquad \hat{\rho}_0 = \Psi\Psi^\dagger, \qquad \hat{\rho}_1 = (\Psi^\dagger\Psi)^T,$$

equivalently the traces of the four 2×2 blocks of the 4×4 matrix $\hat{\rho}$ (10.13). The eigenvalues of $\hat{\rho}_0$ are $\lambda_k = s_k^2$. The entropy of a reduced state measures entanglement only when the whole register is pure (Table 10). The classical correlated coin and one qubit of a Bell pair have the same reduced state.

11 Measurement

A projective measurement has $p_k = \mathrm{Tr}(\hat{P}_k\hat{\rho})$ and leaves $\hat{\rho}_k = \hat{P}_k\hat{\rho}\hat{P}_k/p_k$ (11.1). An unread measurement removes the coherences between different outcome sectors, which is dephasing. The outcome on one qubit of a Bell pair fixes the state of the other, and no signal is transmitted. A Pauli string is measured as a parity, either by rotating each qubit to the $\hat{Z}$ basis or with an ancilla and CNOT gates (Section 11.2). A general measurement is a POVM, positive $\hat{\pi}_k$ with $\sum_k \hat{\pi}_k = \hat{I}$ and $p_k = \mathrm{Tr}(\hat{\pi}_k\hat{\rho})$, for example a detector with readout error (11.7). Informationally complete measurements, the 3^N Pauli settings or the tetrahedral POVM, determine $\hat{\rho}$ (Section 11.8).

12 Qubit errors and noise channels

A channel is $\mathcal{E}(\hat{\rho}) = \sum_k \hat{K}_k\hat{\rho}\hat{K}_k^\dagger$ with $\sum_k \hat{K}_k^\dagger\hat{K}_k = \hat{I}$ (12.4), obtained from a unitary coupling to an environment that is then traced out. The six standard error models of Table 11 deform the Bloch ball,

bit flip, phase flip, bit-phase flip	two axes shrink by $1-2p$, one is kept
depolarising	$\mathbf{r} \to (1-\frac{4}{3}p)\mathbf{r}$
amplitude damping	$\mathbf{r} \to (\sqrt{1-\gamma}\,r_x, \sqrt{1-\gamma}\,r_y, \gamma+(1-\gamma)r_z)$, towards $\lvert g\rangle$
phase damping	$\mathbf{r} \to (\sqrt{1-\kappa}\,r_x, \sqrt{1-\kappa}\,r_y, r_z)$

The Choi state $\hat{\rho}_\mathcal{E} = (\mathcal{E}\otimes\mathrm{id})(|\Phi^+\rangle\langle\Phi^+|)$ (12.16) stores a channel as one two-qubit state. The map is completely positive if and only if $\hat{\rho}_\mathcal{E} \geq 0$ and trace preserving if and only if $\mathrm{Tr}_0\hat{\rho}_\mathcal{E} = \hat{I}/2$, its eigenvectors give Kraus operators, and the transpose, with $\hat{\rho}_\mathcal{E} = \widehat{\mathrm{SWAP}}/2$, is positive but not completely positive. In continuous time the Lindblad equation (12.21) with the decay jump $\sqrt{\Gamma}|g\rangle\langle e|$ gives amplitude damping with $\gamma = 1 - e^{-t/T_1}$, and pure dephasing gives phase damping, with $1/T_2 = 1/(2T_1) + \gamma_\phi$ (12.23).

12.8 Noise in quantum circuits

In the standard model every gate is followed by depolarising noise on the qubits it touches, p_1 after one-qubit gates and p_2 after two-qubit gates (12.24). Waiting qubits decay with T_1 and T_2 in every layer, and readout errors act as a confusion matrix on the bit-string probabilities. Coherent errors add in amplitude and grow as $(n\epsilon)^2$, while stochastic errors add in probability and grow as np. The fidelity of a GHZ state prepared with $N-1$ CNOT gates obeys the exponentially falling bound $F \geq (1-\frac{4}{3}p_1)(1-\frac{16}{15}p_2)^{N-1}$ (12.26), and the full model gives $F = 0.915$ for $N = 4$. The distance to a target is measured by the fidelity $F = \langle\psi|\hat{\sigma}|\psi\rangle$ and the trace distance (Section 13).

15 What quantum information allows and forbids

No unitary copies two states that are neither equal nor orthogonal (15.2). One copy of $\hat{\rho}$ or $\hat{\sigma}$ is identified with probability at most $\frac{1}{2}(1+D)$, reached by projecting on the positive part of $\hat{\rho}-\hat{\sigma}$ (15.4). A measurement on N qubits yields at most $\chi \leq N$ bits of mutual information about the encoded message (15.7). CNOT and rotations build every unitary, controlled gates by (15.11), but a generic unitary needs at least $(4^N-1)/15$ two-qubit gates (15.9). Clifford gates map Pauli strings to Pauli strings (15.12), so Clifford circuits on $|0\ldots0\rangle$ with Pauli measurements are simulated efficiently, and $\hat{T}$ makes the set universal. Haar-random states have $\mathrm{Var}\,\langle\hat{P}\rangle = 1/(2^N+1)$ (15.15), reproduced by the Clifford 2-design. A negative eigenvalue of the partial transpose proves entanglement, and the Werner state (15.19) is entangled for $v > 1/3$. The quantum Fisher information $F_Q = 4\,\mathrm{Var}(\hat{J}_z)$ bounds the phase error by $1/\sqrt{MF_Q}$, with $F_Q \leq N$ for separable states and $F_Q = N^2$ for the GHZ state (15.24).

Part II Physical realisations of qubits

16 Single physical qubits and their drives

A free two-level atom with the gap ω_0, $\hat{\mathcal{H}}_0 = -\frac{\omega_0}{2}\hat{Z}$, is driven by a laser of frequency ω_L and Rabi frequency Ω (16.1). In the rotating frame $|\psi_{\mathrm{RF}}\rangle = \hat{R}_z(\omega_L t)|\psi\rangle$ and after the rotating-wave approximation it evolves with (16.5), $\hat{\mathcal{H}}_{\mathrm{RF}} = \frac{\Delta}{2}\hat{Z} + \frac{\Omega}{2}\hat{X}$ with $\Delta = \omega_L - \omega_0$ (derived in Appendix A). On resonance the excited-state probability oscillates as $\sin^2(\Omega t/2)$, a π pulse is an $\hat{X}$ gate up to a phase, and a $\pi/2$ pulse creates an equal superposition. A resonant pulse of area θ and phase ϕ applies the rotation $\hat{R}_\phi(\theta)$ of (16.11), and rotations about z are applied as phase updates of later pulses, the virtual $\hat{Z}$ gates. These rules hold for any qubit, and the transmon and the spin-1/2 in a magnetic field obey the same rotating-frame Hamiltonian (16.6). The anharmonicity α of the transmon sets the shortest pulse without leakage, $T \gtrsim 12/|\alpha|$ (16.21). The platforms differ in how the two levels are obtained, driven, coupled and read out (Table 13) and in how long their superpositions survive.

17 Interacting qubits and two-qubit gates in the laboratory

Each platform couples its qubits through a physical interaction that is a sum of Pauli strings (Table 14). The Rydberg interaction of two atoms is an Ising coupling plus a detuning of each atom (17.1), the Mølmer–Sørensen drive of two ions applies $\hat{R}_{xx}$ (17.4) after a closed loop of the shared motion (Appendix B), and exchange couplings apply $\widehat{\mathrm{iSWAP}}$ and $\sqrt{\widehat{\mathrm{SWAP}}}$. The angle of every gate is a pulse area or an integrated coupling (Table 15), and a circuit is a timed sequence of device Hamiltonians (17.5). On a chain, distant qubits are joined by SWAP gates. The error of the Rydberg blockade gate falls as $(\Omega/V)^2$ with the interaction strength V.

A Derivation of the qubit Hamiltonian of an atom in laser light

This appendix derives the two-level Hamiltonian (16.1) and the rotating-frame Hamiltonian (16.5) of Section 16.2 from the Hamiltonian of the electrons and the nucleus, following Refs. [2, 3]. We keep $\hbar$ explicit until the end of the derivation, and hydrogen serves as the example wherever numbers are needed.

A.1 The atom and its Hamiltonian

An atom consists of a nucleus with the charge $Z_{\mathrm{at}}e_0$, where Z_{at} is the atomic number, and the mass M, and of N_e electrons with the charge $-e_0$ and the mass m_e. Here e_0 is the elementary charge. The nucleus has the position operator $\hat{\mathbf{R}}$ and the momentum operator $\hat{\mathbf{P}}$, and electron i has $\hat{\mathbf{r}}_i$ and $\hat{\mathbf{p}}_i$. The Hamiltonian of the atom is

$$\hat{\mathcal{H}}_{\mathrm{atom}} = \frac{\hat{\mathbf{P}}^2}{2M} + \sum_{i=1}^{N_e} \frac{\hat{\mathbf{p}}_i^2}{2m_e} - \sum_{i=1}^{N_e} \frac{Z_{\mathrm{at}} e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{R}}|} + \sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}, \tag{A.1}$$

with the vacuum permittivity ε_0. The first two terms are the kinetic energies of the nucleus and of the electrons. The third is the attraction between every electron and the nucleus, and the fourth is the repulsion between pairs of electrons. Two small effects are omitted at this level. Relativistic corrections split levels of equal energy slightly, which is called the fine structure, and the magnetic coupling of the electron spin to the nuclear spin produces the hyperfine splitting described in Section 16.2.

A.2 Centre-of-mass separation

For a single atom the motion of the centre of mass separates from the relative motion of the particles without approximation. The nucleus is heavier than an electron by the factor $M/m_e \approx 1823\,A$, with the mass number A, and by 1836 for the proton, so the centre of mass nearly coincides with the nucleus. We place it at the origin, $\mathbf{R} = 0$, and the electrons obey the *electronic Hamiltonian*

$$\hat{\mathcal{H}}_{\mathrm{el}} = \sum_{i=1}^{N_e} \frac{\hat{\mathbf{p}}_i^2}{2m_e} - \sum_{i=1}^{N_e} \frac{Z_{\mathrm{at}} e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i|} + \sum_{i<j} \frac{e_0^2}{4\pi\varepsilon_0 |\hat{\mathbf{r}}_i - \hat{\mathbf{r}}_j|}, \tag{A.2}$$

whose eigenvalues are the energy levels of the atom. For hydrogen the separation replaces m_e by the reduced mass $\mu = m_e M/(m_e + M)$, which shifts the levels by about $0.05\,\%$.

A.3 Energy eigenstates as a basis

The bound eigenstates of the electronic Hamiltonian form a discrete set,

$$\hat{\mathcal{H}}_{\mathrm{el}} |j\rangle = E_j |j\rangle, \qquad \langle k|j\rangle = \delta_{jk}, \tag{A.3}$$

where the label j collects all quantum numbers of the state. Every bound state of the atom is a superposition

$$|\psi(t)\rangle = \sum_j c_j(t)\, |j\rangle, \qquad c_j = \langle j|\psi\rangle, \tag{A.4}$$

and $|c_j|^2$ is the probability of finding the atom in the state $|j\rangle$, with $\sum_j |c_j|^2 = 1$. Without light each amplitude only acquires a phase, $c_j(t) = c_j(0)\, e^{-iE_j t/\hbar}$. Inserting (A.4) into the Schrödinger equation $i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{\mathcal{H}}|\psi(t)\rangle$ with any Hamiltonian $\hat{\mathcal{H}}$ and projecting on $\langle j|$ gives the linear system

$$i\hbar\, \dot{c}_j = \sum_k H_{jk}\, c_k, \qquad H_{jk} = \langle j|\hat{\mathcal{H}}|k\rangle, \tag{A.5}$$

where the dot denotes the time derivative. The vector of amplitudes is the state vector of Section 2, and (A.5) is the matrix form (3.1) of the Schrödinger equation.

A.4 Coupling to classical light

A laser produces the electric field $\mathbf{E}(t) = E_0\,\mathbf{e}_z\cos(\omega_L t - \phi_L)$, which we take polarised along z, with the amplitude E_0, the angular frequency ω_L and the phase ϕ_L. A point charge q at the position $\mathbf{r}$ in a uniform field $\mathbf{E}$ has the potential energy $-q\,\mathbf{E}\cdot\mathbf{r}$. An optical wavelength of about a hundred nanometres or more is much larger than the atom, so the field is uniform across it, which is the *dipole approximation*. Summing $-q\,\mathbf{E}\cdot\mathbf{r}$ over the electrons, with $q = -e_0$, and over the nucleus at the origin gives

$$\hat{\mathcal{H}}_{\text{int}}(t) = -\hat{\mathbf{d}}\cdot\mathbf{E}(t), \qquad \hat{\mathbf{d}} = -e_0\sum_{i=1}^{N_e}\hat{\mathbf{r}}_i, \tag{A.6}$$

with the *electric dipole operator* $\hat{\mathbf{d}}$ of the atom and its matrix elements $d_{jk} = \langle j|\hat{\mathbf{d}}\cdot\mathbf{e}_z|k\rangle$. The electronic Hamiltonian is unchanged when all electron positions change sign. Its eigenstates can therefore be chosen even or odd under this *parity* operation, whereas $\hat{\mathbf{d}}$ changes sign. Hence $d_{jj} = 0$, and the light couples only levels of opposite parity, the *selection rule* of Section 16.2.

A.5 Two levels near resonance

The Hamiltonian matrix is $H_{jk}(t) = E_j\delta_{jk} - d_{jk}E_0\cos(\omega_L t - \phi_L)$. The free evolution makes every amplitude rotate fast at its own frequency, and we remove this rotation by writing

$$c_j(t) = a_j(t)\,e^{-iE_j t/\hbar}, \tag{A.7}$$

a change of variables called the *interaction picture*. Inserting this into (A.5), the terms $E_j c_j$ cancel and

$$i\hbar\,\dot{a}_j = -E_0\cos(\omega_L t - \phi_L)\sum_k d_{jk}\,e^{i\omega_{jk}t}\,a_k, \qquad \omega_{jk} = \frac{E_j - E_k}{\hbar}. \tag{A.8}$$

The amplitudes a_j now change only because of the light. Writing the cosine as $\frac{1}{2}(e^{i(\omega_L t-\phi_L)} + e^{-i(\omega_L t-\phi_L)})$ splits every coupling into two terms,

$$e^{i\omega_{jk}t}\cos(\omega_L t - \phi_L) = \tfrac{1}{2}\,e^{i\phi_L}e^{i(\omega_{jk}-\omega_L)t} + \tfrac{1}{2}\,e^{-i\phi_L}e^{i(\omega_{jk}+\omega_L)t}. \tag{A.9}$$

With the coupling strength $\Omega_{jk} = -d_{jk}E_0/\hbar$ of the pair j,k, a term that oscillates much faster than Ω_{jk} averages to nearly zero before it can move population. A pair of levels therefore stays uncoupled when

$$|\omega_{jk} - \omega_L| \gg |\Omega_{jk}| \qquad \text{and} \qquad |\omega_{jk} + \omega_L| \gg |\Omega_{jk}|. \tag{A.10}$$

We tune the laser close to the transition between a ground state g and an excited state e of opposite parity, $\omega_L \approx \omega_0 = \omega_{eg}$, and choose the phases of $|g\rangle$ and $|e\rangle$ so that $d_{eg} = d_{ge}$ is real. For $j = e$, $k = g$ the first term of (A.9) varies slowly at the *detuning* $\Delta = \omega_L - \omega_0$, while the second oscillates at nearly twice the optical frequency. For $j = g$, $k = e$, with $\omega_{ge} = -\omega_0$, the roles are exchanged,

$$\begin{aligned} e^{i\omega_0 t}\cos(\omega_L t - \phi_L) &= \tfrac{1}{2}\,e^{i\phi_L}e^{-i\Delta t} + \tfrac{1}{2}\,e^{-i\phi_L}e^{i(\omega_0+\omega_L)t} \qquad (j=e,\ k=g),\\ e^{-i\omega_0 t}\cos(\omega_L t - \phi_L) &= \tfrac{1}{2}\,e^{-i\phi_L}e^{i\Delta t} + \tfrac{1}{2}\,e^{i\phi_L}e^{-i(\omega_0+\omega_L)t} \qquad (j=g,\ k=e). \end{aligned} \tag{A.11}$$

Every other allowed transition satisfies both conditions of (A.10) for a weak field, because its frequency differs from ω_L by much more than its coupling strength. Keeping only the slow terms of the pair g, e, which is the *rotating-wave approximation*, leaves two coupled amplitudes,

$$i\dot{a}_g = \frac{\Omega}{2}\,e^{-i\phi_L}e^{i\Delta t}\,a_e, \qquad i\dot{a}_e = \frac{\Omega}{2}\,e^{i\phi_L}e^{-i\Delta t}\,a_g, \qquad \Omega = \Omega_{eg} = -\frac{d_{eg}E_0}{\hbar}, \tag{A.12}$$

where the first equation is the complex conjugate partner of the second. For $|\Omega|, |\Delta| \ll \omega_0$ all other levels stay nearly empty, and the atom has become a two-level system with the state vector (a_g, a_e). Lab 067 shows the averaging numerically. Of the time-averaged coupling matrix of the ten lowest orbitals of hydrogen only the entry of the pair 1s, 2p survives. The argument uses only discrete levels and dipole couplings, so it holds for every atom and ion with one valence electron.

A.6 The qubit Hamiltonian

The factors $e^{\pm i\Delta t}$ disappear in the amplitudes $b_g = a_g e^{-i\Delta t/2}$ and $b_e = a_e e^{i\Delta t/2}$. Differentiating, $i\dot{b}_g = i\dot{a}_g\, e^{-i\Delta t/2} + \frac{\Delta}{2} b_g$ and $i\dot{b}_e = i\dot{a}_e\, e^{i\Delta t/2} - \frac{\Delta}{2} b_e$, and inserting (A.12) gives a Schrödinger equation with a constant 2×2 matrix,

$$i\frac{d}{dt}\begin{pmatrix} b_g \\ b_e \end{pmatrix} = \begin{pmatrix} \frac{\Delta}{2} & \frac{\Omega}{2} e^{-i\phi_L} \\ \frac{\Omega}{2} e^{i\phi_L} & -\frac{\Delta}{2} \end{pmatrix} \begin{pmatrix} b_g \\ b_e \end{pmatrix}. \tag{A.13}$$

The diagonal is $\frac{\Delta}{2}\hat{Z}$, and the off-diagonal entries $\frac{\Omega}{2}(\cos\phi_L \mp i \sin\phi_L)$ are those of $\frac{\Omega}{2}(\cos\phi_L \hat{X} + \sin\phi_L \hat{Y})$, so the matrix is the rotating-frame Hamiltonian (16.5) with $\hbar = 1$. With the zero of energy halfway between the levels, $E_g = -\hbar\omega_0/2$ and $E_e = +\hbar\omega_0/2$, the amplitudes are $b_g = c_g\, e^{-i\omega_L t/2}$ and $b_e = c_e\, e^{i\omega_L t/2}$, which is the rotating-frame state (16.2). In the original amplitudes c_g, c_e the same two levels obey the laboratory-frame Hamiltonian (16.1), $-\frac{\omega_0}{2}\hat{Z}$ from the energies and $\Omega\cos(\omega_L t - \phi_L)\hat{X}$ from the dipole coupling, so the two routes of Section 16.2 and of this appendix agree.

A.7 Hydrogen as an example

Hydrogen has $Z_{\rm at} = N_e = 1$, and its electronic Hamiltonian is $\hat{\mathcal{H}}_{\rm H} = \hat{\mathbf{p}}^2/2m_e - e_0^2/(4\pi\varepsilon_0|\hat{\mathbf{r}}|)$. Its eigenstates $|nlm\rangle$ carry the principal quantum number $n = 1, 2, \ldots$, the orbital angular momentum $l = 0, \ldots, n-1$ and its projection $m = -l, \ldots, l$ on the z axis, and the energies

$$E_n = -\frac{\mathrm{Ry}}{n^2} \tag{A.14}$$

depend only on n, with the Rydberg energy $\mathrm{Ry} \approx 13.6$ eV. States with $l = 0, 1, 2$ are labelled s, p, d. In the position representation the eigenstates are $\langle \mathbf{r}|nlm\rangle = R_{nl}(r)\, Y_{lm}(\vartheta, \varphi)$, a radial function times a spherical harmonic. A spherical harmonic Y_{lm} is a standard function of the two angles of the position vector that fixes the angular shape of the state. Two examples are

$$\langle \mathbf{r}|1s\rangle = \frac{2}{a_0^{3/2}}\, e^{-r/a_0}\, \frac{1}{\sqrt{4\pi}}, \qquad \langle \mathbf{r}|2p_0\rangle = \frac{1}{2\sqrt{6}\, a_0^{3/2}}\, \frac{r}{a_0}\, e^{-r/2a_0} \sqrt{\frac{3}{4\pi}} \cos\vartheta, \tag{A.15}$$

with the Bohr radius $a_0 \approx 0.053$ nm. For light polarised along z the matrix element of $\hat{z}$ factors into a radial integral and an angular integral,

$$\langle n'l'm'|\hat{z}|nlm\rangle = \int_0^\infty R_{n'l'} R_{nl}\, r^3\, dr \int Y^*_{l'm'} \cos\vartheta\, Y_{lm}\, d\Omega_{\rm s}, \tag{A.16}$$

where $d\Omega_{\rm s}$ is the element of solid angle. The angular integral vanishes unless $m' = m$ and $l' = l \pm 1$, the selection rules for this polarisation, and for $m = 0$ it equals $(l+1)/\sqrt{(2l+1)(2l+3)}$ when $l' = l+1$. The rule $l' \neq l$ is the parity rule, since $|nlm\rangle$ has the parity $(-1)^l$, so 1s does not couple to 2s. For the 1s–2p transition the radial integral and the angular factor $1/\sqrt{3}$ give the standard result

$$\langle 2p_0|\hat{z}|1s\rangle = \frac{128\sqrt{2}}{243}\, a_0 \approx 0.745\, a_0, \qquad \Omega = \frac{e_0 E_0}{\hbar}\, \langle 2p_0|\hat{z}|1s\rangle, \tag{A.17}$$

since $d_{eg} = -e_0\langle 2p_0|\hat{z}|1s\rangle$. The transition energy is $\hbar\omega_0 = E_2 - E_1 = \frac{3}{4}\mathrm{Ry} \approx 10.2$ eV. It is the first line of the ultraviolet Lyman series of hydrogen, called Lyman-α, at the wavelength 121.5 nm for an infinitely heavy nucleus. The other allowed transitions, from 1s to 3p and higher p states and from 2p to 3s and 3d, lie far from this frequency, so the qubit $|g\rangle = |1s\rangle = |0\rangle$, $|e\rangle = |2p_0\rangle = |1\rangle$ is well isolated for a weak field.

A.8 Light shift, Bloch–Siegert shift and decay

Far-detuned states are only nearly empty. A state r coupled to the ground state with the Rabi frequency Ω_r and the detuning $\Delta_r = \omega_L - \omega_{rg}$ forms with g a pair with the rotating-frame Hamiltonian (16.5),

whose eigenvalues are $\pm\frac{1}{2}\sqrt{\Delta_r^2+\Omega_r^2}$. The eigenvalue that tends to the energy $+\Delta_r/2$ of g in this frame for $\Omega_r \to 0$, expanded to second order in Ω_r, is $\frac{\Delta_r}{2}+\frac{\Omega_r^2}{4\Delta_r}$, so the ground state is shifted by

$$\delta E_g = \frac{\Omega_r^2}{4\Delta_r}, \tag{A.18}$$

the *light shift*, which moves the resonance of the qubit transition. The rotating-wave approximation requires $|\Omega|, |\Delta| \ll \omega_L$, and the dropped counter-rotating terms raise the resonance frequency by $\Omega^2/(4\omega_0)$, the *Bloch–Siegert shift*.

An excited atomic level decays by spontaneous emission with the rate

$$\Gamma = \frac{\omega_0^3\,|d_{eg}|^2}{3\pi\varepsilon_0\hbar c^3}, \tag{A.19}$$

with the speed of light c, a result of quantum electrodynamics. With d_{eg} from (A.17) the 2p state of hydrogen lives $1/\Gamma \approx 1.60$ ns.

B Derivation of the Mølmer–Sørensen gate

This appendix derives the gate (17.4) of Section 17.3 from the Hamiltonian (17.2) [6, 7]. The shared motion of the ions is a harmonic oscillator, described by the box in Section 16.4 with the lowering operator $\hat{a}$.

B.1 Displacements in phase space

A classical oscillator is described by its position and momentum, a point in the *phase space* spanned by them. For the quantum oscillator the complex number $\langle\hat{a}\rangle = \sqrt{m\omega/2}\,\langle\hat{x}\rangle + i\langle\hat{p}\rangle/\sqrt{2m\omega}$ plays the role of this point, with the scaled mean position as its real part and the scaled mean momentum as its imaginary part. The *displacement operator* $\hat{D}(\beta) = e^{\beta\hat{a}^\dagger - \beta^*\hat{a}}$ shifts the motion in phase space by the complex number β, $\hat{D}(\beta)^\dagger\hat{a}\hat{D}(\beta) = \hat{a}+\beta$, without changing its shape, and two displacements combine as $\hat{D}(\beta_2)\hat{D}(\beta_1) = e^{i\,\mathrm{Im}(\beta_2\beta_1^*)}\hat{D}(\beta_1+\beta_2)$. A displacement around a closed loop therefore returns every motional state to itself up to a phase.

B.2 The closed loop and its phase

The collective spin $\hat{S}_x = \frac{1}{2}(\hat{X}_0+\hat{X}_1)$ commutes with the Hamiltonian (17.2) at all times, so we treat each of its eigenvalues $m = 1, 0, -1$ separately. For a fixed m the Hamiltonian $mg(\hat{a}\,e^{-i\delta t} + \hat{a}^\dagger e^{i\delta t})$ is a force on the oscillator. The mean amplitude obeys $\frac{d}{dt}\langle\hat{a}\rangle = i\langle[\hat{\mathcal{H}},\hat{a}]\rangle = -img\,e^{i\delta t}$ by (16.23), and integration gives the displacement $m\,\alpha(t)$ with $\alpha(t)$ of (17.3), a circle of radius g/δ that returns to the origin at $t = 2\pi K/\delta$. The evolution operator for the eigenvalue m is

$$\hat{U}_m(t) = e^{i\Phi(t)\,m^2}\,\hat{D}\big(m\,\alpha(t)\big), \qquad \Phi(t) = \int_0^t \mathrm{Im}\big[\alpha^*(t')\,\dot{\alpha}(t')\big]\,dt' = \frac{g^2}{\delta}\Big(t - \frac{\sin\delta t}{\delta}\Big), \tag{B.1}$$

which one verifies by differentiating with the composition rule of the displacements. The phase is twice the area enclosed by the path in phase space and grows as m^2. At $t = 2\pi K/\delta$ the displacement vanishes, $\Phi = 2\pi K g^2/\delta^2$, and the motion has returned to its initial state, whatever that state was. Since m^2 is the eigenvalue of $\hat{S}_x^2$, the evolution on the qubits is $e^{i\Phi\hat{S}_x^2}$, and $\hat{S}_x^2 = \frac{1}{2}(\hat{I}+\hat{X}_0\hat{X}_1)$ gives (17.4). Lab 068 checks (B.1) against the numerical evolution of two qubits and one motional mode.

B.3 Effective coupling and sign

Averaged over one loop the phase grows as $\Phi \approx Jt$ with $J = g^2/\delta$, so the qubits evolve as under the effective Hamiltonian

$$\hat{\mathcal{H}}_{\text{eff}} = -J\hat{S}_x^2 = -\frac{J}{2}(\hat{I} + \hat{X}_0\hat{X}_1), \tag{B.2}$$

an $\hat{X}_0\hat{X}_1$ coupling of strength $-J$ plus a constant. Since $e^{i\Phi\hat{X}_0\hat{X}_1/2} = \hat{R}_{xx}(-\Phi)$, the gate rotates by $-\Phi$ for $\delta > 0$. Part I creates a Bell state with $\hat{R}_{xx}(+\pi/2)$, which takes $|00\rangle$ to $(|00\rangle - i|11\rangle)/\sqrt{2}$. The Mølmer–Sørensen loop with $\delta > 0$ applies $\hat{R}_{xx}(-\pi/2)$ and produces $(|00\rangle + i|11\rangle)/\sqrt{2}$, and the opposite sign of δ reverses the rotation.

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